Comparative efficacy and safety of Janus kinase inhibitors and biologic disease-modifying antirheumatic drugs in rheumatoid arthritis: a systematic review and network meta-analysis

Supplementary appendix

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Appendix 1. PubMed Search Strategy

1. exp Arthritis, Rheumatoid/
2. ("rheumatoid arthritis" or rheumatoid or rheumatism). tiab.
3. 1 or 2
4. exp Baricitinib/
5. (Baricitinib or Olumiant or LY3009104 or "LY 3009104" or INCB028050 or "INCB 028050" or "INCB 28050"). all.
6. exp Tofacitinib/
7. (Tofacitinib or Tasocitinib or Xeljanz or CP690550 or "CP 690550" or "CP 690 550"). all.
8. exp Upadacitinib/
9. (Upadacitinib or Rinvoq or ABT-494). all.
10. exp Adalimumab/
11. (Adalimumab or Humira or Exemptia). all.
12. exp Etanercept/
13. (Etanercept or Enbrel or Erelzi or "TNFR-Fc Fusion Protein" or "TNFR Fc Fusion Protein" or TNR001 or "TNR 001"). all.
14. exp Infliximab/
15. (Infliximab or Remicade or Renflexis or Inflectra or "Monoclonal Antibody cA2"). all.
16. exp Golimumab/
17. (Golimumab or Simponi or CNTO148 or "CNTO 148"). all.
18. exp Certolizumab Pegol/
19. (Certolizumab or Cimzia or CDP870 or "CDP 870"). all.
20. exp Tocilizumab/
21. (Tocilizumab or Actemra or Atlizumab or Roactemra or RHPM-1 or RG-1569 or R-1569 or MSB11456 or "MSB 11456" or "Monoclonal Antibody, MRA" or RO-4877533). all.
22. exp Rituximab/
23. (Rituximab or Mabthera or Rituxan or "IDEC C2B8" or GP2013 or "GP 2013"). all.
24. exp Abatacept/
25. (Abatacept or Orencia or Belatacept or Nulojix or BMS224818 or "BMS 224818" or BMS188667 or "BMS 188667" or CTLA4-Ig or CTLA4-Fc). all.
26. or/4-25
27. randomized controlled trial. pt.
28. controlled clinical trial. pt.
29. randomized. tiab.
30. placebo. tiab.
31. clinical trials as topic/
32. randomly. tiab.
33. trial. ti.
34. or/27-33
35. animals. mh.
36. humans. mh.
37. 35 not 36
38. 34 not 37
39. 3 and 26 and 38

Appendix 2. Currently recommended doses of JAK inhibitors and bDMARDs

Drugs	Currently recommended doses
Baricitinib 2mg	2 mg orally once daily
Baricitinib 4mg	4 mg orally once daily
Tofacitinib 5mg	5 mg orally twice daily
Tofacitinib 10mg	10 mg orally twice daily
Upadacitinib 15mg	15 mg orally once daily
Upadacitinib 30mg	30 mg orally once daily
Adalimumab	40 mg subcutaneous every two weeks
Etanercept	50 mg subcutaneous once weekly or 25 mg subcutaneous twice weekly
Infliximab	3 mg/kg intravenous every eight weeks after initial dosing at 0, 2 and 6 weeks
Golimumab	50 mg subcutaneous once a month
Certolizumab pegol	400 mg subcutaneous initially and at weeks 2 and 4, followed by 200 mg every other week (for maintenance dosing, 400 mg every four weeks can be considered)
Tocilizumab	8 mg/kg intravenous every four weeks or 162 mg subcutaneous every other week
Rituximab	Two 1000 mg intravenous doses two weeks apart
Abatacept	~10 mg/kg intravenous every four weeks after the initial dosing regimen of baseline, two- and four-week infusions or 125 mg subcutaneous weekly

bDMARDs, biologic disease-modifying antirheumatic drugs; JAK, Janus kinase.

Appendix 3. Computer code (gemtc package in R)

```
#Set up the labels
treatments <- read.table(textConnection(
   'id description
   1 Placebo
   2 "Baricitinib 2mg"
   3 "Baricitinib 4mg"
   4 "Tofacitinib 5mg"
   5 "Tofacitinib 10mg"
   6 "Upadacitinib 15mg"
   7 "Upadacitinib 30mg"
   8 Adalimumab
   9 Etanercept
   10 Infliximab
   11 Golimumab
   12 "Certolizumab pegol"
   13 Tocilizumab
   14 Rituximab
   15 Abatacept
   16 csDMARDs
   '), header=TRUE)

#Build a model (ACR20, for example)
network <- mtc.network(ACR20, description = "ACR20 network", treatments = treatments)
model.ran <- mtc.model(network, likelihood="binom", link="logit", type = "consistency",
linearModel='random',dic=TRUE)
result.ran <- mtc.run(model.ran, n.adapt=20000, n.iter=50000, thin=10)
print(summary(result.ran))

#Build a model (DAS28, for example)
network <- mtc.network(DAS28, description = "DAS28 network", treatments = treatments)
model.ran <- mtc.model(network, likelihood="normal", link="identity", type = "consistency",
linearModel='random',dic=TRUE)
result.ran <- mtc.run(model.ran, n.adapt=20000, n.iter=50000, thin=10)
print(summary(result.ran))

#Gelman-Rubin diagnostic and trace plots
gelman.diag(result.ran)
gelman.plot(result.ran)
summary(result.ran)
plot(result.ran)

#Results of network meta-analysis (ACR20, for example)
```

```
tb<- relative.effect.table(result.ran)
print(tb)
tb1<-round(exp(tb),2)
tb1
write.csv(tb1, "ACR20.csv")

# Evaluation of heterogeneity
result.anohe <- mtc.anohe(network, n.adapt=20000, n.iter=50000)
summary.anohe<-summary(result.anohe)
summary.anohe

#Relative ranking (ACR20, for example)
ranks <- rank.probability(result.ran,preferredDirection=1)
print(ranks)
write.csv(ranks, "ACR20-rank.csv")
```

Appendix 4. References for included trials

Abe 2006[1], Baek 2019[2], Bi 2019[3], Blanco 2017[4], Burmester 2013[5], Burmester 2018[6], Chen 2009[7], Chen 2016[8], Choy 2012[9], Cohen 2006[10], Combe 2006[11], Conaghan 2013[12], Dougados 2017[13], Elmedany 2019[14], Emery 2006[15], Emery 2008[16], Emery 2010[17], Fleischmann 2017[18], Fleischmann 2019[19], Furst 2003[20], Gao 2010[21], Genovese 2005[22], Genovese 2008[23], Genovese 2016a[24], Genovese 2016b[25], Genovese 2018[26], Gerlag 2010[27], Greenwald 2010[28], Hobbs 2015[29], Huang 2009[30], Jobanputra 2012[31], Kang 2018[32], Kay 2008[33], Kennedy 2014[34], Keystone 2004[35], Keystone 2008[36], Keystone 2009[37], Keystone 2012[38], Keystone 2015[39], Kim 2007[40], Kim 2012[41], Kim 2013[42], Kivitz 2014[43], Kremer 2003[44], Kremer 2006[45], Kremer 2011[46], Kremer 2012[47], Kremer 2013[48], Kremer 2016[49], Lan 2004[50], Li 2016[51], Li 2018[52], Lipsky 2000[53], Machado 2014[54], Maini 1998[55], Maini 2006[56], NCT01034397[57], NCT01283971[58], NCT02720523[59], O'Dell 2013[60], Schiff 2008[61], Schiff 2014[62], Shi 2013[63], Smolen 2008[64], Smolen 2009a[65], Smolen 2009b[66], Smolen 2015[67], Smolen 2016[68], Strand 2006[69], Takeuchi 2013[70], Tanaka 2011[71], Tanaka 2012[72], Tanaka 2016[73], Tanaka 2019[74], Taylor 2017[75], Van der Heijde 2013[76], Van Vollenhoven 2012[77], Weinblatt 1999[78], Weinblatt 2003[79], Weinblatt 2006[80], Weinblatt 2012[81], Weinblatt 2013[82], Westhovens 2006[83], Wijesinghe 2017[84], Yamamoto 2014a[85], Yamamoto 2014b[86], Yazici 2012[87], Zhang 2006[88].

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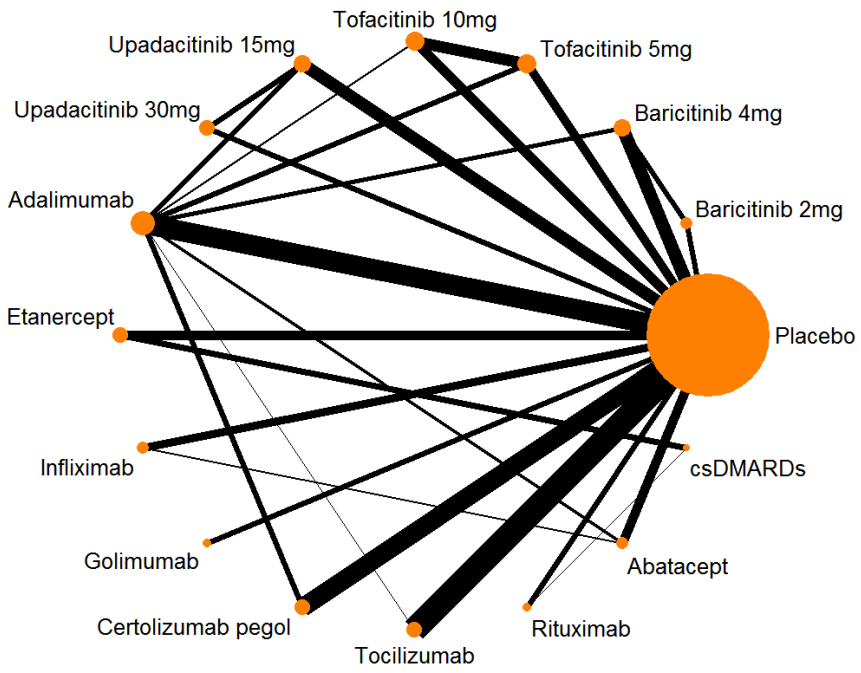
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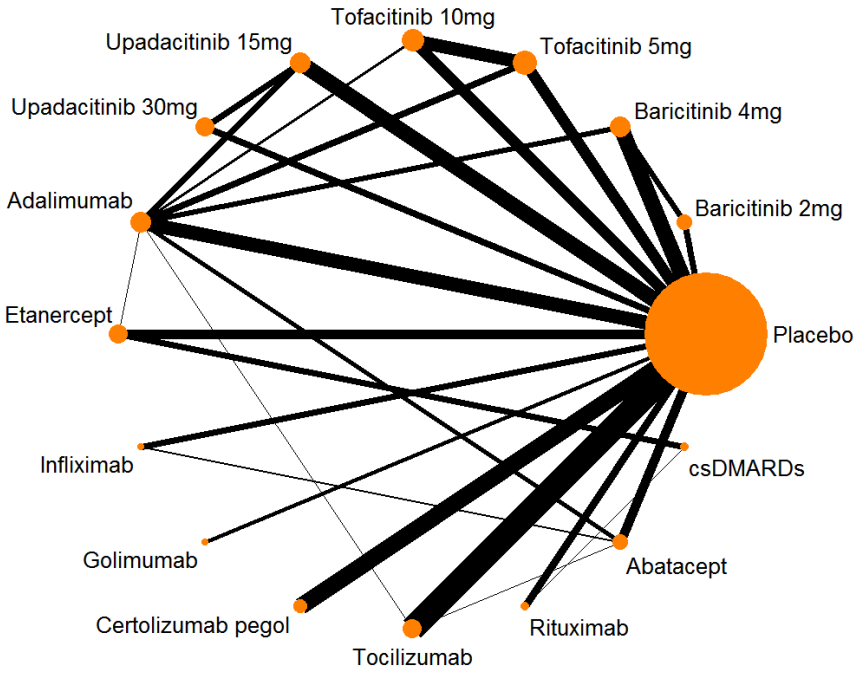
Appendix 5. Evidence structure of eligible comparisons for network meta-analysis

5.1 ACR20



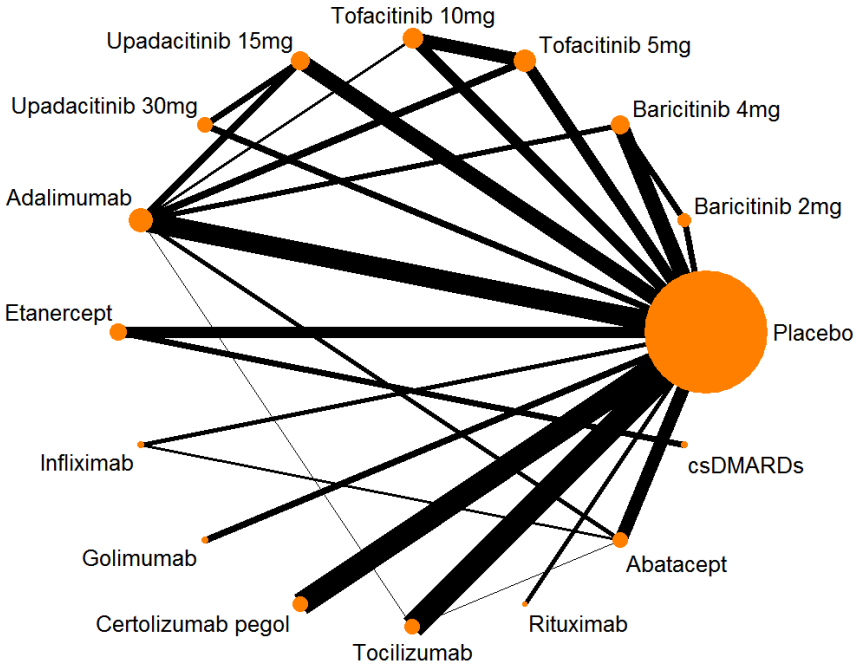
Each circle represents a drug. The connected circles represent the two drugs that have been compared in studies. The width of the lines is proportional to the number of trials. The size of each circle is proportional to the number of randomly assigned patients. ACR20, American College of Rheumatology 20% response; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs.

5.2 DAS28



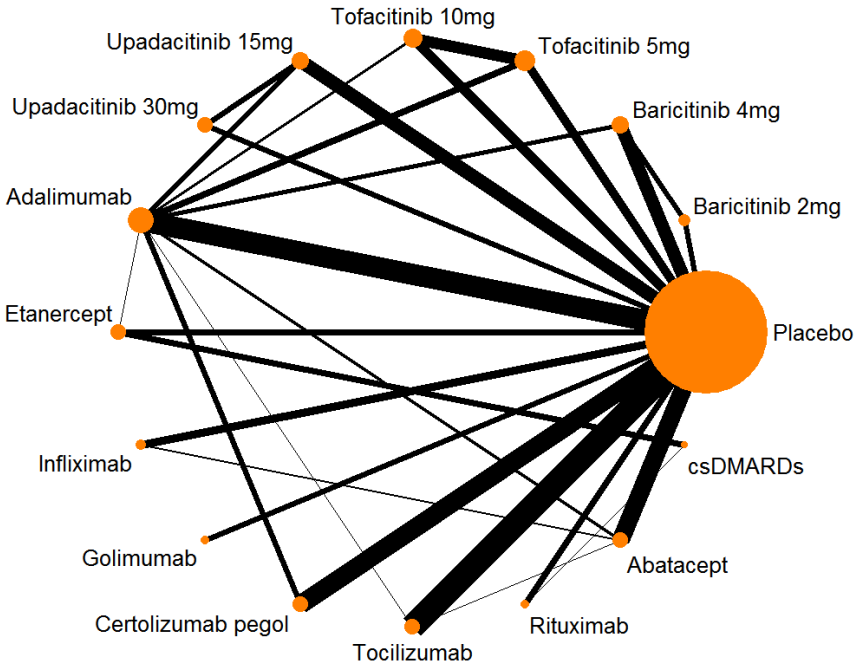
Each circle represents a drug. The connected circles represent the two drugs that have been compared in studies. The width of the lines is proportional to the number of trials. The size of each circle is proportional to the number of randomly assigned patients. csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; DAS28, Disease Activity Score in 28 joints.

5.3 HAQ-DI



Each circle represents a drug. The connected circles represent the two drugs that have been compared in studies. The width of the lines is proportional to the number of trials. The size of each circle is proportional to the number of randomly assigned patients. csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; HAQ-DI, Health Assessment Questionnaire–Disability Index.

5.4 Discontinuations for AEs



Each circle represents a drug. The connected circles represent the two drugs that have been compared in studies. The width of the lines is proportional to the number of trials. The size of each circle is proportional to the number of randomly assigned patients. AEs, adverse events; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs.

Appendix 6. Characteristics of included studies

Study	Treatment	Background therapy	Subjects	Number of patients	Mean age (years)	Outcomes	Follow-up time point to assess efficacy/safety (weeks)
Abe 2006	Infliximab vs placebo	MTX	MTX-IR	96	55	①④	14/14
Baek 2019	Tocilizumab vs placebo	csDMARDs	csDMARDs-IR	99	52	①②③④	24/24
Bi 2019	Certolizumab pegol vs placebo	MTX	MTX-IR	429	48	①②③④	24/24
Blanco 2017	Abatacept vs placebo	csDMARDs	TNFi-IR	276	54	①②③④	12/16
Burmester 2013	Tofacitinib 5mg vs tofacitinib 10mg vs placebo	MTX	TNFi-IR	399	55	①②③④	13/13
Burmester 2018	Upadacitinib 15mg vs Upadacitinib 30mg vs placebo	csDMARDs	csDMARDs-IR	661	56	①②③④	12/12
Chen 2009	Adalimumab vs placebo	MTX	MTX-IR	47	53	①③④	12/12
Chen 2016	Etanercept vs placebo	MTX	MTX-IR	598	49	①②③	12/-
Choy 2012	Certolizumab pegol vs placebo	MTX	MTX-IR	243	54	①②③④	24/24
Cohen 2006	Rituximab vs placebo	MTX	TNFi-IR	517	52	①②③④	24/24
Combe 2006	Etanercept vs placebo	SSZ	SSZ-IR	151	51	①②③④	24/24
Conaghan 2013	Abatacept vs placebo	MTX	MTX-IR	50	52	④	-/16
Dougados 2017	Baricitinib 2mg vs baricitinib 4mg vs placebo	csDMARDs	csDMARDs-IR	684	52	①②③④	12/12
Elmedany 2019	Tocilizumab vs abatacept	MTX	TNFi-IR	132	50	②③④	24/24
Emery 2006	Rituximab vs placebo	MTX	csDMARDs-IR or bDMARDs-IR	341	51	①②③④	24/24
Emery 2008	Tocilizumab vs placebo	MTX	TNFi-IR	335	54	①③④	24/24
Emery 2010	Rituximab vs placebo	MTX	MTX-IR	342	52	①②③④	24/24
Fleischmann 2017	Tofacitinib 5mg vs adalimumab	MTX	MTX-IR	762	51	①②③④	26/52
Fleischmann 2019	Upadacitinib 15mg vs adalimumab vs placebo	MTX	MTX-IR	1629	54	①②③④	12/14
Furst 2003	Adalimumab vs placebo	csDMARDs	csDMARDs-IR	636	56	①④	24/24
Gao 2010	Infliximab vs placebo	MTX	MTX-IR	24	NR	①	14/-
Genovese 2005	Abatacept vs placebo	csDMARDs or anakinra	TNFi-IR	391	53	①③④	13/24
Genovese 2008	Tocilizumab vs placebo	csDMARDs	csDMARDs-IR	1216	53	①②③④	24/24
Genovese 2016a	Baricitinib 2mg vs baricitinib 4mg vs placebo	csDMARDs	bDMARDs-IR	527	56	①②③④	12/12
Genovese 2016b	Upadacitinib 15mg vs Upadacitinib 30mg vs placebo	MTX	MTX-IR	150	55	①②③④	12/12
Genovese 2018	Upadacitinib 15mg vs Upadacitinib 30mg vs placebo	csDMARDs	bDMARDs-IR	498	57	①②③④	12/12
Gerlag 2010	Etanercept vs placebo	MTX	MTX-IR	140	54	①②③④	12/12
Greenwald 2010	Baricitinib 4mg vs placebo	csDMARDs	csDMARDs-IR or bDMARDs-IR	62	56	①②③④	12/12
Hobbs 2015	Etanercept vs placebo	csDMARDs	csDMARDs-IR	210	56	①②③④	12/12
Huang 2009	Adalimumab vs placebo	MTX	csDMARDs-IR	181	49	①③④	12/12
Jobanputra 2012	Adalimumab vs etanercept	csDMARDs	csDMARDs-IR	120	54	②④	12/52
Kang 2018	Certolizumab pegol vs placebo	MTX	MTX-IR	121	51	①②③④	12/24
Kay 2008	Golimumab vs	MTX	MTX-IR	70	55	①②④	16/16

	placebo						
Kennedy 2014	Adalimumab vs placebo	csDMARDs	csDMARDs-IR	129	50	①②③④	12/12
Keystone 2004	Adalimumab vs placebo	MTX	MTX-IR	407	56	①③④	24/52
Keystone 2008	Certolizumab pegol vs placebo	MTX	MTX-IR	592	52	①②④	12/12
Keystone 2009	Golimumab vs placebo	MTX	MTX-IR	222	52	①②④	14/14
Keystone 2012	Etanercept vs placebo	MTX or SSZ	MTX-IR or SSZ-IR	129	51	①④	24/24
Keystone 2015	Baricitinib 2mg vs baricitinib 4mg vs placebo	MTX	MTX-IR	202	51	①②③④	12/12
Kim 2007	Adalimumab vs placebo	MTX	MTX-IR	128	49	①③④	24/24
Kim 2012	Etanercept vs csDMARDs	MTX	MTX-IR	300	48	①②③④	16/16
Kim 2013	Infliximab vs placebo	MTX	MTX-IR	138	50	①③	30/-
Kivitz 2014	Tocilizumab vs placebo	csDMARDs	csDMARDs-IR	656	52	①②③④	24/24
Kremer 2003	Abatacept vs placebo	MTX	MTX-IR	234	55	①③④	26/26
Kremer 2006	Abatacept vs placebo	MTX	MTX-IR	652	51	①③④	52/52
Kremer 2011	Tocilizumab vs placebo	MTX	MTX-IR	791	52	①②③④	24/52
Kremer 2012	Tofacitinib 5mg vs tofacitinib 10mg vs placebo	MTX	MTX-IR	214	54	①②③④	12/12
Kremer 2013	Tofacitinib 5mg vs tofacitinib 10mg vs placebo	csDMARDs	csDMARDs-IR	792	52	①②③④	13/26
Kremer 2016	Upadacitinib 15mg vs Upadacitinib 30mg vs placebo	MTX	TNFi-IR	166	58	①②③④	12/12
Lan 2004	Etanercept vs placebo	MTX	MTX-IR	58	49	①③	12/-
Li 2016	Golimumab vs placebo	MTX	MTX-IR	264	48	①③④	14/24
Li 2018	Baricitinib 4mg vs placebo	MTX	MTX-IR	290	49	①②③④	12/24
Lipsky 2000	Infliximab vs placebo	MTX	MTX-IR	174	52	①④	54/54
Machado 2014	Etanercept vs csDMARDs	MTX	MTX-IR	423	48	①②③④	24/24
Maini 1998	Infliximab vs placebo	MTX	MTX-IR	29	54	④	-/14
Maini 2006	Tocilizumab vs placebo	MTX	MTX-IR	99	50	①②③④	20/20
NCT 01034397	Tocilizumab vs placebo	csDMARDs	csDMARDs-IR	54	54	②③④	12/24
NCT 01283971	Adalimumab vs tocilizumab	MTX	TNFi-IR	94	53	①②③④	24/24
NCT 02720523	Upadacitinib 15mg vs Upadacitinib 30mg vs placebo	csDMARDs	csDMARDs-IR	148	55	①②③④	12/12
O'Dell 2013	Etanercept vs csDMARDs	MTX	MTX-IR	353	57	①②③④	24/48
Schiff 2008	Infliximab vs abatacept vs placebo	MTX	MTX-IR	431	49	①②③④	28/28
Schiff 2014	Certolizumab pegol vs placebo	csDMARDs	TNFi-IR	37	57	①④	12/12
Shi 2013	Tocilizumab vs placebo	csDMARDs	csDMARDs-IR	208	47	①②③	24/-
Smolen 2008	Tocilizumab vs placebo	MTX	MTX-IR	409	51	①②③④	24/16
Smolen 2009a	Golimumab vs placebo	csDMARDs	TNFi-IR	308	54	①②③④	14/24
Smolen 2009b	Certolizumab pegol vs placebo	MTX	MTX-IR	373	52	①②③④	24/24
Smolen 2015	Certolizumab pegol vs placebo	csDMARDs	csDMARDs-IR	194	54	①③④	24/24
Smolen 2016	Adalimumab vs certolizumab pegol	MTX	MTX-IR	915	53	①④	12/12
Strand 2006	Rituximab vs placebo	MTX	MTX-IR	80	54	①③④	24/24
Takeuchi 2013	Abatacept vs placebo	MTX	MTX-IR	127	53	①②③④	24/24

Tanaka 2011	Tofacitinib 5mg vs tofacitinib 10mg vs placebo	MTX	MTX-IR	81	50	①②③④	12/12
Tanaka 2012	Golimumab vs placebo	MTX	MTX-IR	174	51	①②③④	14/16
Tanaka 2016	Baricitinib 2mg vs baricitinib 4mg vs placebo	MTX	MTX-IR	97	54	①②③④	12/12
Tanaka 2019	Etanercept vs placebo	csDMARDs	csDMARDs-IR or bDMARDs-IR	301	56	①②③④	12/12
Taylor 2017	Baricitinib 4mg vs adalimumab vs placebo	MTX	MTX-IR	1305	53	①②③④	12/24
Van der Heijde 2013	Tofacitinib 5mg vs tofacitinib 10mg vs placebo	MTX	MTX-IR	797	53	①②③④	13/52
Van Vollenhoven 2012	Tofacitinib 5mg vs tofacitinib 10mg vs adalimumab vs placebo	MTX	MTX-IR	717	53	①②③④	13/13
Weinblatt 1999	Etanercept vs placebo	MTX	MTX-IR	89	50	①③④	12/24
Weinblatt 2003	Adalimumab vs placebo	MTX	MTX-IR	129	57	①③④	24/16
Weinblatt 2006	Abatacept vs placebo	csDMARDs	csDMARDs-IR	1274	52	④	-/52
Weinblatt 2012	Certolizumab pegol vs placebo	csDMARDs	csDMARDs-IR or TNFi-IR	1063	55	①②③④	12/12
Weinblatt 2013	Adalimumab vs abatacept	MTX	MTX-IR	646	51	①②③④	52/52
Westhovens 2006	Infliximab vs placebo	MTX	MTX-IR	721	52	①②④	22/22
Wijesinghe 2017	Rituximab vs csDMARDs	MTX	csDMARDs-IR	39	46	①②④	24/24
Yamamoto 2014a	Certolizumab pegol vs placebo	csDMARDs	csDMARDs-IR	230	56	①②③④	12/24
Yamamoto 2014b	Certolizumab pegol vs placebo	MTX	MTX-IR	159	51	①②③④	12/24
Yazici 2012	Tocilizumab vs placebo	csDMARDs	csDMARDs-IR or TNFi-IR	614	55	①②③④	12/24
Zhang 2006	Infliximab vs placebo	MTX	MTX-IR	173	48	①③④	18/18

Outcomes: ①: ACR20, ②: DAS28, ③: HAQ-DI, ④: Discontinuations for AEs.

ACR20, American College of Rheumatology 20% response; AEs, adverse events; bDMARDs, biologic DMARDs; csDMARDs, conventional synthetic DMARDs; DAS28, Disease Activity Score in 28 joints; DMARDs, disease-modifying antirheumatic drugs; HAQ-DI, Health Assessment Questionnaire–Disability Index; IR, inadequate response; MTX, methotrexate; NR, not reported; SSZ, sulfasalazine; TNFi, tumour necrosis factor inhibitors.

Appendix 7. Risk of bias table of included studies

Study ID	Random sequence generation	Allocation concealment	Blinding of participants & personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Other bias*
Abe 2006	Unclear	Unclear	Low	Low	Low	Low	Unclear
Baek 2019	Unclear	Unclear	Low	Low	Low	Low	Unclear
Bi 2019	Low	Low	Low	Low	High	Low	Low
Blanco 2017	Unclear	Unclear	Low	Low	Low	Low	Low
Burmester 2013	Low	Low	Low	Low	Low	Low	Low
Burmester 2018	Low	Low	Low	Low	Low	Low	Low
Chen 2009	Unclear	Unclear	Low	Low	Unclear	Low	High
Chen 2016	Unclear	Unclear	Low	Low	Unclear	Unclear	Low
Choy 2012	Low	Low	Low	Low	High	Low	Low
Cohen 2006	Low	Low	Low	Low	High	Low	Low
Combe 2006	Unclear	Unclear	Low	Low	Unclear	Low	Low
Conaghan 2013	Low	Low	Low	Low	Low	Unclear	Unclear
Dougados 2017	Unclear	Unclear	Low	Low	Low	Low	Low
Elmedany 2019	Low	Low	High	High	Unclear	Unclear	Low
Emery 2006	Low	Low	Low	Low	High	Low	Low
Emery 2008	Unclear	Unclear	Low	Low	Low	Low	Low
Emery 2010	Unclear	Unclear	Low	Low	Low	Low	Low
Fleischmann 2017	Low	Low	Low	Low	Low	Low	Low
Fleischmann 2019	Low	Low	Low	Low	Low	Low	Low
Furst 2003	Unclear	Unclear	Low	Low	Low	Low	Low
Gao 2010	Unclear	Unclear	Low	Low	Low	Unclear	High
Genovese 2005	Low	Low	Low	Low	Low	Low	Low
Genovese 2008	Unclear	Unclear	Low	Low	Low	Low	Low
Genovese 2016	Unclear	Unclear	Low	Low	Low	Low	Low
Genovese 2016	Low	Low	Low	Low	Low	Low	Low
Genovese 2018	Low	Low	Low	Low	Low	Low	Low
Gerlag 2010	Unclear	Unclear	High	High	Low	Low	Low
Greenwald 2010	Unclear	Unclear	Low	Low	Low	Low	Unclear
Hobbs 2015	Low	Low	Low	Low	Low	Low	Low
Huang 2009	Unclear	Unclear	Low	Low	Low	Unclear	Low
Jobanputra 2012	Low	Low	High	High	Low	Unclear	Low
Kang 2018	Unclear	Unclear	Low	Low	Low	Low	Low
Kay 2008	Unclear	Unclear	Low	Low	Low	Low	Unclear
Kennedy 2014	Unclear	Unclear	Low	Low	Low	Low	Low
Keystone 2004	Unclear	Unclear	Low	Low	Low	Low	Low
Keystone 2008	Unclear	Unclear	Low	Low	Unclear	Low	Low
Keystone 2009	Low	Low	Low	Low	Low	Low	Low
Keystone 2012	Unclear	Unclear	High	High	Low	Low	Low
Keystone 2015	Unclear	Unclear	Low	Low	Low	Low	Low
Kim 2007	Unclear	Unclear	Low	Low	Low	Low	Low
Kim 2012	Low	Low	High	High	Low	Low	Low

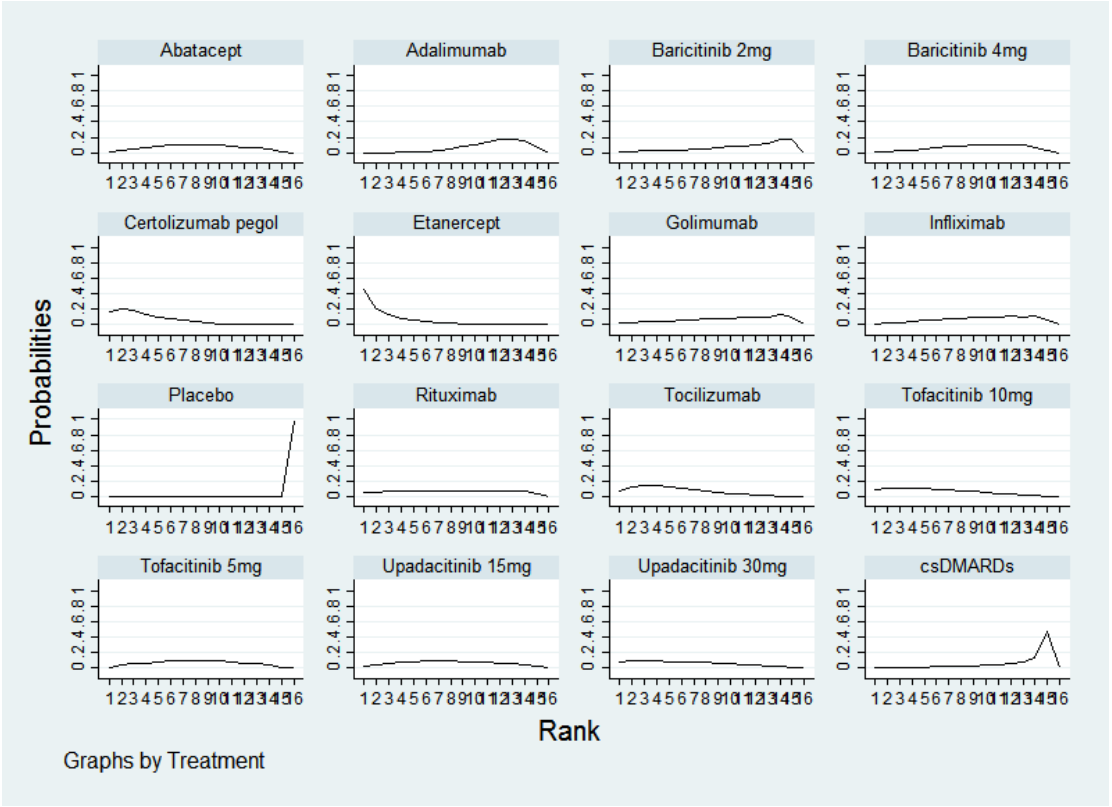
Kim 2013	Unclear	Unclear	Low	Low	Low	Unclear	Low
Kivitz 2014	Unclear	Unclear	Low	Low	Low	Low	Low
Kremer 2003	Low	Low	Low	Low	High	Low	Low
Kremer 2006	Low	Low	Low	Low	Low	Low	Low
Kremer 2011	Unclear	Unclear	Low	Low	Low	Low	Low
Kremer 2012	Unclear	Unclear	Low	Low	Low	Low	Low
Kremer 2013	Unclear	Unclear	Low	Low	Low	Low	Low
Kremer 2016	Low	Low	Low	Low	Low	Low	Low
Lan 2004	Unclear	Unclear	Low	Low	Low	Unclear	Unclear
Li 2016	Low	Low	Low	Low	Low	Low	Low
Li 2018	Unclear	Unclear	Low	Low	Low	Low	Low
Lipsky 2000	Unclear	Unclear	Low	Low	High	Low	Low
Machado 2014	Low	Low	High	High	Low	Low	Low
Maini 1998	Low	Low	Low	Low	High	Unclear	High
Maini 2006	Low	Low	Low	Low	Low	Low	Unclear
NCT01034397	Unclear	Unclear	Low	Low	Low	Unclear	Unclear
NCT01283971	Unclear	Unclear	High	High	Low	Low	Unclear
NCT02720523	Unclear	Unclear	Low	Low	Low	Low	Low
O'Dell 2013	Unclear	Unclear	Low	Low	Low	Low	Low
Schiff 2008	Low	Low	Low	Low	Low	Low	Low
Schiff 2014	Unclear	Unclear	Low	Low	Low	Low	High
Shi 2013	Unclear	Unclear	Low	Low	Low	Unclear	Low
Smolen 2008	Low	Low	Low	Low	Low	Low	Low
Smolen 2009a	Low	Low	Low	Low	Low	Low	Low
Smolen 2009b	Unclear	Unclear	Low	Low	High	Low	Low
Smolen 2015	Low	Low	Low	Low	Low	Low	Low
Smolen 2016	Low	Low	Low	Low	Low	Low	Low
Strand 2006	Unclear	Unclear	Low	Low	Low	Low	Unclear
Takeuchi 2013	Unclear	Unclear	Low	Low	Low	Low	Low
Tanaka 2011	Unclear	Unclear	Low	Low	Low	Low	Unclear
Tanaka 2012	Unclear	Unclear	Low	Low	Low	Low	Low
Tanaka 2016	Low	Low	Low	Low	Low	Low	Unclear
Tanaka 2019	Unclear	Unclear	High	High	Low	Low	Low
Taylor 2017	Unclear	Unclear	Low	Low	Low	Low	Low
Van der Heijde 2013	Low	Low	Low	Low	Low	Low	Low
Van Vollenhoven 2012	Low	Low	Low	Low	Low	Low	Low
Weinblatt 1999	Low	Unclear	Low	Low	Low	Unclear	Unclear
Weinblatt 2003	Low	Low	Low	Low	Unclear	Low	Low
Weinblatt 2006	Unclear	Unclear	Low	Low	Low	Unclear	Low
Weinblatt 2012	Low	Low	Low	Low	Low	Low	Low
Weinblatt 2013	Unclear	Unclear	High	Low	Low	Low	Low
Westhovens 2006	Unclear	Unclear	Low	Low	Low	Low	Low
Wijesinghe 2017	Low	Low	Low	Low	Low	Low	High

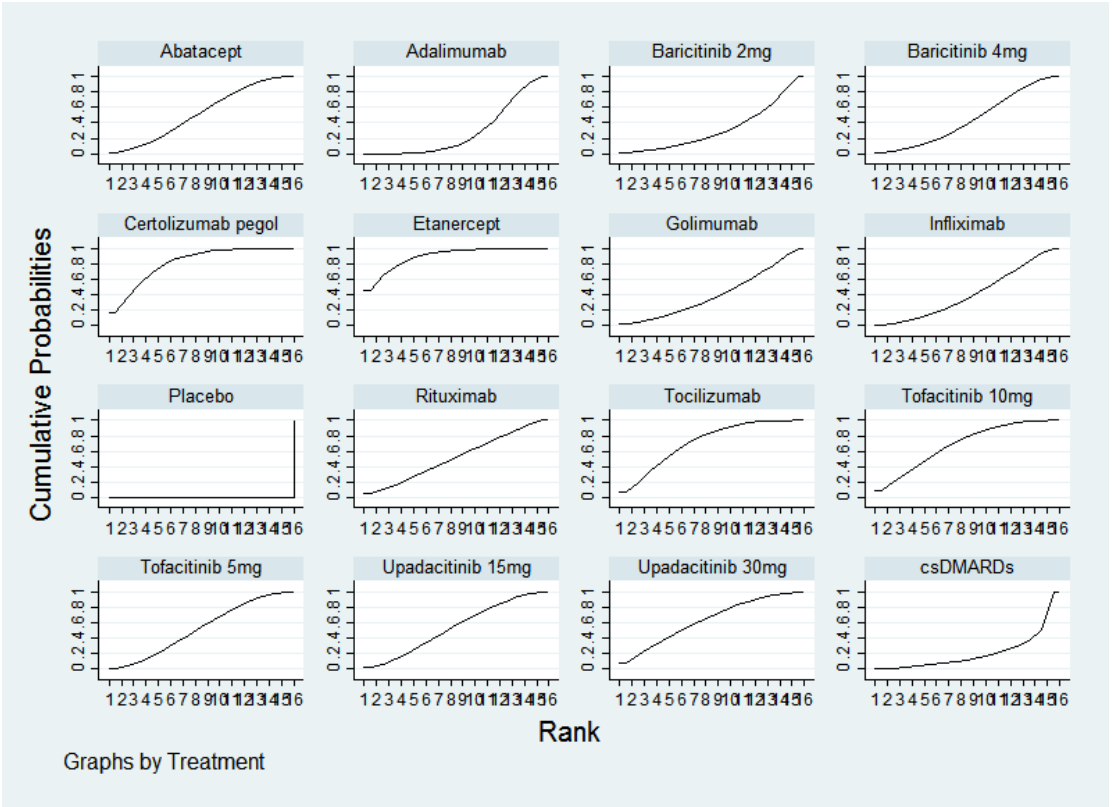
Yamamoto 2014a	Low	Low	Low	Low	Unclear	Low	Low
Yamamoto 2014b	Low	Low	Low	Low	Unclear	Low	Low
Yazici 2012	Unclear	Unclear	Low	Low	Low	Low	Low
Zhang 2006	Unclear	Unclear	Low	Low	Low	Low	Low

* Other bias was evaluated according to the sample size. We defined it as low risk of bias when the sample size >100, and high risk of bias when the sample size <50. It will be rated as unclear risk of bias if the sample size is between 50 and 100.

Appendix 8. Probability plots, cumulative probability plots and SUCRA.

8.1 ACR20

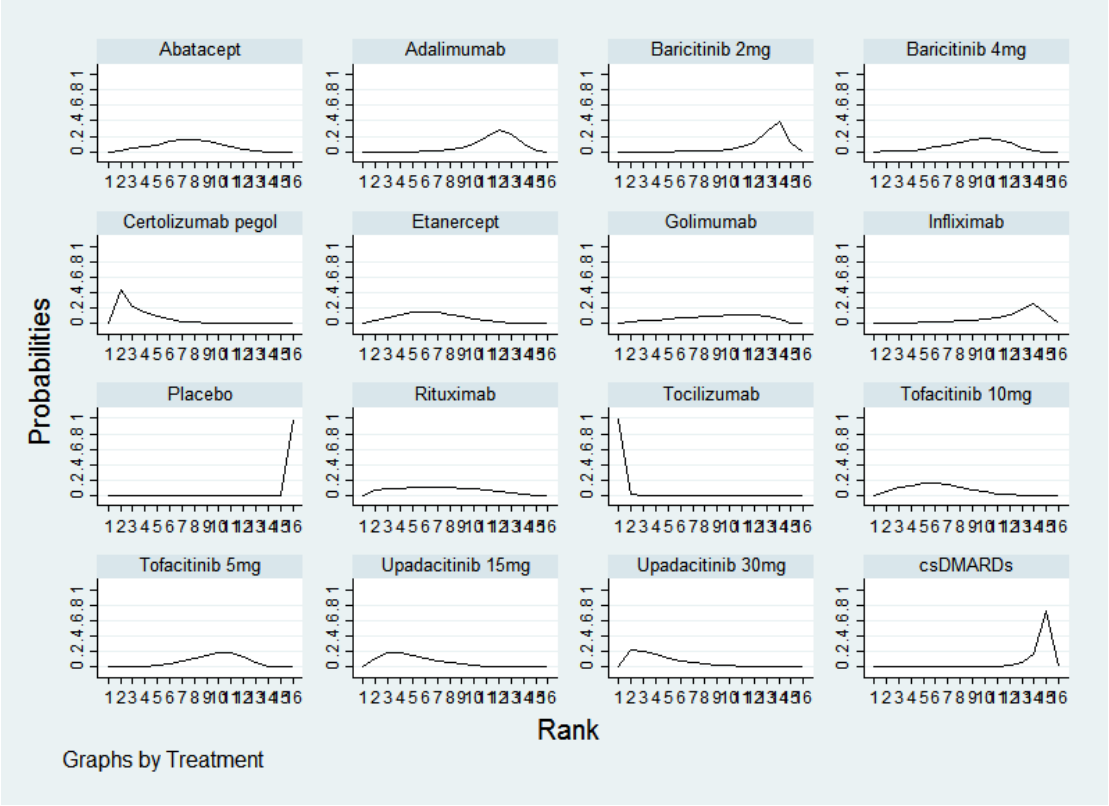


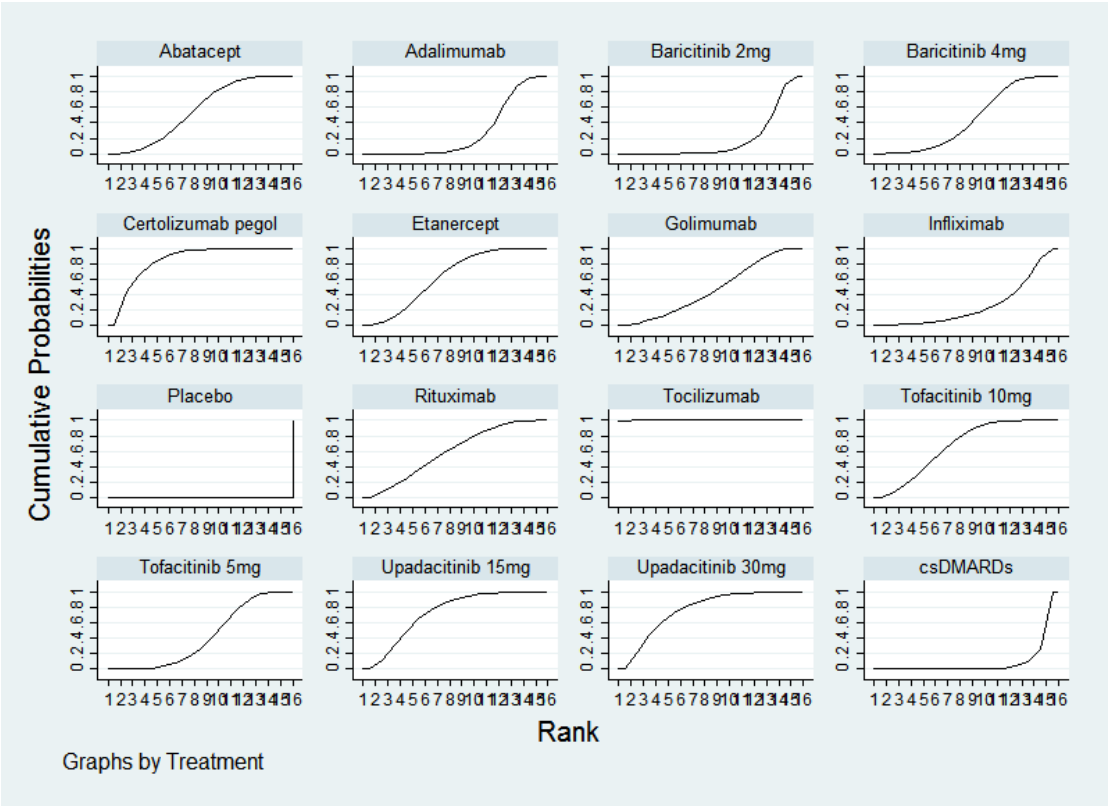


Treatment	SUCRA	PrBest	MeanRank
Placebo	0.0	0.0	16.0
Baricitinib 2mg	32.5	0.8	11.1
Baricitinib 4mg	44.5	0.8	9.3
Tofacitinib 5mg	53.1	1.6	8.0
Tofacitinib 10mg	68.6	8.5	5.7
Upadacitinib 15mg	54.7	2.3	7.8
Upadacitinib 30mg	64.0	8.3	6.4
Adalimumab	29.7	0.0	11.5
Etanercept	89.7	44.8	2.5
Infliximab	41.7	0.9	9.7
Golimumab	41.0	1.6	9.8
Certolizumab pegol	81.6	16.3	3.8
Tocilizumab	72.1	7.8	5.2
Rituximab	52.7	4.5	8.1
Abatacept	52.6	1.6	8.1
csDMARDs	21.3	0.2	12.8

ACR20, American College of Rheumatology 20% response; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; SUCRA: surface under the cumulative ranking curve.

8.2 DAS28

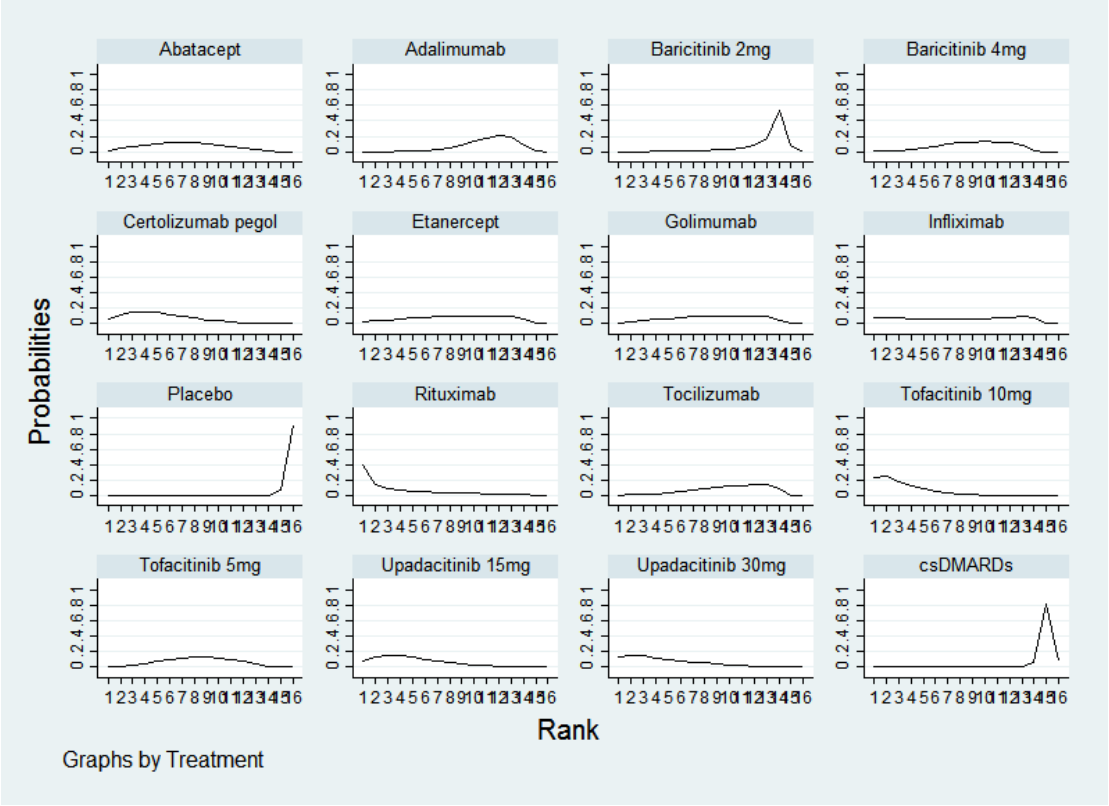

Abatacept
Adalimumab
Baricitinib 2mg
Baricitinib 4mg
Certolizumab pegol
Etanercept
Golimumab
Infliximab
Placebo
Rituximab
Tocilizumab
Tofacitinib 10mg
Tofacitinib 5mg
Upadacitinib 15mg
Upadacitinib 30mg
csDMARDs
Probabilities
Rank
Graphs by Treatment

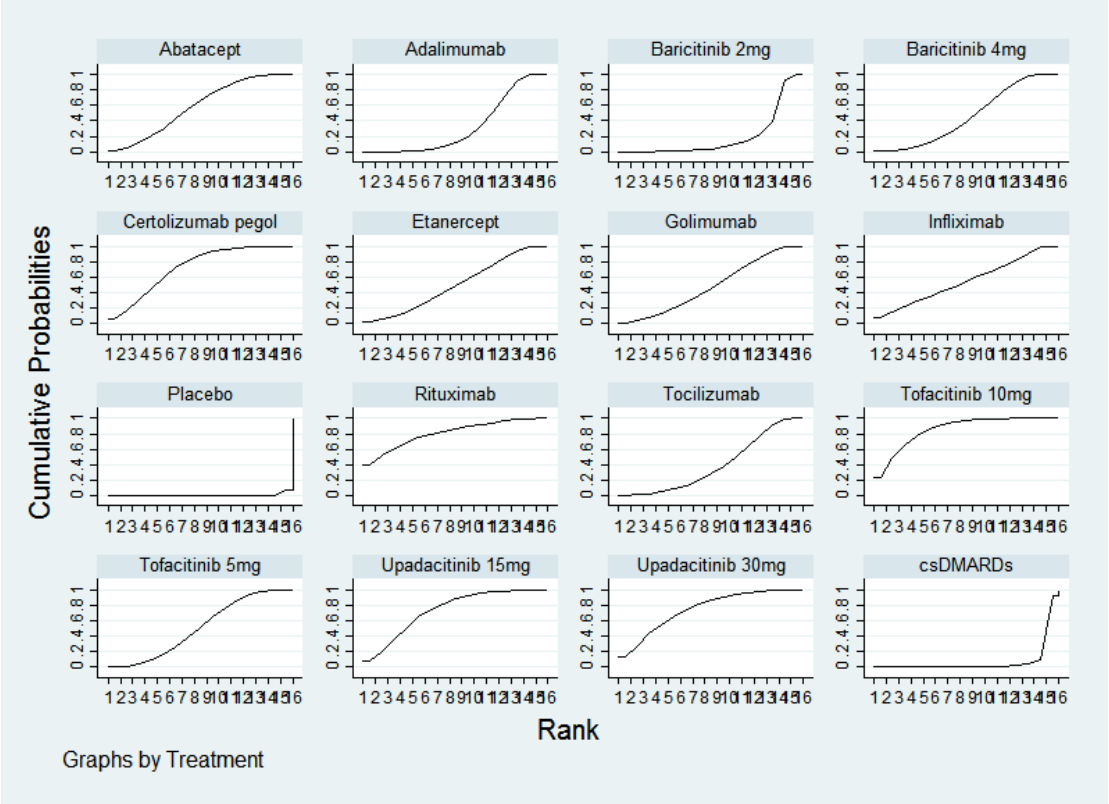

Abatacept
Adalimumab
Baricitinib 2mg
Baricitinib 4mg
Certolizumab pegol
Etanercept
Golimumab
Infliximab
Placebo
Rituximab
Tocilizumab
Tofacitinib 10mg
Tofacitinib 5mg
Upadacitinib 15mg
Upadacitinib 30mg
csDMARDs
Cumulative Probabilities
Rank
Graphs by Treatment

Treatment	SUCRA	PrBest	MeanRank
Placebo	0.0	0.0	16.0
Baricitinib 2mg	19.5	0.0	13.1
Baricitinib 4mg	44.6	0.0	9.3
Tofacitinib 5mg	41.9	0.0	9.7
Tofacitinib 10mg	66.7	0.0	6.0
Upadacitinib 15mg	73.7	0.0	5.0
Upadacitinib 30mg	77.2	0.3	4.4
Adalimumab	28.3	0.0	11.8
Etanercept	64.0	0.0	6.4
Infliximab	26.8	0.0	12.0
Golimumab	47.2	0.0	8.9
Certolizumab pegol	84.4	0.3	3.3
Tocilizumab	99.9	99.3	1.0
Rituximab	60.0	0.1	7.0
Abatacept	56.5	0.0	7.5
csDMARDs	9.4	0.0	14.6

csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; DAS28, Disease Activity Score in 28 joints; SUCRA: surface under the cumulative ranking curve.

8.3 HAQ-DI

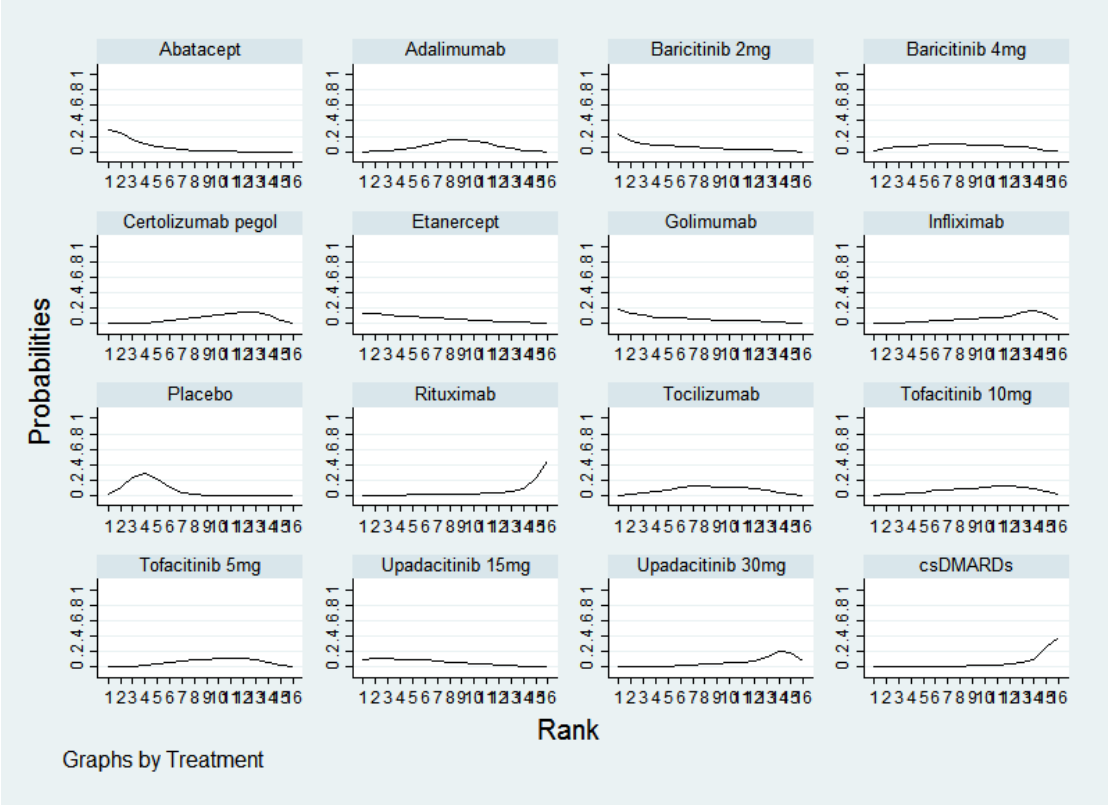

Abatacept
Adalimumab
Baricitinib 2mg
Baricitinib 4mg
Certolizumab pegol
Etanercept
Golimumab
Infliximab
Placebo
Rituximab
Tocilizumab
Tofacitinib 10mg
Tofacitinib 5mg
Upadacitinib 15mg
Upadacitinib 30mg
csDMARDs
Probabilities
Rank
Graphs by Treatment

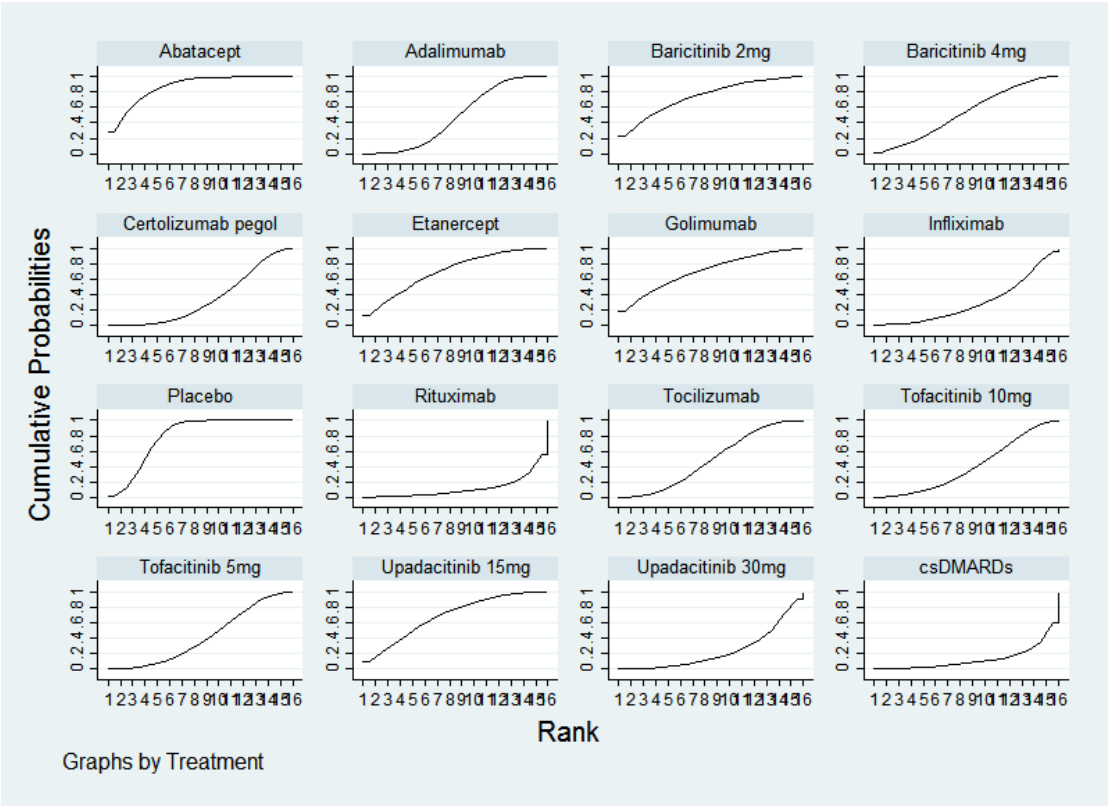

Abatacept
Adalimumab
Baricitinib 2mg
Baricitinib 4mg
Certolizumab pegol
Etanercept
Golimumab
Infliximab
Placebo
Rituximab
Tocilizumab
Tofacitinib 10mg
Tofacitinib 5mg
Upadacitinib 15mg
Upadacitinib 30mg
csDMARDs
Cumulative Probabilities
Rank
Graphs by Treatment

Treatment	SUCRA	PrBest	MeanRank
Placebo	0.5	0.0	15.9
Baricitinib 2mg	19.5	0.0	13.1
Baricitinib 4mg	46.1	0.2	9.1
Tofacitinib 5mg	51.5	0.2	8.3
Tofacitinib 10mg	85.5	22.6	3.2
Upadacitinib 15mg	74.3	7.4	4.9
Upadacitinib 30mg	75.3	12.7	4.7
Adalimumab	32.7	0.0	11.1
Etanercept	50.5	1.9	8.4
Infliximab	55.1	7.5	7.7
Golimumab	49.2	1.3	8.6
Certolizumab pegol	72.7	5.4	5.1
Tocilizumab	39.8	0.3	10.0
Rituximab	81.2	38.8	3.8
Abatacept	58.8	1.6	7.2
csDMARDs	7.5	0.0	14.9

csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; HAQ-DI, Health Assessment Questionnaire–Disability Index; SUCRA: surface under the cumulative ranking curve.

8.4 Discontinuations for AEs





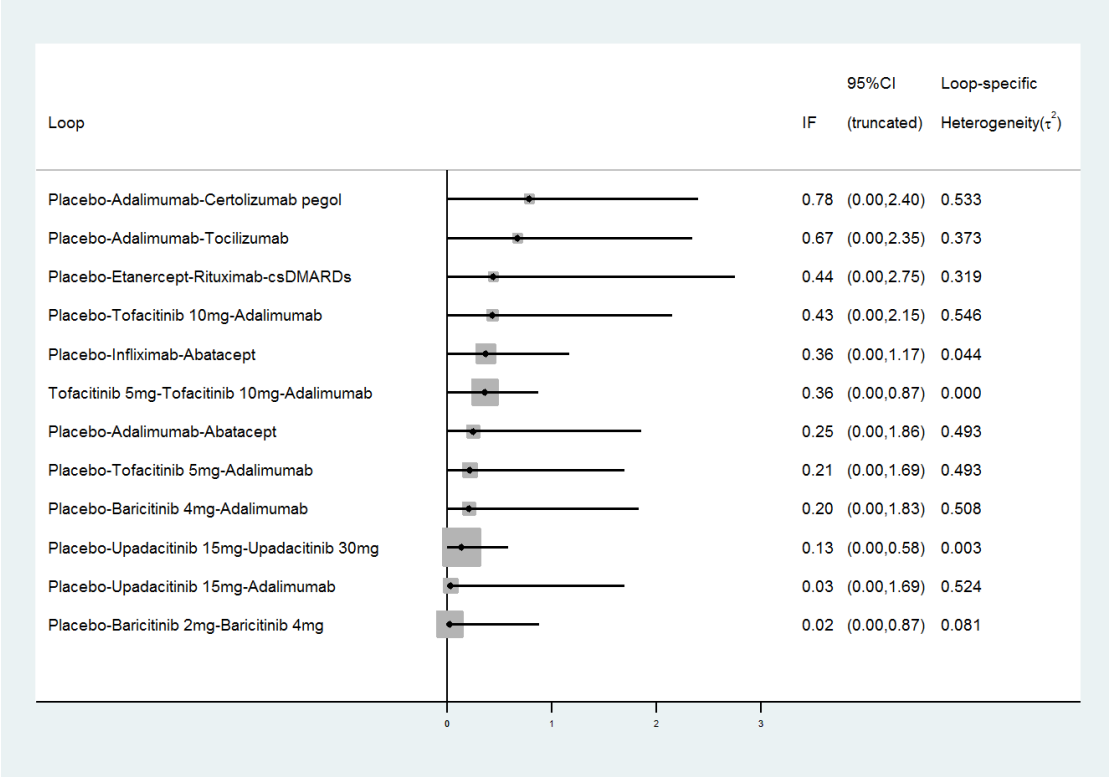
Treatment	SUCRA	PrBest	MeanRank
Placebo	79.6	1.7	4.1
Baricitinib 2mg	74.3	22.9	4.9
Baricitinib 4mg	53.2	1.6	8.0
Tofacitinib 5mg	42.1	0.4	9.7
Tofacitinib 10mg	40.4	0.5	9.9
Upadacitinib 15mg	69.5	10.5	5.6
Upadacitinib 30mg	24.2	0.2	12.4
Adalimumab	48.2	0.1	8.8
Etanercept	70.7	13.1	5.4
Infliximab	30.6	0.4	11.4
Golimumab	70.6	18.8	5.4
Certolizumab pegol	35.0	0.1	10.8
Tocilizumab	49.2	0.4	8.6
Rituximab	11.8	0.3	14.2
Abatacept	87.1	28.8	2.9
csDMARDs	13.3	0.3	14.0

AEs, adverse events; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; SUCRA: surface under the cumulative ranking curve.

Appendix 9. Evaluation of inconsistency

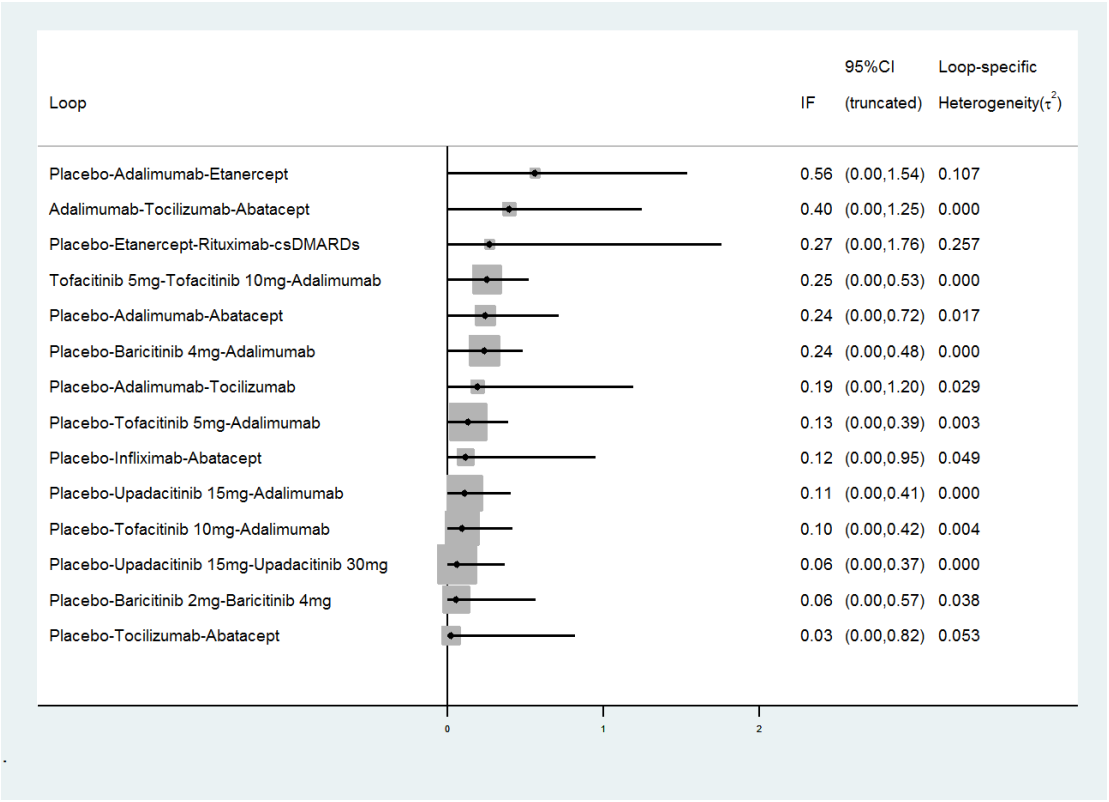
9.1 Loop-specific approach

9.1.1 ACR20



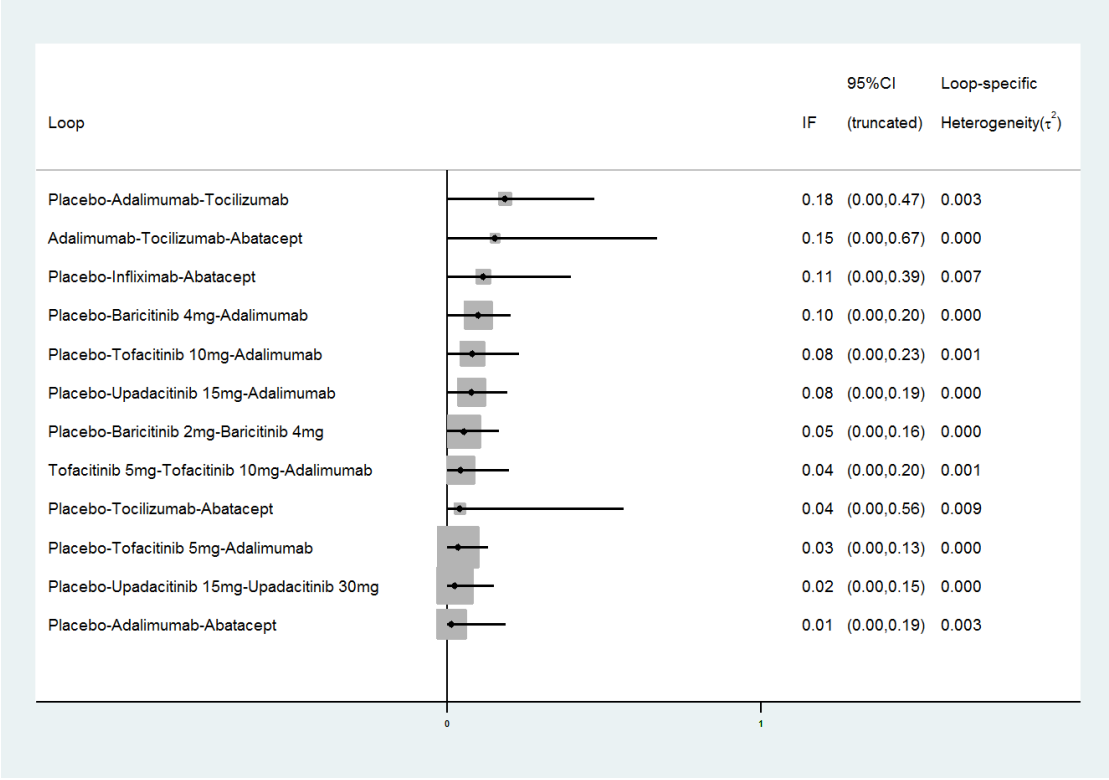
ACR20, American College of Rheumatology 20% response; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CI, confidence interval.

9.1.2 DAS28



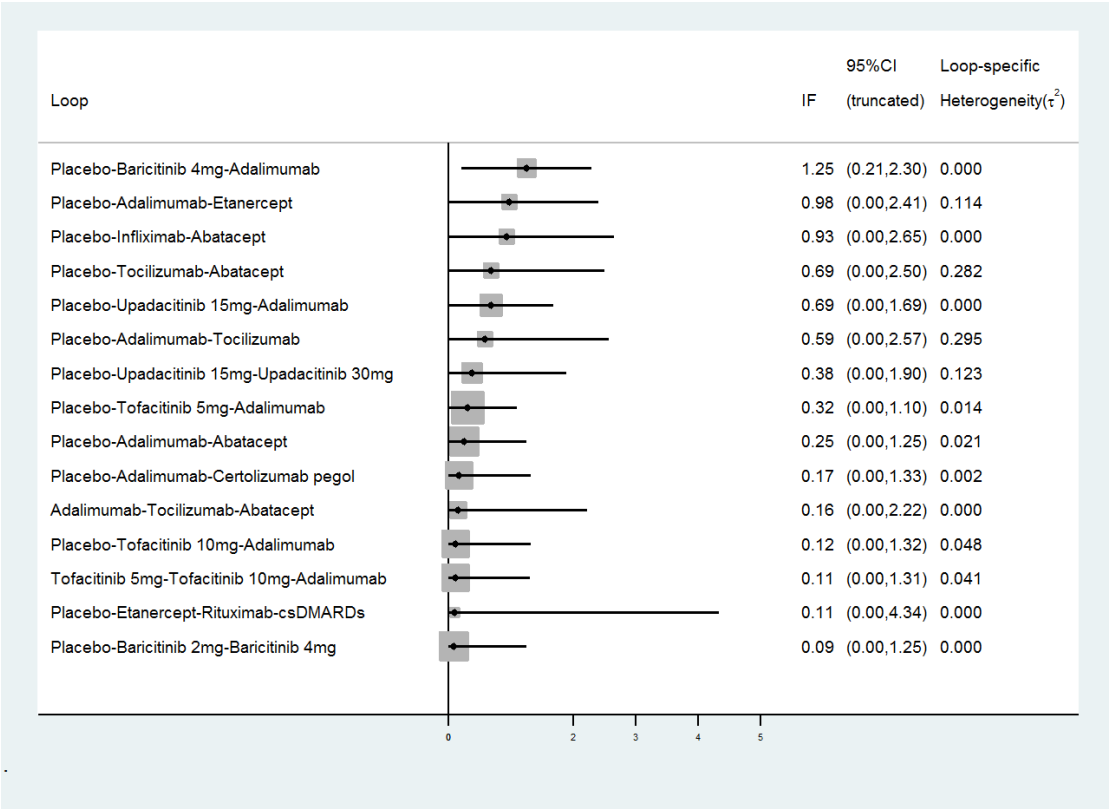
CI, confidence interval; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; DAS28, Disease Activity Score in 28 joints.

9.1.3 HAQ-DI



CI, confidence interval; HAQ-DI, Health Assessment Questionnaire–Disability Index.

9.1.4 Discontinuations for AEs



AEs, adverse events; CI, confidence interval; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs.

9.2 Node-splitting method

9.2.1 ACR20

Comparison	Direct		Indirect		Difference		
	Log OR	SE	Log OR	SE	Log OR	SE	P-value
PBO vs Bar 2mg	1.083	0.274	1.080	0.750	0.003	0.793	0.997
PBO vs Bar 4mg	1.201	0.203	1.767	0.857	-0.566	0.880	0.521
PBO vs Tof 5mg	1.266	0.225	1.124	0.447	0.143	0.501	0.776
PBO vs Tof 10mg	1.461	0.221	1.184	0.718	0.277	0.756	0.714
PBO vs Upa 15mg	1.322	0.218	1.450	0.858	-0.128	0.885	0.885
PBO vs Upa 30mg	1.397	0.249	1.685	0.914	-0.288	0.945	0.760
PBO vs ADA	0.971	0.168	1.345	0.243	-0.374	0.295	0.204
PBO vs ETN	1.679	0.203	2.121	1.063	-0.442	1.083	0.683
PBO vs IFX	1.198	0.217	1.264	0.966	-0.066	0.992	0.947
PBO vs CZP	1.670	0.173	0.936	0.474	0.734	0.504	0.146
PBO vs TCZ	1.489	0.170	0.902	0.685	0.588	0.705	0.405
PBO vs RTX	1.338	0.263	0.896	1.050	0.442	1.083	0.683
PBO vs ABT	1.296	0.211	1.350	0.440	-0.055	0.489	0.911
Bar 2mg vs Bar 4mg	0.149	0.281	0.146	0.742	0.003	0.793	0.997
Bar 4mg vs ADA	-0.372	0.464	-0.060	0.268	-0.311	0.536	0.561
Tof 5mg vs Tof 10mg	0.194	0.217	0.290	1.036	-0.095	1.059	0.928
Tof 5mg vs ADA	-0.143	0.337	-0.145	0.286	0.002	0.442	0.996
Tof 10mg vs ADA	-0.098	0.483	-0.417	0.267	0.319	0.552	0.563
Upa 15mg vs Upa 30mg	0.107	0.252	-0.181	0.912	0.288	0.945	0.760
Upa 15mg vs ADA	-0.341	0.463	-0.197	0.285	-0.144	0.544	0.791
ADA vs CZP	-0.096	0.452	0.638	0.224	-0.734	0.504	0.146
ADA vs TCZ	-0.167	0.670	0.421	0.221	-0.588	0.705	0.405
ADA vs ABT	0.059	0.469	0.257	0.252	-0.198	0.532	0.710
ETN vs csDMARDs	-0.843	0.288	-0.401	1.043	-0.442	1.083	0.683
IFX vs ABT	0.317	0.496	0.017	0.318	0.300	0.589	0.611
RTX vs csDMARDs	-0.061	0.989	-0.503	0.440	0.442	1.083	0.683

ACR20, American College of Rheumatology 20% response; OR, odds ratio; SE, standard error; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

9.2.2 DAS28

Comparison	Direct		Indirect		Difference		
	WMD	SE	WMD	SE	WMD	SE	P-value
PBO vs Bar 2mg	-0.788	0.167	-0.843	0.466	0.055	0.492	0.911
PBO vs Bar 4mg	-1.060	0.126	-1.275	0.571	0.215	0.585	0.713
PBO vs Tof 5mg	-1.030	0.132	-1.197	0.304	0.167	0.332	0.614
PBO vs Tof 10mg	-1.226	0.133	-1.465	0.484	0.240	0.503	0.634
PBO vs Upa 15mg	-1.297	0.144	-1.575	0.579	0.278	0.597	0.642
PBO vs Upa 30mg	-1.336	0.165	-1.523	0.600	0.188	0.621	0.763
PBO vs ADA	-0.909	0.158	-1.010	0.165	0.101	0.229	0.659
PBO vs ETN	-1.262	0.136	-1.039	0.299	-0.223	0.328	0.496
PBO vs IFX	-0.826	0.256	-1.033	0.632	0.207	0.689	0.764
PBO vs TCZ	-1.908	0.121	-1.970	0.291	0.061	0.315	0.845
PBO vs RTX	-1.249	0.196	-0.909	0.451	-0.340	0.492	0.489
PBO vs ABT	-1.200	0.143	-1.081	0.235	-0.120	0.275	0.664
Bar 2mg vs Bar 4mg	-0.268	0.172	-0.324	0.460	0.055	0.492	0.911
Bar 4mg vs ADA	0.292	0.293	0.038	0.187	0.253	0.348	0.467
Tof 5mg vs Tof 10mg	-0.196	0.132	0.067	0.667	-0.264	0.679	0.698
Tof 5mg vs ADA	0.200	0.210	0.012	0.195	0.188	0.287	0.513
Tof 10mg vs ADA	0.210	0.300	0.313	0.182	-0.103	0.351	0.770
Upa 15mg vs Upa 30mg	-0.049	0.166	0.138	0.599	-0.188	0.621	0.763
Upa 15mg vs ADA	0.469	0.300	0.302	0.206	0.168	0.364	0.645
ADA vs ETN	0.200	0.363	-0.374	0.174	0.574	0.403	0.154
ADA vs TCZ	-1.150	0.466	-0.936	0.162	-0.214	0.493	0.665
ADA vs ABT	-0.030	0.301	-0.273	0.178	0.243	0.350	0.487
ETN vs csDMARDs	0.733	0.181	0.392	0.457	0.340	0.492	0.489
IFX vs ABT	-0.280	0.328	-0.351	0.368	0.071	0.493	0.885
TCZ vs ABT	0.720	0.339	0.757	0.174	-0.037	0.381	0.922
RTX vs csDMARDs	0.440	0.393	0.780	0.296	-0.340	0.492	0.489

DAS28, Disease Activity Score in 28 joints; SE, standard error; WMD, weighted mean difference; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; ETN, etanercept; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

9.2.3 HAQ-DI

Comparison	Direct		Indirect		Difference		
	WMD	SE	WMD	SE	WMD	SE	P-value
PBO vs Bar 2mg	-0.197	0.041	-0.268	0.111	0.071	0.115	0.538
PBO vs Bar 4mg	-0.258	0.030	-0.367	0.121	0.109	0.124	0.380
PBO vs Tof 5mg	-0.271	0.033	-0.294	0.066	0.023	0.074	0.761
PBO vs Tof 10mg	-0.329	0.033	-0.369	0.110	0.041	0.116	0.725
PBO vs Upa 15mg	-0.305	0.035	-0.406	0.129	0.101	0.134	0.450
PBO vs Upa 30mg	-0.309	0.042	-0.387	0.135	0.078	0.142	0.583
PBO vs ADA	-0.268	0.029	-0.196	0.042	-0.072	0.051	0.155
PBO vs ETN	-0.272	0.040	-0.062	222.930	-0.210	222.930	0.999
PBO vs IFX	-0.313	0.062	-0.075	0.163	-0.238	0.177	0.180
PBO vs TCZ	-0.255	0.032	-0.391	0.116	0.136	0.120	0.257
PBO vs ABT	-0.291	0.036	-0.267	0.067	-0.024	0.076	0.756
Bar 2mg vs Bar 4mg	-0.050	0.043	-0.122	0.108	0.071	0.115	0.538
Bar 4mg vs ADA	0.091	0.060	-0.015	0.043	0.106	0.074	0.153
Tof 5mg vs Tof 10mg	-0.058	0.032	-0.033	0.148	-0.024	0.151	0.872
Tof 5mg vs ADA	0.040	0.049	0.023	0.044	0.017	0.066	0.793
Tof 10mg vs ADA	0.138	0.073	0.071	0.041	0.067	0.084	0.425
Upa 15mg vs Upa 30mg	-0.012	0.042	0.066	0.135	-0.078	0.142	0.583
Upa 15mg vs ADA	0.109	0.066	0.043	0.049	0.066	0.083	0.426
ADA vs TCZ	-0.170	0.130	-0.006	0.040	-0.164	0.136	0.227
ADA vs ABT	-0.010	0.069	-0.053	0.043	0.043	0.081	0.596
ETN vs csDMARDs	0.175	0.055	0.529	338.450	-0.353	338.450	0.999
IFX vs ABT	-0.080	0.084	0.072	0.085	-0.152	0.120	0.204
TCZ vs ABT	0.010	0.234	-0.022	0.044	0.032	0.239	0.893

HAQ-DI, Health Assessment Questionnaire–Disability Index; SE, standard error; WMD, weighted mean difference; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; ETN, etanercept; IFX, infliximab; PBO, placebo; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

9.2.4 Discontinuations for AEs

Comparison	Direct		Indirect		Difference		
	Log OR	SE	Log OR	SE	Log OR	SE	P-value
PBO vs Bar 2mg	-0.087	0.413	0.680	1.010	-0.767	1.089	0.481
PBO vs Bar 4mg	0.163	0.274	2.332	1.023	-2.169	1.064	0.042
PBO vs Tof 5mg	0.576	0.280	-0.030	0.431	0.606	0.522	0.246
PBO vs Tof 10mg	0.529	0.285	-0.238	0.795	0.768	0.877	0.381
PBO vs Upa 15mg	0.238	0.319	-1.332	0.923	1.570	0.990	0.113
PBO vs Upa 30mg	0.669	0.374	0.356	1.125	0.312	1.209	0.796
PBO vs ADA	0.389	0.225	0.273	0.266	0.116	0.347	0.738
PBO vs ETN	-0.242	0.426	0.711	0.531	-0.953	0.681	0.161
PBO vs IFX	0.483	0.333	1.677	1.767	-1.194	1.841	0.517
PBO vs CZP	0.458	0.235	0.190	0.638	0.268	0.680	0.693
PBO vs TCZ	0.197	0.212	0.880	0.552	-0.683	0.593	0.249
PBO vs RTX	0.861	0.548	1.131	2.116	-0.269	2.186	0.902
PBO vs ABT	0.030	0.276	-0.537	0.413	0.567	0.500	0.257
Bar 2mg vs Bar 4mg	0.186	0.405	0.953	1.018	-0.767	1.089	0.481
Bar 4mg vs ADA	-0.794	0.510	0.510	0.382	-1.304	0.654	0.046
Tof 5mg vs Tof 10mg	-0.035	0.232	1.187	0.977	-1.222	1.002	0.223
Tof 5mg vs ADA	0.076	0.329	-0.201	0.354	0.278	0.482	0.564
Tof 10mg vs ADA	-0.016	0.561	-0.101	0.319	0.085	0.645	0.895
Upa 15mg vs Upa 30mg	0.541	0.375	0.853	1.124	-0.312	1.209	0.796
Upa 15mg vs ADA	0.710	0.484	-0.040	0.427	0.750	0.646	0.246
ADA vs ETN	0.357	0.520	-0.647	0.452	1.004	0.689	0.145
ADA vs CZP	-0.134	0.613	0.135	0.294	-0.268	0.680	0.693
ADA vs TCZ	0.511	0.822	-0.116	0.270	0.627	0.865	0.469
ADA vs ABT	-0.595	0.502	-0.449	0.311	-0.145	0.591	0.806
ETN vs csDMARDs	0.999	0.405	0.730	2.148	0.269	2.186	0.902
IFX vs ABT	-1.499	0.841	-0.493	0.420	-1.006	0.945	0.287
TCZ vs ABT	-0.950	0.692	-0.326	0.318	-0.624	0.762	0.413
RTX vs csDMARDs	0.000	2.048	0.269	0.762	-0.269	2.186	0.902

AEs, adverse events; OR, odds ratio; SE, standard error; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

9.3 Design-by-treatment test

9.3.1 ACR20

chi2 (14) = 4.94

Prob > chi2 = 0.9926

ACR20, American College of Rheumatology 20% response.

9.3.2 DAS28

chi2 (15) = 3.96

Prob > chi2 = 0.9990

DAS28, Disease Activity Score in 28 joints.

9.3.3 HAQ-DI

chi2 (13) = 9.21

Prob > chi2 = 0.8175

HAQ-DI, Health Assessment Questionnaire–Disability Index.

9.3.4 Discontinuations for AEs

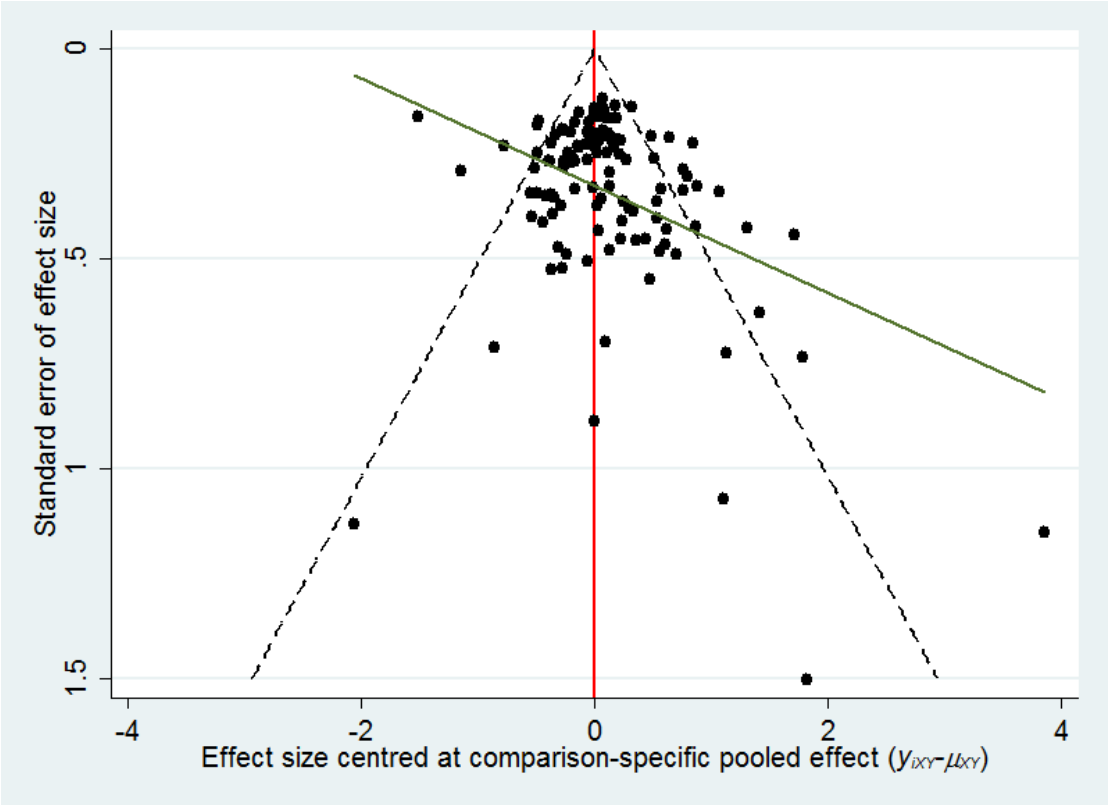
chi2 (16) = 13.62

Prob > chi2 = 0.6937

AEs, adverse events.

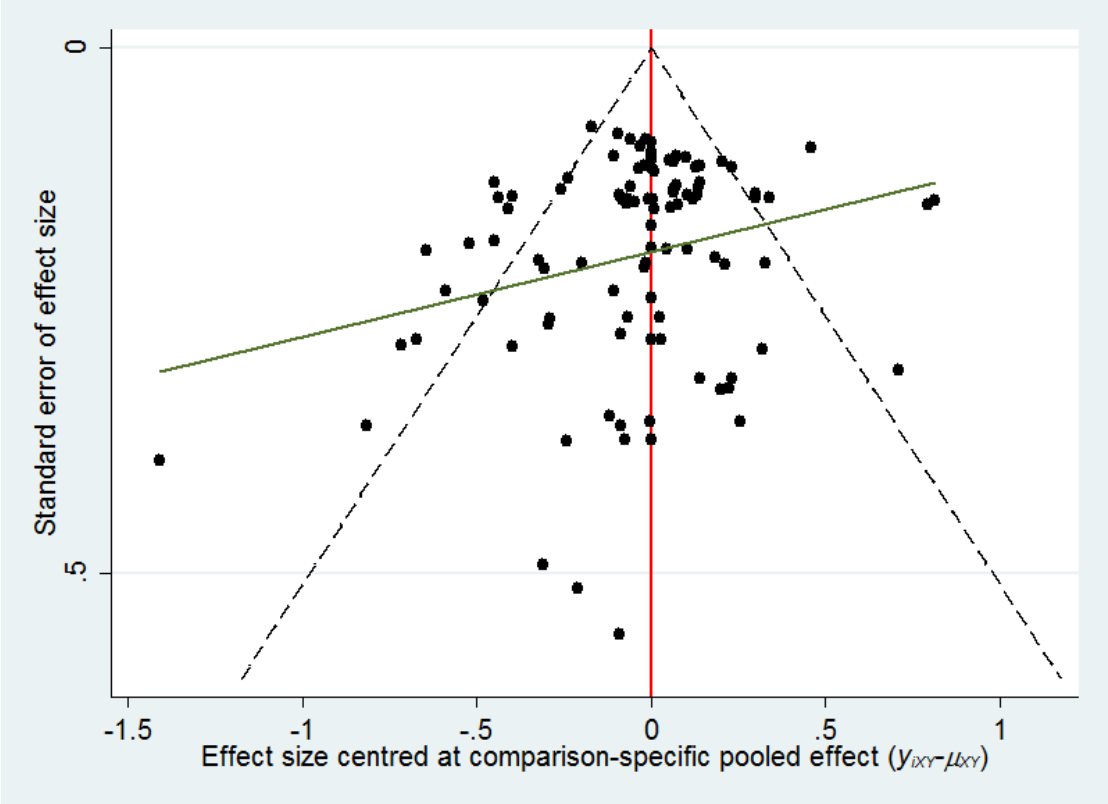
Appendix 10. Comparison-adjusted funnel plots

10.1 ACR20



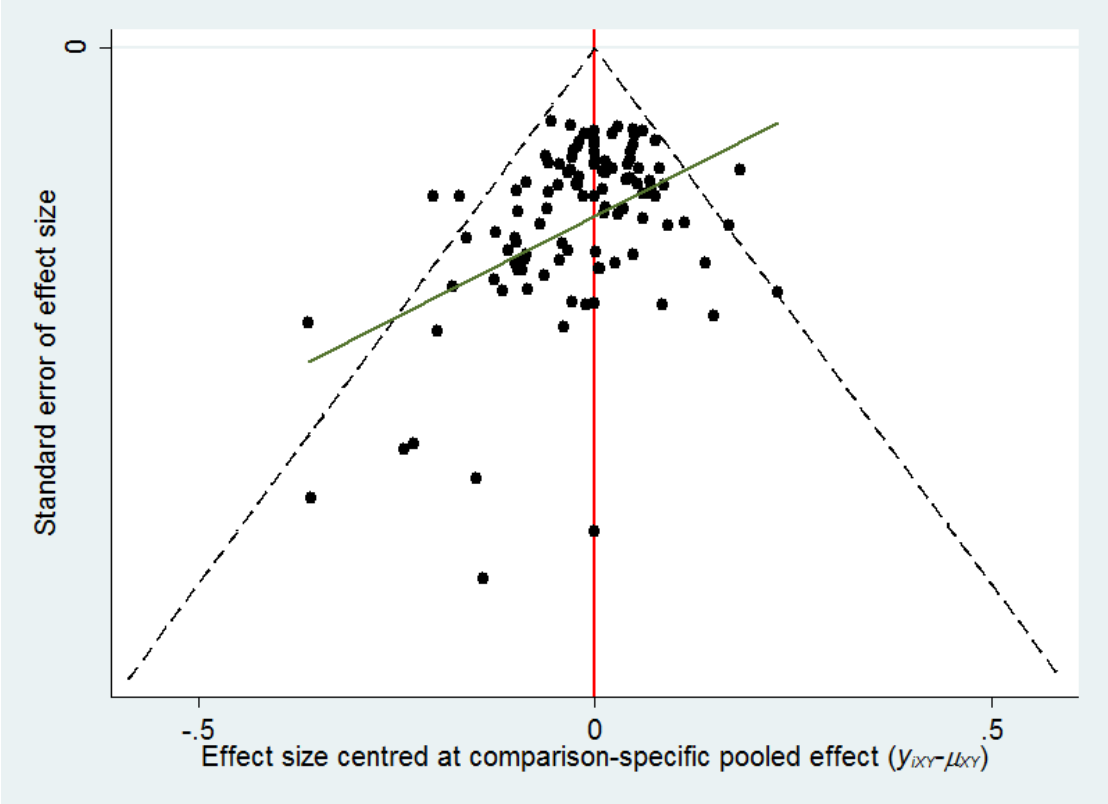
ACR20, American College of Rheumatology 20% response.

10.2 DAS28



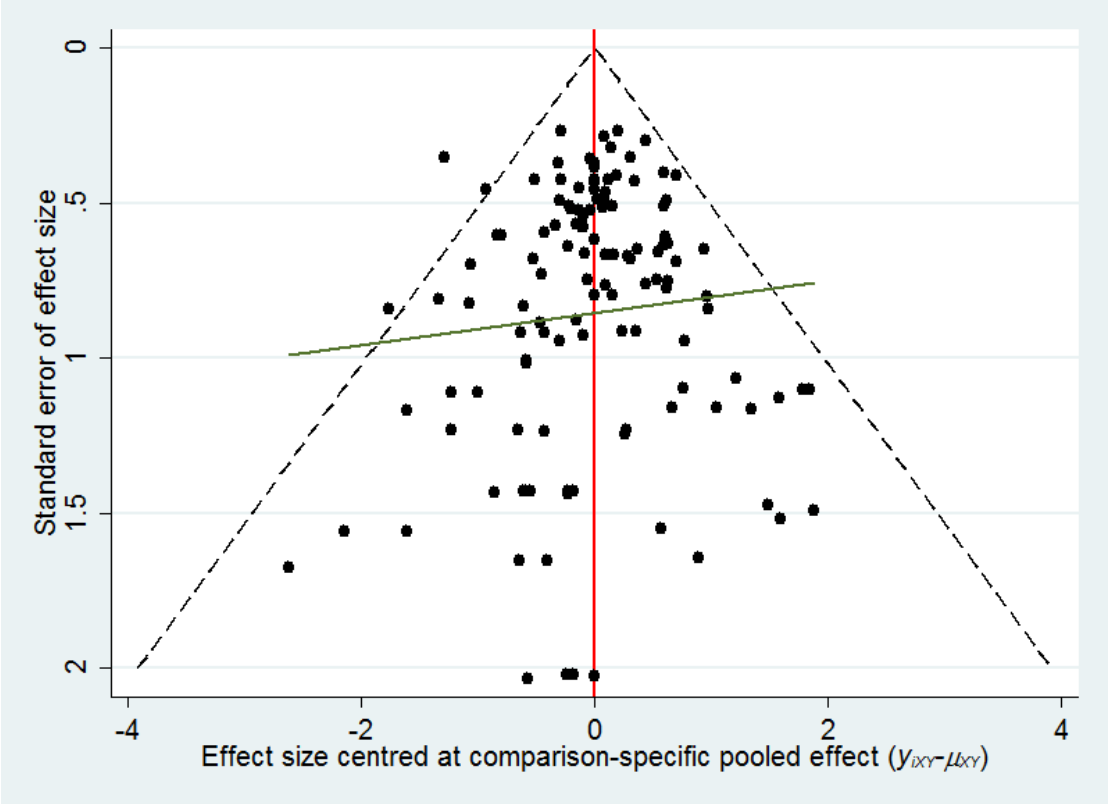
DAS28, Disease Activity Score in 28 joints.

10.3 HAQ-DI



HAQ-DI, Health Assessment Questionnaire–Disability Index.

10.4 Discontinuations for AEs



AEs, adverse events.

Appendix 11. Subgroup analysis by subjects

11.1 Summary of subgroup analysis

11.1.1 ACR20

Drug	OR with 95%CrI compared with placebo		
	Primary results	csDMARDs-IR	bDMARDs-IR
Baricitinib 2mg	3.05 (1.78, 5.27)	3.39 (1.70, 6.75)	2.56 (1.65, 4.04)
Baricitinib 4mg	3.50 (2.31, 5.29)	3.71 (2.24, 6.19)	3.33 (2.15, 5.23)
Tofacitinib 5mg	3.79 (2.52, 5.80)	4.23 (2.64, 6.90)	2.22 (1.33, 3.80)
Tofacitinib 10mg	4.37 (2.85, 6.77)	4.73 (2.86, 7.92)	2.88 (1.72, 4.93)
Upadacitinib 15mg	3.85 (2.47, 6.04)	3.91 (2.22, 6.90)	4.04 (2.73, 6.04)
Upadacitinib 30mg	4.21 (2.55, 6.94)	4.51 (2.27, 9.04)	3.57 (2.43, 5.29)
Adalimumab	3.10 (2.32, 4.18)	3.11 (2.27, 4.34)	-
Etanercept	5.61 (3.72, 8.51)	5.02 (3.16, 8.04)	-
Infliximab	3.41 (2.20, 5.29)	3.43 (2.17, 5.52)	-
Golimumab	3.35 (2.03, 5.59)	3.65 (1.98, 6.76)	2.49 (1.49, 4.24)
Certolizumab pegol	4.95 (3.54, 6.98)	5.27 (3.56, 7.80)	-
Tocilizumab	4.50 (3.15, 6.49)	4.57 (2.96, 7.13)	9.05 (5.03, 17.22)
Rituximab	3.77 (2.20, 6.43)	3.55 (1.57, 8.04)	4.88 (3.22, 7.51)
Abatacept	3.77 (2.54, 5.64)	3.96 (2.41, 6.62)	3.44 (2.44, 4.91)
csDMARDs	2.48 (1.24, 4.99)	2.22 (1.05, 4.86)	-

Some data is missing because there are no or too few samples. ACR20, American College of Rheumatology 20% response; bDMARDs, biologic DMARDs; CrI, credible interval; csDMARDs, conventional synthetic DMARDs; DMARDs, disease-modifying antirheumatic drugs; IR, inadequate response; OR, odds ratio.

11.1.2 DAS28

Drug	WMD with 95%CrI compared with placebo		
	Primary results	csDMARDs-IR	bDMARDs-IR
Baricitinib 2mg	-0.80 (-1.11, -0.48)	-0.84 (-1.23, -0.45)	-0.68 (-0.95, -0.42)
Baricitinib 4mg	-1.07 (-1.31, -0.82)	-1.12 (-1.42, -0.82)	-0.96 (-1.24, -0.69)
Tofacitinib 5mg	-1.05 (-1.29, -0.81)	-1.07 (-1.35, -0.79)	-1.03 (-1.32, -0.74)
Tofacitinib 10mg	-1.24 (-1.50, -0.98)	-1.25 (-1.55, -0.95)	-1.25 (-1.54, -0.97)
Upadacitinib 15mg	-1.31 (-1.58, -1.03)	-1.35 (-1.71, -1.00)	-1.26 (-1.53, -0.98)
Upadacitinib 30mg	-1.35 (-1.66, -1.03)	-1.37 (-1.78, -0.95)	-1.28 (-1.56, -1.00)
Adalimumab	-0.93 (-1.16, -0.70)	-0.98 (-1.24, -0.72)	-
Etanercept	-1.22 (-1.47, -0.97)	-1.13 (-1.41, -0.85)	-
Infliximab	-0.85 (-1.33, -0.38)	-0.87 (-1.37, -0.38)	-
Golimumab	-1.08 (-1.48, -0.69)	-1.31 (-1.86, -0.78)	-0.73 (-1.01, -0.45)
Certolizumab pegol	-1.42 (-1.68, -1.16)	-1.49 (-1.80, -1.16)	-
Tocilizumab	-1.91 (-2.14, -1.68)	-1.96 (-2.24, -1.69)	-1.79 (-2.20, -1.37)
Rituximab	-1.19 (-1.55, -0.83)	-0.89 (-1.44, -0.34)	-1.50 (-1.74, -1.25)
Abatacept	-1.16 (-1.41, -0.92)	-1.22 (-1.56, -0.88)	-1.07 (-1.26, -0.88)
csDMARDs	-0.53 (-0.93, -0.13)	-0.41 (-0.85, 0.03)	-

Some data is missing because there are no or too few samples. bDMARDs, biologic DMARDs; CrI, credible interval; csDMARDs, conventional synthetic DMARDs; DAS28, Disease Activity Score in 28 joints; DMARDs, disease-modifying antirheumatic drugs; IR, inadequate response; WMD, weighted mean difference.

11.1.3 HAQ-DI

Drug	WMD with 95%CrI compared with placebo		
	Primary results	csDMARDs-IR	bDMARDs-IR
Baricitinib 2mg	-0.21 (-0.28, -0.13)	-0.21 (-0.31, -0.11)	-0.18 (-0.29, -0.07)
Baricitinib 4mg	-0.27 (-0.32, -0.21)	-0.28 (-0.35, -0.22)	-0.22 (-0.32, -0.12)
Tofacitinib 5mg	-0.27 (-0.33, -0.22)	-0.28 (-0.35, -0.22)	-0.24 (-0.35, -0.13)
Tofacitinib 10mg	-0.33 (-0.39, -0.27)	-0.35 (-0.42, -0.29)	-0.24 (-0.35, -0.13)
Upadacitinib 15mg	-0.31 (-0.38, -0.25)	-0.34 (-0.41, -0.26)	-0.24 (-0.35, -0.13)
Upadacitinib 30mg	-0.32 (-0.39, -0.24)	-0.34 (-0.44, -0.24)	-0.26 (-0.37, -0.16)
Adalimumab	-0.25 (-0.30, -0.20)	-0.25 (-0.31, -0.20)	-
Etanercept	-0.27 (-0.35, -0.19)	-0.26 (-0.34, -0.17)	-
Infliximab	-0.28 (-0.39, -0.17)	-0.27 (-0.39, -0.16)	-
Golimumab	-0.27 (-0.34, -0.20)	-0.30 (-0.40, -0.21)	-0.21 (-0.29, -0.13)
Certolizumab pegol	-0.31 (-0.36, -0.25)	-0.32 (-0.39, -0.26)	-
Tocilizumab	-0.25 (-0.32, -0.19)	-0.25 (-0.32, -0.18)	-0.34 (-0.53, -0.16)
Rituximab	-0.34 (-0.45, -0.22)	-0.25 (-0.47, -0.03)	-0.37 (-0.46, -0.28)
Abatacept	-0.28 (-0.35, -0.22)	-0.26 (-0.34, -0.19)	-0.35 (-0.44, -0.26)
csDMARDs	-0.10 (-0.23, 0.04)	-0.08 (-0.22, 0.06)	-

Some data is missing because there are no or too few samples. bDMARDs, biologic DMARDs; CrI, credible interval; csDMARDs, conventional synthetic DMARDs; DMARDs, disease-modifying antirheumatic drugs; HAQ-DI, Health Assessment Questionnaire–Disability Index; IR, inadequate response; WMD, weighted mean difference.

11.1.4 Discontinuations for AEs

Drug	OR with 95%CrI compared with placebo		
	Primary results	csDMARDs-IR	bDMARDs-IR
Baricitinib 2mg	0.98 (0.42, 2.16)	0.75 (0.23, 2.17)	1.87 (0.53, 7.63)
Baricitinib 4mg	1.33 (0.73, 2.38)	1.17 (0.57, 2.31)	2.41 (0.74, 9.19)
Tofacitinib 5mg	1.52 (0.92, 2.54)	1.58 (0.91, 2.79)	1.00 (0.10, 9.57)
Tofacitinib 10mg	1.55 (0.91, 2.74)	1.68 (0.92, 3.14)	0.41 (0.01, 5.28)
Upadacitinib 15mg	1.07 (0.57, 2.05)	1.09 (0.48, 2.48)	1.03 (0.38, 2.71)
Upadacitinib 30mg	2.01 (0.98, 4.17)	2.51 (0.90, 7.55)	1.78 (0.77, 4.36)
Adalimumab	1.41 (0.97, 2.02)	1.42 (0.95, 2.11)	-
Etanercept	1.07 (0.53, 2.09)	1.02 (0.47, 2.15)	-
Infliximab	1.82 (0.93, 3.58)	1.82 (0.91, 3.71)	-
Golimumab	1.02 (0.47, 2.29)	1.70 (0.63, 4.88)	0.37 (0.10, 1.17)
Certolizumab pegol	1.65 (1.06, 2.61)	1.74 (1.03, 2.97)	-
Tocilizumab	1.39 (0.92, 2.18)	1.40 (0.82, 2.56)	1.56 (0.69, 3.69)
Rituximab	3.17 (1.11, 10.80)	1.39 (0.24, 8.77)	3.14 (0.74, 23.26)
Abatacept	0.85 (0.52, 1.37)	0.82 (0.43, 1.46)	0.90 (0.40, 2.15)
csDMARDs	3.02 (1.00, 8.97)	2.86 (0.89, 9.27)	-

Some data is missing because there are no or too few samples. AEs, adverse events; bDMARDs, biologic DMARDs; CrI, credible interval; csDMARDs, conventional synthetic DMARDs; DMARDs, disease-modifying antirheumatic drugs; IR, inadequate response; OR, odds ratio.

11.2 csDMARDs-IR

11.2.1 ACR20

Bar 2mg															
0.91 (0.45, 1.85)	Bar 4mg														
0.80 (0.35, 1.83)	0.88 (0.44, 1.74)	Tof 5mg													
0.72 (0.30, 1.69)	0.79 (0.38, 1.62)	0.90 (0.53, 1.52)	Tof 10mg												
0.86 (0.36, 2.13)	0.95 (0.45, 2.02)	1.08 (0.52, 2.26)	1.21 (0.57, 2.58)	Upa 15mg											
0.75 (0.29, 2.00)	0.82 (0.35, 1.95)	0.94 (0.41, 2.16)	1.05 (0.44, 2.49)	0.87 (0.43, 1.74)	Upa 30mg										
1.08 (0.51, 2.32)	1.19 (0.67, 2.10)	1.35 (0.81, 2.29)	1.52 (0.86, 2.67)	1.25 (0.68, 2.32)	1.44 (0.68, 3.06)	ADA									
0.68 (0.29, 1.57)	0.74 (0.37, 1.47)	0.84 (0.43, 1.66)	0.94 (0.48, 1.88)	0.78 (0.37, 1.62)	0.90 (0.39, 2.05)	0.62 (0.35, 1.10)	ETN								
0.99 (0.43, 2.23)	1.08 (0.54, 2.14)	1.23 (0.63, 2.37)	1.38 (0.69, 2.73)	1.14 (0.54, 2.35)	1.31 (0.57, 2.98)	0.91 (0.51, 1.59)	1.46 (0.75, 2.84)	IFX							
0.92 (0.37, 2.36)	1.01 (0.46, 2.25)	1.16 (0.53, 2.55)	1.29 (0.58, 2.87)	1.06 (0.47, 2.48)	1.23 (0.49, 3.10)	0.85 (0.43, 1.71)	1.37 (0.63, 2.99)	0.94 (0.44, 2.07)	GOL						
0.64 (0.29, 1.43)	0.70 (0.37, 1.33)	0.80 (0.43, 1.48)	0.90 (0.48, 1.71)	0.74 (0.37, 1.46)	0.86 (0.39, 1.90)	0.59 (0.37, 0.96)	0.95 (0.52, 1.77)	0.65 (0.36, 1.19)	0.70 (0.33, 1.44)	CZP					
0.74 (0.33, 1.68)	0.81 (0.41, 1.60)	0.92 (0.48, 1.77)	1.03 (0.53, 2.04)	0.86 (0.42, 1.75)	0.99 (0.43, 2.23)	0.68 (0.40, 1.17)	1.10 (0.58, 2.08)	0.75 (0.40, 1.44)	0.80 (0.37, 1.70)	1.16 (0.64, 2.06)	TCZ				
0.95 (0.33, 2.81)	1.04 (0.40, 2.67)	1.19 (0.46, 3.12)	1.33 (0.51, 3.55)	1.10 (0.41, 2.96)	1.27 (0.43, 3.62)	0.88 (0.36, 2.10)	1.42 (0.57, 3.49)	0.97 (0.38, 2.48)	1.03 (0.37, 2.87)	1.49 (0.60, 3.69)	1.28 (0.51, 3.27)	RTX			
0.85 (0.36, 2.00)	0.93 (0.46, 1.91)	1.07 (0.54, 2.13)	1.19 (0.58, 2.44)	0.99 (0.46, 2.11)	1.14 (0.48, 2.66)	0.79 (0.45, 1.37)	1.27 (0.63, 2.50)	0.87 (0.46, 1.65)	0.92 (0.41, 2.03)	1.33 (0.71, 2.50)	1.15 (0.59, 2.26)	0.90 (0.34, 2.33)	ABT		
1.52 (0.54, 4.21)	1.67 (0.66, 4.16)	1.89 (0.76, 4.65)	2.12 (0.85, 5.29)	1.76 (0.67, 4.52)	2.01 (0.72, 5.63)	1.39 (0.60, 3.19)	2.26 (1.20, 4.20)	1.54 (0.63, 3.77)	1.64 (0.61, 4.30)	2.37 (0.99, 5.56)	2.06 (0.84, 4.91)	1.59 (0.56, 4.48)	1.79 (0.70, 4.51)	csDM ARDs	
3.39 (1.70, 6.75)	3.71 (2.24, 6.19)	4.23 (2.64, 6.90)	4.73 (2.86, 7.92)	3.91 (2.22, 6.90)	4.51 (2.27, 9.04)	3.11 (2.27, 4.34)	5.02 (3.16, 8.04)	3.43 (2.17, 5.52)	3.65 (1.98, 6.76)	5.27 (3.56, 7.80)	4.57 (2.96, 7.13)	3.55 (1.57, 8.04)	3.96 (2.41, 6.62)	2.22 (1.05, 4.86)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of ACR20. ORs higher than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). ACR20, American College of Rheumatology 20% response; CrI, credible interval; ORs, odds ratios; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

11.2.2 DAS28

Bar 2mg															
0.28 (-0.12, 0.68)	Bar 4mg														
0.23 (-0.25, 0.71)	-0.05 (-0.45, 0.35)	Tof 5mg													
0.41 (-0.08, 0.91)	0.13 (-0.29, 0.55)	0.18 (-0.12, 0.49)	Tof 10mg												
0.51 (-0.01, 1.04)	0.23 (-0.22, 0.69)	0.28 (-0.15, 0.73)	0.11 (-0.35, 0.56)	Upa 15mg											
0.53 (-0.05, 1.10)	0.25 (-0.27, 0.76)	0.30 (-0.20, 0.78)	0.12 (-0.39, 0.63)	0.01 (-0.41, 0.44)	Upa 30mg										
0.14 (-0.33, 0.59)	-0.14 (-0.51, 0.22)	-0.09 (-0.41, 0.23)	-0.27 (-0.62, 0.08)	-0.38 (-0.78, 0.03)	-0.39 (-0.87, 0.09)	ADA									
0.29 (-0.19, 0.77)	0.01 (-0.40, 0.41)	0.06 (-0.33, 0.45)	-0.12 (-0.52, 0.29)	-0.22 (-0.67, 0.22)	-0.24 (-0.74, 0.28)	0.15 (-0.20, 0.51)	ETN								
0.04 (-0.60, 0.67)	-0.24 (-0.82, 0.34)	-0.19 (-0.76, 0.37)	-0.37 (-0.95, 0.20)	-0.48 (-1.09, 0.12)	-0.49 (-1.14, 0.17)	-0.10 (-0.65, 0.45)	-0.26 (-0.82, 0.32)	IFX							
0.47 (-0.18, 1.14)	0.20 (-0.42, 0.81)	0.24 (-0.37, 0.84)	0.06 (-0.56, 0.67)	-0.04 (-0.68, 0.60)	-0.05 (-0.72, 0.63)	0.34 (-0.26, 0.93)	0.19 (-0.42, 0.79)	0.44 (-0.29, 1.17)	GOL						
0.65 (0.14, 1.15)	0.37 (-0.08, 0.80)	0.42 (-0.01, 0.83)	0.24 (-0.21, 0.66)	0.13 (-0.34, 0.60)	0.12 (-0.41, 0.65)	0.51 (0.10, 0.91)	0.36 (-0.07, 0.78)	0.61 (0.02, 1.20)	0.17 (-0.45, 0.80)	CZP					
1.12 (0.64, 1.60)	0.85 (0.45, 1.26)	0.90 (0.50, 1.29)	0.72 (0.31, 1.12)	0.61 (0.16, 1.06)	0.60 (0.09, 1.11)	0.99 (0.61, 1.37)	0.83 (0.45, 1.23)	1.09 (0.52, 1.66)	0.65 (0.06, 1.26)	0.48 (0.06, 0.91)	TCZ				
0.06 (-0.62, 0.72)	-0.22 (-0.86, 0.40)	-0.18 (-0.80, 0.43)	-0.36 (-0.99, 0.27)	-0.46 (-1.13, 0.19)	-0.48 (-1.17, 0.21)	-0.08 (-0.70, 0.51)	-0.24 (-0.81, 0.33)	0.02 (-0.73, 0.75)	-0.42 (-1.19, 0.34)	-0.59 (-1.22, 0.04)	-1.07 (-1.70, -0.46)	RTX			
0.38 (-0.14, 0.90)	0.10 (-0.35, 0.55)	0.15 (-0.28, 0.58)	-0.03 (-0.48, 0.41)	-0.14 (-0.61, 0.35)	-0.15 (-0.68, 0.39)	0.24 (-0.14, 0.63)	0.09 (-0.35, 0.53)	0.34 (-0.19, 0.89)	-0.10 (-0.74, 0.54)	-0.27 (-0.73, 0.22)	-0.74 (-1.19, -0.30)	0.33 (-0.31, 0.99)	ABT		
-0.44 (-1.02, 0.16)	-0.71 (-1.24, -0.19)	-0.66 (-1.17, -0.15)	-0.84 (-1.36, -0.31)	-0.95 (-1.51, -0.39)	-0.96 (-1.56, -0.34)	-0.57 (-1.06, -0.08)	-0.72 (-1.08, -0.36)	-0.47 (-1.12, 0.19)	-0.91 (-1.60, -0.21)	-1.08 (-1.62, -0.53)	-1.55 (-2.08, -1.04)	-0.48 (-1.07, 0.10)	-0.81 (-1.37, -0.27)	csDM ARDs	
-0.84 (-1.23, -0.45)	-1.12 (-1.42, -0.82)	-1.07 (-1.35, -0.79)	-1.25 (-1.55, -0.95)	-1.35 (-1.71, -1.00)	-1.37 (-1.78, -0.95)	-0.98 (-1.24, -0.72)	-1.13 (-1.41, -0.85)	-0.87 (-1.37, -0.38)	-1.31 (-1.86, -0.78)	-1.49 (-1.80, -1.16)	-1.96 (-2.24, -1.69)	-0.89 (-1.44, -0.34)	-1.22 (-1.56, -0.88)	-0.41 (-0.85, 0.03)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of DAS28. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; DAS28, Disease Activity Score in 28 joints; WMDs, weighted mean differences; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

11.2.3 HAQ-DI

Bar 2mg															
0.07 (-0.03, 0.17)	Bar 4mg														
0.07 (-0.04, 0.19)	0.00 (-0.08, 0.10)	Tof 5mg													
0.14 (0.03, 0.26)	0.07 (-0.02, 0.17)	0.07 (0.00, 0.13)	Tof 10mg												
0.12 (0.00, 0.25)	0.06 (-0.04, 0.16)	0.05 (-0.05, 0.15)	-0.02 (-0.12, 0.09)	Upa 15mg											
0.13 (-0.01, 0.27)	0.06 (-0.06, 0.18)	0.06 (-0.06, 0.17)	-0.01 (-0.13, 0.11)	0.00 (-0.10, 0.11)	Upa 30mg										
0.04 (-0.06, 0.15)	-0.03 (-0.10, 0.05)	-0.03 (-0.10, 0.04)	-0.10 (-0.17, -0.02)	-0.09 (-0.17, 0.01)	-0.09 (-0.20, 0.02)	ADA									
0.04 (-0.08, 0.18)	-0.02 (-0.13, 0.09)	-0.03 (-0.13, 0.08)	-0.10 (-0.20, 0.01)	-0.08 (-0.19, 0.04)	-0.08 (-0.21, 0.05)	0.01 (-0.09, 0.10)	ETN								
0.06 (-0.08, 0.21)	-0.01 (-0.14, 0.13)	-0.01 (-0.14, 0.12)	-0.08 (-0.21, 0.05)	-0.07 (-0.20, 0.07)	-0.07 (-0.22, 0.09)	0.02 (-0.10, 0.14)	0.01 (-0.13, 0.16)	IFX							
0.09 (-0.04, 0.23)	0.02 (-0.09, 0.14)	0.02 (-0.09, 0.13)	-0.05 (-0.16, 0.07)	-0.03 (-0.15, 0.09)	-0.04 (-0.17, 0.10)	0.05 (-0.05, 0.16)	0.05 (-0.08, 0.17)	0.03 (-0.12, 0.18)	GOL						
0.11 (0.00, 0.23)	0.04 (-0.05, 0.13)	0.04 (-0.05, 0.13)	-0.03 (-0.12, 0.06)	-0.01 (-0.11, 0.09)	-0.02 (-0.14, 0.10)	0.07 (-0.01, 0.15)	0.07 (-0.04, 0.17)	0.05 (-0.08, 0.18)	0.02 (-0.10, 0.13)	CZP					
0.03 (-0.08, 0.15)	-0.03 (-0.12, 0.06)	-0.04 (-0.13, 0.05)	-0.11 (-0.20, -0.01)	-0.09 (-0.19, 0.02)	-0.09 (-0.21, 0.03)	-0.01 (-0.09, 0.08)	-0.01 (-0.12, 0.10)	-0.02 (-0.16, 0.11)	-0.06 (-0.17, 0.06)	-0.08 (-0.17, 0.02)	TCZ				
0.04 (-0.2, 0.28)	-0.03 (-0.25, 0.20)	-0.03 (-0.26, 0.19)	-0.10 (-0.33, 0.13)	-0.08 (-0.32, 0.15)	-0.09 (-0.33, 0.15)	0.00 (-0.23, 0.22)	0.00 (-0.24, 0.23)	-0.02 (-0.27, 0.23)	-0.05 (-0.29, 0.18)	-0.07 (-0.3, 0.16)	0.01 (-0.22, 0.23)	RTX			
0.05 (-0.07, 0.17)	-0.02 (-0.12, 0.08)	-0.02 (-0.12, 0.07)	-0.09 (-0.19, 0.01)	-0.08 (-0.18, 0.03)	-0.08 (-0.21, 0.05)	0.01 (-0.07, 0.09)	0.00 (-0.11, 0.12)	-0.01 (-0.13, 0.11)	-0.04 (-0.16, 0.08)	-0.06 (-0.16, 0.04)	0.01 (-0.09, 0.11)	0.01 (-0.22, 0.24)	ABT		
-0.13 (-0.30, 0.04)	-0.20 (-0.35, -0.04)	-0.20 (-0.35, -0.05)	-0.27 (-0.42, -0.12)	-0.26 (-0.41, -0.10)	-0.26 (-0.43, -0.09)	-0.17 (-0.32, -0.03)	-0.18 (-0.29, -0.07)	-0.19 (-0.37, -0.01)	-0.22 (-0.39, -0.06)	-0.24 (-0.40, -0.09)	-0.17 (-0.32, -0.01)	-0.17 (-0.43, 0.09)	-0.18 (-0.34, -0.02)	csDM ARDs	
-0.21 (-0.31, -0.11)	-0.28 (-0.35, -0.22)	-0.28 (-0.35, -0.22)	-0.35 (-0.42, -0.29)	-0.34 (-0.41, -0.26)	-0.34 (-0.44, -0.24)	-0.25 (-0.31, -0.20)	-0.26 (-0.34, -0.17)	-0.27 (-0.39, -0.16)	-0.30 (-0.40, -0.21)	-0.32 (-0.39, -0.26)	-0.25 (-0.32, -0.18)	-0.25 (-0.47, -0.03)	-0.26 (-0.34, -0.19)	-0.08 (-0.22, 0.06)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of HAQ-DI. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; HAQ-DI, Health Assessment Questionnaire–Disability Index; WMDs, weighted mean differences; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

11.2.4 Discontinuations for AEs

Bar 2mg															
0.64 (0.20, 1.92)	Bar 4mg														
0.47 (0.13, 1.56)	0.74 (0.30, 1.76)	Tof 5mg													
0.45 (0.12, 1.50)	0.70 (0.27, 1.68)	0.94 (0.53, 1.64)	Tof 10mg												
0.69 (0.16, 2.67)	1.09 (0.36, 3.10)	1.45 (0.54, 3.86)	1.55 (0.57, 4.28)	Upa 15mg											
0.30 (0.06, 1.31)	0.47 (0.12, 1.58)	0.63 (0.19, 1.98)	0.67 (0.20, 2.20)	0.43 (0.15, 1.20)	Upa 30mg										
0.53 (0.16, 1.61)	0.83 (0.38, 1.72)	1.11 (0.63, 2.03)	1.18 (0.62, 2.32)	0.77 (0.32, 1.83)	1.77 (0.60, 5.54)	ADA									
0.74 (0.18, 2.74)	1.16 (0.42, 3.17)	1.55 (0.64, 4.01)	1.66 (0.65, 4.45)	1.07 (0.36, 3.33)	2.48 (0.69, 9.43)	1.40 (0.64, 3.12)	ETN								
0.41 (0.11, 1.49)	0.65 (0.23, 1.72)	0.87 (0.36, 2.15)	0.92 (0.36, 2.39)	0.60 (0.20, 1.75)	1.38 (0.41, 5.07)	0.78 (0.35, 1.76)	0.56 (0.19, 1.56)	IFX							
0.44 (0.09, 1.88)	0.69 (0.19, 2.30)	0.93 (0.28, 2.98)	0.99 (0.30, 3.21)	0.64 (0.17, 2.40)	1.49 (0.34, 6.29)	0.84 (0.27, 2.42)	0.60 (0.16, 2.06)	1.07 (0.30, 3.62)	GOL						
0.43 (0.12, 1.41)	0.67 (0.28, 1.60)	0.91 (0.43, 1.96)	0.97 (0.44, 2.14)	0.63 (0.24, 1.64)	1.45 (0.45, 4.80)	0.82 (0.43, 1.52)	0.58 (0.23, 1.44)	1.04 (0.43, 2.52)	0.98 (0.31, 3.20)	CZP					
0.54 (0.14, 1.73)	0.84 (0.31, 2.02)	1.13 (0.50, 2.43)	1.20 (0.51, 2.68)	0.77 (0.28, 2.05)	1.79 (0.55, 5.93)	1.01 (0.48, 1.98)	0.73 (0.27, 1.83)	1.30 (0.50, 3.10)	1.21 (0.38, 3.92)	1.24 (0.55, 2.64)	TCZ				
0.54 (0.06, 4.22)	0.85 (0.12, 5.55)	1.15 (0.17, 7.16)	1.23 (0.17, 7.94)	0.79 (0.11, 5.50)	1.85 (0.22, 14.15)	1.03 (0.15, 6.29)	0.74 (0.10, 4.87)	1.32 (0.19, 8.79)	1.25 (0.15, 9.67)	1.28 (0.19, 7.98)	1.03 (0.15, 6.73)	RTX			
0.92 (0.25, 3.15)	1.45 (0.58, 3.54)	1.94 (0.88, 4.42)	2.07 (0.90, 4.94)	1.33 (0.49, 3.72)	3.10 (0.96, 10.72)	1.75 (0.90, 3.46)	1.25 (0.48, 3.28)	2.24 (0.93, 5.51)	2.09 (0.65, 7.11)	2.13 (0.98, 4.88)	1.72 (0.78, 4.25)	1.70 (0.26, 11.82)	ABT		
0.26 (0.05, 1.27)	0.41 (0.10, 1.55)	0.55 (0.15, 1.98)	0.59 (0.16, 2.17)	0.38 (0.09, 1.57)	0.89 (0.19, 4.28)	0.50 (0.15, 1.62)	0.36 (0.14, 0.86)	0.63 (0.16, 2.57)	0.59 (0.13, 2.81)	0.61 (0.17, 2.17)	0.49 (0.14, 1.84)	0.48 (0.06, 4.15)	0.28 (0.08, 1.05)	csDM ARDs	
0.75 (0.23, 2.17)	1.17 (0.57, 2.31)	1.58 (0.91, 2.79)	1.68 (0.92, 3.14)	1.09 (0.48, 2.48)	2.51 (0.90, 7.55)	1.42 (0.95, 2.11)	1.02 (0.47, 2.15)	1.82 (0.91, 3.71)	1.70 (0.63, 4.88)	1.74 (1.03, 2.97)	1.40 (0.82, 2.56)	1.39 (0.24, 8.77)	0.82 (0.43, 1.46)	2.86 (0.89, 9.27)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of discontinuations for AEs. ORs lower than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). AEs, adverse events; CrI, credible interval; ORs, odds ratios; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

11.3 bDMARDs-IR

11.3.1 ACR20

Bar 2mg										
0.77 (0.50, 1.17)	Bar 4mg									
1.16 (0.58, 2.32)	1.50 (0.75, 3.00)	Tof 5mg								
0.89 (0.44, 1.75)	1.15 (0.57, 2.29)	0.77 (0.47, 1.25)	Tof 10mg							
0.63 (0.35, 1.16)	0.83 (0.45, 1.51)	0.55 (0.28, 1.07)	0.72 (0.37, 1.38)	Upa 15mg						
0.72 (0.40, 1.30)	0.94 (0.52, 1.69)	0.62 (0.32, 1.21)	0.81 (0.42, 1.56)	1.13 (0.77, 1.67)	Upa 30mg					
1.03 (0.52, 2.07)	1.33 (0.67, 2.70)	0.89 (0.42, 1.88)	1.16 (0.55, 2.44)	1.62 (0.84, 3.14)	1.43 (0.74, 2.77)	GOL				
0.28 (0.13, 0.59)	0.37 (0.17, 0.77)	0.24 (0.11, 0.54)	0.32 (0.14, 0.71)	0.45 (0.21, 0.91)	0.39 (0.19, 0.80)	0.27 (0.12, 0.61)	TCZ			
0.53 (0.28, 0.98)	0.68 (0.37, 1.26)	0.45 (0.23, 0.90)	0.59 (0.30, 1.16)	0.83 (0.46, 1.48)	0.73 (0.41, 1.31)	0.51 (0.26, 1.01)	1.86 (0.90, 3.96)	RTX		
0.74 (0.42, 1.32)	0.97 (0.55, 1.72)	0.64 (0.35, 1.23)	0.84 (0.45, 1.60)	1.18 (0.69, 1.99)	1.04 (0.61, 1.74)	0.72 (0.39, 1.37)	2.63 (1.32, 5.46)	1.42 (0.82, 2.48)	ABT	
2.56 (1.65, 4.04)	3.33 (2.15, 5.23)	2.22 (1.33, 3.80)	2.88 (1.72, 4.93)	4.04 (2.73, 6.04)	3.57 (2.43, 5.29)	2.49 (1.49, 4.24)	9.05 (5.03, 17.22)	4.88 (3.22, 7.51)	3.44 (2.44, 4.91)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of ACR20. ORs higher than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). ACR20, American College of Rheumatology 20% response; CrI, credible interval; ORs, odds ratios; ABT, abatacept; Bar, baricitinib; GOL, golimumab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

11.3.2 DAS28

Bar 2mg										
0.28 (-0.01, 0.57)	Bar 4mg									
0.35 (-0.04, 0.74)	0.07 (-0.33, 0.47)	Tof 5mg								
0.57 (0.18, 0.96)	0.29 (-0.11, 0.68)	0.22 (-0.06, 0.50)	Tof 10mg							
0.57 (0.19, 0.96)	0.29 (-0.09, 0.68)	0.23 (-0.17, 0.62)	0.01 (-0.39, 0.40)	Upa 15mg						
0.60 (0.22, 0.98)	0.32 (-0.07, 0.71)	0.25 (-0.14, 0.65)	0.03 (-0.37, 0.43)	0.03 (-0.25, 0.30)	Upa 30mg					
0.05 (-0.33, 0.44)	-0.23 (-0.62, 0.16)	-0.30 (-0.70, 0.11)	-0.52 (-0.92, -0.12)	-0.53 (-0.91, -0.13)	-0.55 (-0.94, -0.15)	GOL				
1.11 (0.61, 1.60)	0.83 (0.32, 1.33)	0.76 (0.26, 1.27)	0.54 (0.04, 1.04)	0.53 (0.03, 1.03)	0.51 (0.00, 1.01)	1.06 (0.55, 1.55)	TCZ			
0.82 (0.46, 1.17)	0.54 (0.17, 0.90)	0.47 (0.09, 0.84)	0.25 (-0.12, 0.63)	0.24 (-0.12, 0.61)	0.22 (-0.15, 0.59)	0.77 (0.40, 1.14)	-0.29 (-0.77, 0.19)	RTX		
0.39 (0.06, 0.71)	0.11 (-0.23, 0.45)	0.04 (-0.31, 0.39)	-0.18 (-0.52, 0.16)	-0.19 (-0.52, 0.15)	-0.21 (-0.55, 0.12)	0.34 (0.00, 0.68)	-0.72 (-1.09, -0.35)	-0.43 (-0.74, -0.11)	ABT	
-0.68 (-0.95, -0.42)	-0.96 (-1.24, -0.69)	-1.03 (-1.32, -0.74)	-1.25 (-1.54, -0.97)	-1.26 (-1.53, -0.98)	-1.28 (-1.56, -1.00)	-0.73 (-1.01, -0.45)	-1.79 (-2.20, -1.37)	-1.50 (-1.74, -1.25)	-1.07 (-1.26, -0.88)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of DAS28. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; DAS28, Disease Activity Score in 28 joints; WMDs, weighted mean differences; ABT, abatacept; Bar, baricitinib; GOL, golimumab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

11.3.3 HAQ-DI

Bar 2mg										
0.04 (-0.06, 0.14)	Bar 4mg									
0.06 (-0.10, 0.21)	0.02 (-0.13, 0.17)	Tof 5mg								
0.06 (-0.09, 0.21)	0.02 (-0.14, 0.17)	0.00 (-0.13, 0.13)	Tof 10mg							
0.06 (-0.09, 0.21)	0.02 (-0.13, 0.17)	0.00 (-0.15, 0.16)	0.00 (-0.15, 0.16)	Upa 15mg						
0.08 (-0.07, 0.23)	0.04 (-0.11, 0.19)	0.02 (-0.14, 0.18)	0.02 (-0.14, 0.18)	0.02 (-0.09, 0.13)	Upa 30mg					
0.03 (-0.10, 0.16)	-0.01 (-0.15, 0.12)	-0.03 (-0.17, 0.11)	-0.03 (-0.17, 0.11)	-0.03 (-0.17, 0.10)	-0.05 (-0.19, 0.08)	GOL				
0.16 (-0.05, 0.38)	0.12 (-0.09, 0.34)	0.10 (-0.11, 0.32)	0.10 (-0.11, 0.32)	0.10 (-0.11, 0.32)	0.08 (-0.13, 0.30)	0.13 (-0.07, 0.34)	TCZ			
0.19 (0.05, 0.33)	0.15 (0.01, 0.29)	0.13 (-0.02, 0.28)	0.13 (-0.02, 0.27)	0.13 (-0.02, 0.27)	0.11 (-0.04, 0.25)	0.16 (0.04, 0.28)	0.03 (-0.18, 0.23)	RTX		
0.17 (0.03, 0.31)	0.13 (-0.01, 0.27)	0.11 (-0.03, 0.25)	0.11 (-0.03, 0.25)	0.11 (-0.03, 0.25)	0.09 (-0.05, 0.23)	0.14 (0.02, 0.26)	0.01 (-0.19, 0.20)	-0.02 (-0.15, 0.11)	ABT	
-0.18 (-0.29, -0.07)	-0.22 (-0.32, -0.12)	-0.24 (-0.35, -0.13)	-0.24 (-0.35, -0.13)	-0.24 (-0.35, -0.13)	-0.26 (-0.37, -0.16)	-0.21 (-0.29, -0.13)	-0.34 (-0.53, -0.16)	-0.37 (-0.46, -0.28)	-0.35 (-0.44, -0.26)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of HAQ-DI. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; HAQ-DI, Health Assessment Questionnaire–Disability Index; WMDs, weighted mean differences; ABT, abatacept; Bar, baricitinib; GOL, golimumab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

11.3.4 Discontinuations for AEs

Bar 2mg										
0.78 (0.27, 2.17)	Bar 4mg									
1.90 (0.14, 27.09)	2.46 (0.19, 34.32)	Tof 5mg								
4.74 (0.26, 181.90)	6.14 (0.35, 229.60)	2.42 (0.18, 76.15)	Tof 10mg							
1.83 (0.37, 9.92)	2.37 (0.51, 12.20)	0.96 (0.08, 11.30)	0.39 (0.01, 6.16)	Upa 15mg						
1.05 (0.22, 5.17)	1.35 (0.31, 6.37)	0.56 (0.05, 6.10)	0.23 (0.01, 3.36)	0.58 (0.23, 1.35)	Upa 30mg					
5.14 (0.91, 34.34)	6.62 (1.21, 42.69)	2.72 (0.21, 37.77)	1.10 (0.03, 20.6)	2.81 (0.60, 14.54)	4.87 (1.15, 24.19)	GOL				
1.18 (0.25, 5.98)	1.53 (0.36, 7.35)	0.63 (0.06, 6.89)	0.26 (0.01, 3.88)	0.65 (0.18, 2.32)	1.14 (0.34, 3.72)	0.23 (0.05, 0.98)	TCZ			
0.58 (0.06, 4.52)	0.75 (0.08, 5.50)	0.30 (0.02, 4.83)	0.12 (0.00, 2.43)	0.32 (0.04, 1.92)	0.56 (0.07, 3.20)	0.11 (0.01, 0.78)	0.49 (0.06, 2.79)	RTX		
2.06 (0.46, 10.39)	2.64 (0.62, 12.74)	1.09 (0.09, 11.91)	0.45 (0.01, 6.65)	1.12 (0.31, 4.03)	1.97 (0.60, 6.55)	0.41 (0.08, 1.68)	1.72 (0.72, 4.32)	3.53 (0.63, 29.7)	ABT	
1.87 (0.53, 7.63)	2.41 (0.74, 9.19)	1.00 (0.10, 9.57)	0.41 (0.01, 5.28)	1.03 (0.38, 2.71)	1.78 (0.77, 4.36)	0.37 (0.10, 1.17)	1.56 (0.69, 3.69)	3.14 (0.74, 23.26)	0.90 (0.40, 2.15)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of discontinuations for AEs. ORs lower than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). AEs, adverse events; CrI, credible interval; ORs, odds ratios; ABT, abatacept; Bar, baricitinib; GOL, golimumab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

Appendix 12. Sensitivity analyses

12.1 Sensitivity analysis by excluding trials with the follow-up time point for evaluation ≥52 weeks.

12.1.1 ACR20

Bar 2mg															
0.87 (0.50, 1.52)	Bar 4mg														
0.80 (0.40, 1.59)	0.92 (0.51, 1.65)	Tof 5mg													
0.70 (0.35, 1.41)	0.80 (0.43, 1.45)	0.87 (0.56, 1.37)	Tof 10mg												
0.79 (0.39, 1.60)	0.91 (0.49, 1.65)	0.99 (0.54, 1.84)	1.14 (0.60, 2.15)	Upa 15mg											
0.72 (0.35, 1.51)	0.83 (0.43, 1.61)	0.91 (0.47, 1.77)	1.04 (0.52, 2.03)	0.91 (0.54, 1.53)	Upa 30mg										
0.99 (0.54, 1.83)	1.14 (0.69, 1.87)	1.24 (0.78, 1.98)	1.43 (0.86, 2.35)	1.25 (0.75, 2.10)	1.38 (0.77, 2.46)	ADA									
0.54 (0.27, 1.07)	0.62 (0.34, 1.12)	0.68 (0.37, 1.21)	0.77 (0.42, 1.43)	0.68 (0.37, 1.25)	0.75 (0.39, 1.44)	0.54 (0.32, 0.92)	ETN								
0.90 (0.44, 1.86)	1.03 (0.54, 1.93)	1.12 (0.60, 2.13)	1.29 (0.67, 2.47)	1.13 (0.58, 2.19)	1.24 (0.62, 2.49)	0.90 (0.51, 1.59)	1.66 (0.88, 3.14)	IFX							
0.91 (0.43, 1.93)	1.04 (0.54, 2.02)	1.13 (0.58, 2.21)	1.30 (0.66, 2.57)	1.15 (0.57, 2.28)	1.26 (0.61, 2.59)	0.91 (0.50, 1.66)	1.68 (0.87, 3.24)	1.01 (0.50, 2.04)	GOL						
0.62 (0.33, 1.17)	0.70 (0.41, 1.20)	0.77 (0.46, 1.31)	0.88 (0.51, 1.53)	0.78 (0.45, 1.35)	0.85 (0.46, 1.55)	0.62 (0.41, 0.94)	1.14 (0.67, 1.94)	0.69 (0.38, 1.23)	0.68 (0.37, 1.26)	CZP					
0.67 (0.35, 1.29)	0.77 (0.44, 1.34)	0.84 (0.48, 1.48)	0.97 (0.55, 1.72)	0.85 (0.48, 1.50)	0.93 (0.50, 1.74)	0.68 (0.42, 1.09)	1.25 (0.72, 2.14)	0.75 (0.41, 1.37)	0.74 (0.39, 1.39)	1.09 (0.67, 1.80)	TCZ				
0.81 (0.37, 1.74)	0.93 (0.46, 1.84)	1.02 (0.51, 2.02)	1.16 (0.57, 2.34)	1.02 (0.50, 2.08)	1.12 (0.53, 2.34)	0.81 (0.43, 1.53)	1.49 (0.77, 2.94)	0.90 (0.44, 1.86)	0.89 (0.42, 1.89)	1.31 (0.69, 2.50)	1.20 (0.63, 2.31)	RTX			
0.79 (0.39, 1.58)	0.90 (0.49, 1.64)	0.98 (0.54, 1.80)	1.12 (0.61, 2.10)	0.99 (0.53, 1.86)	1.09 (0.55, 2.12)	0.79 (0.46, 1.34)	1.45 (0.8, 2.65)	0.87 (0.48, 1.59)	0.86 (0.44, 1.69)	1.28 (0.73, 2.2)	1.16 (0.66, 2.06)	0.97 (0.48, 1.94)	ABT		
1.23 (0.50, 2.99)	1.40 (0.61, 3.20)	1.54 (0.67, 3.51)	1.76 (0.76, 4.05)	1.54 (0.66, 3.61)	1.70 (0.71, 4.09)	1.23 (0.57, 2.68)	2.26 (1.25, 4.11)	1.37 (0.58, 3.21)	1.34 (0.56, 3.22)	1.99 (0.90, 4.37)	1.82 (0.82, 4.06)	1.51 (0.64, 3.55)	1.56 (0.68, 3.60)	csDM ARDs	
3.05 (1.78, 5.27)	3.50 (2.30, 5.28)	3.81 (2.51, 5.86)	4.36 (2.82, 6.86)	3.84 (2.46, 6.02)	4.22 (2.55, 7.00)	3.07 (2.25, 4.20)	5.64 (3.75, 8.53)	3.40 (2.12, 5.51)	3.36 (2.01, 5.62)	4.95 (3.55, 6.95)	4.52 (3.16, 6.50)	3.75 (2.20, 6.49)	3.88 (2.50, 6.07)	2.49 (1.23, 5.05)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of ACR20. ORs higher than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). ACR20, American College of Rheumatology 20% response; CrI, credible interval; ORs, odds ratios; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.1.2. DAS28

Bar 2mg															
0.27 (-0.05, 0.60)	Bar 4mg														
0.25 (-0.15, 0.65)	-0.02 (-0.37, 0.32)	Tof 5mg													
0.44 (0.03, 0.85)	0.17 (-0.19, 0.52)	0.19 (-0.07, 0.45)	Tof 10mg												
0.51 (0.09, 0.94)	0.24 (-0.14, 0.61)	0.26 (-0.10, 0.63)	0.07 (-0.31, 0.45)	Upa 15mg											
0.55 (0.11, 1.01)	0.28 (-0.12, 0.69)	0.30 (-0.1, 0.70)	0.11 (-0.30, 0.53)	0.04 (-0.29, 0.36)	Upa 30mg										
0.10 (-0.28, 0.49)	-0.17 (-0.50, 0.16)	-0.15 (-0.44, 0.15)	-0.34 (-0.65, -0.02)	-0.41 (-0.76, -0.06)	-0.45 (-0.84, -0.05)	ADA									
0.42 (0.02, 0.83)	0.15 (-0.20, 0.50)	0.18 (-0.17, 0.52)	-0.01 (-0.37, 0.34)	-0.09 (-0.46, 0.29)	-0.13 (-0.53, 0.28)	0.32 (-0.01, 0.65)	ETN								
0.07 (-0.50, 0.65)	-0.20 (-0.73, 0.35)	-0.17 (-0.71, 0.36)	-0.37 (-0.90, 0.18)	-0.44 (-0.99, 0.11)	-0.48 (-1.06, 0.09)	-0.03 (-0.56, 0.50)	-0.35 (-0.89, 0.19)	IFX							
0.29 (-0.22, 0.81)	0.02 (-0.45, 0.50)	0.04 (-0.43, 0.51)	-0.15 (-0.62, 0.33)	-0.22 (-0.7, 0.27)	-0.26 (-0.77, 0.25)	0.19 (-0.27, 0.66)	-0.13 (-0.6, 0.34)	0.22 (-0.41, 0.84)	GOL						
0.62 (0.21, 1.04)	0.35 (-0.01, 0.71)	0.38 (0.02, 0.72)	0.18 (-0.18, 0.54)	0.11 (-0.27, 0.48)	0.07 (-0.34, 0.48)	0.52 (0.18, 0.87)	0.20 (-0.16, 0.56)	0.55 (0.01, 1.09)	0.33 (-0.15, 0.80)	CZP					
1.12 (0.73, 1.52)	0.85 (0.51, 1.19)	0.87 (0.54, 1.21)	0.68 (0.34, 1.03)	0.61 (0.25, 0.98)	0.56 (0.17, 0.97)	1.02 (0.68, 1.35)	0.69 (0.36, 1.04)	1.04 (0.52, 1.57)	0.83 (0.37, 1.29)	0.49 (0.15, 0.85)	TCZ				
0.40 (-0.09, 0.89)	0.13 (-0.31, 0.57)	0.15 (-0.29, 0.58)	-0.04 (-0.48, 0.40)	-0.11 (-0.57, 0.34)	-0.15 (-0.64, 0.33)	0.30 (-0.14, 0.72)	-0.02 (-0.44, 0.39)	0.33 (-0.27, 0.93)	0.11 (-0.44, 0.64)	-0.22 (-0.66, 0.22)	-0.72 (-1.15, -0.30)	RTX			
0.41 (-0.01, 0.83)	0.14 (-0.23, 0.51)	0.16 (-0.20, 0.52)	-0.03 (-0.41, 0.34)	-0.10 (-0.49, 0.29)	-0.14 (-0.56, 0.27)	0.30 (-0.06, 0.66)	-0.02 (-0.38, 0.35)	0.33 (-0.17, 0.83)	0.12 (-0.37, 0.59)	-0.21 (-0.59, 0.16)	-0.71 (-1.04, -0.39)	0.01 (-0.44, 0.46)	ABT		
-0.26 (-0.77, 0.26)	-0.53 (-1.01, -0.07)	-0.51 (-0.98, -0.04)	-0.70 (-1.17, -0.22)	-0.77 (-1.27, -0.28)	-0.81 (-1.33, -0.30)	-0.36 (-0.82, 0.10)	-0.69 (-1.03, -0.35)	-0.34 (-0.97, 0.29)	-0.55 (-1.13, 0.02)	-0.88 (-1.36, -0.41)	-1.38 (-1.84, -0.91)	-0.66 (-1.13, -0.18)	-0.67 (-1.15, -0.18)	csDM ARDs	
-0.79 (-1.11, -0.47)	-1.06 (-1.31, -0.81)	-1.04 (-1.29, -0.80)	-1.23 (-1.49, -0.97)	-1.30 (-1.58, -1.02)	-1.34 (-1.66, -1.02)	-0.89 (-1.14, -0.65)	-1.22 (-1.47, -0.96)	-0.86 (-1.34, -0.39)	-1.08 (-1.48, -0.69)	-1.41 (-1.67, -1.15)	-1.91 (-2.14, -1.68)	-1.19 (-1.56, -0.82)	-1.20 (-1.47, -0.94)	-0.53 (-0.94, -0.13)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of DAS28. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; DAS28, Disease Activity Score in 28 joints; WMDs, weighted mean differences; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.1.3. HAQ-DI

Bar 2mg															
0.06 (-0.02, 0.14)	Bar 4mg														
0.07 (-0.03, 0.17)	0.01 (-0.07, 0.09)	Tof 5mg													
0.13 (0.03, 0.23)	0.07 (-0.02, 0.15)	0.06 (-0.01, 0.12)	Tof 10mg												
0.11 (0.00, 0.21)	0.05 (-0.04, 0.13)	0.04 (-0.05, 0.12)	-0.02 (-0.11, 0.07)	Upa 15mg											
0.11 (0.00, 0.22)	0.05 (-0.04, 0.15)	0.04 (-0.06, 0.14)	-0.01 (-0.12, 0.09)	0.00 (-0.07, 0.08)	Upa 30mg										
0.04 (-0.05, 0.13)	-0.02 (-0.09, 0.05)	-0.03 (-0.1, 0.03)	-0.09 (-0.16, -0.02)	-0.07 (-0.15, 0.01)	-0.07 (-0.17, 0.02)	ADA									
0.07 (-0.04, 0.18)	0.01 (-0.09, 0.11)	0.00 (-0.10, 0.09)	-0.06 (-0.16, 0.04)	-0.04 (-0.14, 0.07)	-0.04 (-0.15, 0.07)	0.03 (-0.06, 0.12)	ETN								
0.08 (-0.06, 0.22)	0.02 (-0.11, 0.15)	0.01 (-0.12, 0.14)	-0.05 (-0.18, 0.08)	-0.03 (-0.16, 0.10)	-0.03 (-0.17, 0.11)	0.04 (-0.08, 0.16)	0.01 (-0.13, 0.15)	IFX							
0.06 (-0.04, 0.17)	0.00 (-0.09, 0.10)	0.00 (-0.10, 0.09)	-0.06 (-0.16, 0.04)	-0.04 (-0.14, 0.06)	-0.05 (-0.15, 0.06)	0.03 (-0.06, 0.12)	0.00 (-0.11, 0.10)	-0.01 (-0.15, 0.12)	GOL						
0.10 (0.01, 0.20)	0.04 (-0.04, 0.12)	0.03 (-0.05, 0.11)	-0.02 (-0.11, 0.06)	0.00 (-0.09, 0.08)	-0.01 (-0.11, 0.09)	0.07 (-0.01, 0.14)	0.04 (-0.06, 0.13)	0.02 (-0.10, 0.15)	0.04 (-0.05, 0.13)	CZP					
0.05 (-0.05, 0.16)	-0.01 (-0.09, 0.08)	-0.02 (-0.10, 0.07)	-0.08 (-0.16, 0.01)	-0.06 (-0.15, 0.04)	-0.06 (-0.16, 0.04)	0.01 (-0.07, 0.10)	-0.02 (-0.12, 0.08)	-0.03 (-0.16, 0.10)	-0.01 (-0.11, 0.08)	-0.05 (-0.14, 0.03)	TCZ				
0.13 (-0.01, 0.27)	0.07 (-0.06, 0.20)	0.06 (-0.07, 0.19)	0.01 (-0.13, 0.14)	0.03 (-0.11, 0.16)	0.02 (-0.13, 0.16)	0.10 (-0.04, 0.22)	0.07 (-0.08, 0.20)	0.06 (-0.11, 0.22)	0.07 (-0.08, 0.21)	0.03 (-0.11, 0.16)	0.08 (-0.06, 0.21)	RTX			
0.09 (-0.01, 0.19)	0.03 (-0.06, 0.12)	0.02 (-0.07, 0.11)	-0.04 (-0.13, 0.05)	-0.02 (-0.11, 0.08)	-0.02 (-0.13, 0.08)	0.05 (-0.03, 0.14)	0.02 (-0.08, 0.12)	0.01 (-0.11, 0.13)	0.02 (-0.08, 0.12)	-0.02 (-0.10, 0.07)	0.04 (-0.06, 0.13)	-0.04 (-0.18, 0.10)	ABT		
-0.11 (-0.26, 0.05)	-0.17 (-0.31, -0.02)	-0.18 (-0.32, -0.03)	-0.23 (-0.38, -0.09)	-0.21 (-0.36, -0.06)	-0.22 (-0.37, -0.06)	-0.14 (-0.29, 0.00)	-0.18 (-0.28, -0.07)	-0.18 (-0.36, -0.01)	-0.17 (-0.33, -0.02)	-0.21 (-0.36, -0.06)	-0.16 (-0.31, -0.01)	-0.24 (-0.42, -0.06)	-0.20 (-0.35, -0.04)	csDM ARDs	
-0.21 (-0.28, -0.13)	-0.26 (-0.32, -0.21)	-0.27 (-0.33, -0.22)	-0.33 (-0.39, -0.27)	-0.31 (-0.38, -0.24)	-0.32 (-0.40, -0.24)	-0.24 (-0.29, -0.19)	-0.27 (-0.35, -0.20)	-0.28 (-0.40, -0.17)	-0.27 (-0.35, -0.20)	-0.31 (-0.36, -0.25)	-0.26 (-0.32, -0.19)	-0.34 (-0.45, -0.22)	-0.29 (-0.36, -0.23)	-0.10 (-0.23, 0.04)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of HAQ-DI. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; HAQ-DI, Health Assessment Questionnaire–Disability Index; WMDs, weighted mean differences; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.1.4 Discontinuations for AEs

Bar 2mg															
0.72 (0.35, 1.45)	Bar 4mg														
0.57 (0.22, 1.40)	0.79 (0.37, 1.64)	Tof 5mg													
0.51 (0.20, 1.26)	0.70 (0.33, 1.48)	0.88 (0.55, 1.48)	Tof 10mg												
0.96 (0.39, 2.29)	1.34 (0.64, 2.68)	1.69 (0.79, 3.65)	1.88 (0.88, 3.98)	Upa 15mg											
0.51 (0.19, 1.26)	0.69 (0.31, 1.57)	0.87 (0.38, 2.09)	0.99 (0.43, 2.34)	0.52 (0.28, 0.97)	Upa 30mg										
0.75 (0.34, 1.64)	1.04 (0.59, 1.81)	1.32 (0.70, 2.48)	1.49 (0.79, 2.73)	0.78 (0.42, 1.47)	1.51 (0.70, 3.02)	ADA									
1.34 (0.48, 3.71)	1.84 (0.78, 4.41)	2.34 (0.93, 5.98)	2.63 (1.04, 6.68)	1.39 (0.57, 3.41)	2.64 (1.06, 6.90)	1.77 (0.79, 3.96)	ETN								
0.42 (0.16, 1.11)	0.58 (0.24, 1.33)	0.73 (0.29, 1.75)	0.82 (0.33, 1.94)	0.43 (0.18, 1.01)	0.84 (0.32, 2.04)	0.56 (0.25, 1.19)	0.31 (0.11, 0.83)	IFX							
0.99 (0.34, 2.78)	1.38 (0.56, 3.26)	1.72 (0.69, 4.53)	1.93 (0.76, 5.15)	1.03 (0.40, 2.64)	1.96 (0.75, 5.23)	1.32 (0.58, 2.96)	0.74 (0.26, 2.09)	2.37 (0.90, 6.48)	GOL						
0.62 (0.27, 1.39)	0.86 (0.45, 1.59)	1.08 (0.54, 2.17)	1.22 (0.61, 2.38)	0.64 (0.34, 1.23)	1.24 (0.56, 2.52)	0.82 (0.50, 1.33)	0.46 (0.20, 1.07)	1.47 (0.70, 3.32)	0.62 (0.27, 1.48)	CZP					
0.54 (0.23, 1.24)	0.74 (0.38, 1.40)	0.93 (0.47, 1.93)	1.05 (0.52, 2.13)	0.56 (0.29, 1.11)	1.07 (0.50, 2.23)	0.71 (0.41, 1.23)	0.40 (0.17, 0.94)	1.28 (0.60, 2.95)	0.54 (0.23, 1.24)	0.87 (0.5, 1.52)	TCZ				
0.31 (0.08, 1.06)	0.44 (0.12, 1.29)	0.55 (0.15, 1.74)	0.62 (0.17, 1.89)	0.33 (0.09, 1.01)	0.63 (0.17, 2.01)	0.42 (0.12, 1.18)	0.24 (0.06, 0.79)	0.75 (0.20, 2.41)	0.32 (0.08, 1.06)	0.51 (0.15, 1.46)	0.59 (0.17, 1.69)	RTX			
1.20 (0.47, 2.98)	1.66 (0.77, 3.49)	2.09 (0.96, 4.78)	2.36 (1.07, 5.23)	1.24 (0.57, 2.72)	2.38 (1.02, 5.57)	1.59 (0.80, 3.17)	0.90 (0.35, 2.22)	2.84 (1.30, 6.92)	1.21 (0.47, 3.10)	1.95 (0.99, 3.81)	2.23 (1.15, 4.34)	3.79 (1.24, 13.78)	ABT		
0.48 (0.13, 1.65)	0.67 (0.20, 2.05)	0.83 (0.26, 2.73)	0.94 (0.29, 2.99)	0.49 (0.15, 1.57)	0.94 (0.29, 3.12)	0.63 (0.21, 1.87)	0.36 (0.17, 0.73)	1.13 (0.33, 3.92)	0.48 (0.13, 1.65)	0.77 (0.25, 2.30)	0.90 (0.29, 2.62)	1.55 (0.37, 7.10)	0.40 (0.12, 1.30)	csDM ARDs	
1.00 (0.48, 2.02)	1.39 (0.84, 2.25)	1.75 (0.99, 3.18)	1.98 (1.11, 3.47)	1.04 (0.61, 1.79)	2.00 (1.05, 3.68)	1.33 (0.91, 1.92)	0.75 (0.36, 1.55)	2.38 (1.24, 4.85)	1.01 (0.48, 2.12)	1.62 (1.10, 2.39)	1.87 (1.23, 2.83)	3.13 (1.20, 10.06)	0.84 (0.48, 1.47)	2.09 (0.76, 6.12)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of discontinuations for AEs. ORs lower than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). AEs, adverse events; CrI, credible interval; ORs, odds ratios; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.2 Sensitivity analysis by excluding trials with sample size <50

12.2.1 ACR20

Bar 2mg															
0.87 (0.51, 1.50)	Bar 4mg														
0.80 (0.41, 1.56)	0.92 (0.51, 1.62)	Tof 5mg													
0.70 (0.35, 1.39)	0.80 (0.44, 1.45)	0.87 (0.57, 1.35)	Tof 10mg												
0.79 (0.40, 1.58)	0.91 (0.50, 1.68)	0.99 (0.55, 1.81)	1.14 (0.61, 2.13)	Upa 15mg											
0.72 (0.35, 1.49)	0.83 (0.43, 1.58)	0.90 (0.48, 1.72)	1.03 (0.53, 2.00)	0.91 (0.55, 1.53)	Upa 30mg										
0.98 (0.53, 1.78)	1.12 (0.69, 1.82)	1.23 (0.78, 1.92)	1.41 (0.86, 2.27)	1.24 (0.74, 2.03)	1.36 (0.77, 2.40)	ADA									
0.55 (0.28, 1.08)	0.63 (0.35, 1.14)	0.69 (0.38, 1.23)	0.79 (0.43, 1.44)	0.69 (0.38, 1.27)	0.76 (0.39, 1.47)	0.56 (0.33, 0.94)	ETN								
0.94 (0.46, 1.88)	1.07 (0.59, 1.98)	1.17 (0.64, 2.16)	1.34 (0.71, 2.52)	1.18 (0.63, 2.20)	1.29 (0.66, 2.54)	0.95 (0.55, 1.62)	1.7 (0.91, 3.14)	IFX							
0.91 (0.43, 1.94)	1.04 (0.54, 2.02)	1.14 (0.59, 2.20)	1.30 (0.66, 2.57)	1.15 (0.58, 2.27)	1.26 (0.61, 2.58)	0.92 (0.51, 1.67)	1.65 (0.85, 3.21)	0.97 (0.49, 1.91)	GOL						
0.64 (0.34, 1.20)	0.74 (0.44, 1.23)	0.80 (0.48, 1.35)	0.92 (0.53, 1.60)	0.81 (0.47, 1.40)	0.89 (0.49, 1.62)	0.65 (0.44, 0.99)	1.16 (0.68, 2.00)	0.69 (0.39, 1.20)	0.71 (0.38, 1.31)	CZP					
0.67 (0.35, 1.30)	0.77 (0.44, 1.33)	0.84 (0.49, 1.45)	0.97 (0.55, 1.70)	0.85 (0.48, 1.48)	0.94 (0.50, 1.72)	0.69 (0.43, 1.09)	1.22 (0.70, 2.14)	0.72 (0.41, 1.27)	0.74 (0.40, 1.38)	1.05 (0.64, 1.72)	TCZ				
0.78 (0.36, 1.70)	0.90 (0.45, 1.82)	0.98 (0.49, 1.97)	1.12 (0.55, 2.31)	0.99 (0.48, 2.01)	1.09 (0.52, 2.28)	0.80 (0.43, 1.51)	1.43 (0.71, 2.87)	0.83 (0.41, 1.71)	0.87 (0.40, 1.84)	1.22 (0.64, 2.34)	1.16 (0.60, 2.27)	RTX			
0.81 (0.41, 1.57)	0.93 (0.53, 1.63)	1.01 (0.58, 1.78)	1.16 (0.65, 2.07)	1.02 (0.56, 1.84)	1.12 (0.59, 2.12)	0.82 (0.52, 1.31)	1.47 (0.82, 2.62)	0.87 (0.49, 1.52)	0.89 (0.46, 1.71)	1.26 (0.75, 2.1)	1.20 (0.71, 2.06)	1.03 (0.52, 2.04)	ABT		
1.29 (0.51, 3.22)	1.47 (0.64, 3.45)	1.61 (0.68, 3.77)	1.85 (0.78, 4.31)	1.62 (0.69, 3.91)	1.77 (0.73, 4.45)	1.31 (0.59, 2.99)	2.34 (1.27, 4.30)	1.38 (0.58, 3.28)	1.42 (0.57, 3.52)	2.00 (0.89, 4.51)	1.91 (0.84, 4.37)	1.64 (0.65, 4.20)	1.59 (0.69, 3.68)	csDM ARDs	
3.05 (1.78, 5.20)	3.49 (2.32, 5.27)	3.81 (2.54, 5.77)	4.36 (2.83, 6.79)	3.84 (2.46, 5.94)	4.22 (2.55, 7.00)	3.10 (2.32, 4.18)	5.53 (3.63, 8.46)	3.26 (2.09, 5.13)	3.36 (2.00, 5.62)	4.75 (3.40, 6.69)	4.51 (3.19, 6.52)	3.89 (2.23, 6.79)	3.77 (2.55, 5.64)	2.36 (1.12, 4.98)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of ACR20. ORs higher than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). ACR20, American College of Rheumatology 20% response; CrI, credible interval; ORs, odds ratios; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.2.2 DAS28

Bar 2mg															
0.27 (-0.05, 0.59)	Bar 4mg														
0.25 (-0.15, 0.65)	-0.02 (-0.37, 0.33)	Tof 5mg													
0.44 (0.04, 0.86)	0.17 (-0.19, 0.53)	0.19 (-0.07, 0.45)	Tof 10mg												
0.51 (0.09, 0.94)	0.24 (-0.13, 0.62)	0.26 (-0.11, 0.62)	0.07 (-0.31, 0.44)	Upa 15mg											
0.55 (0.10, 0.99)	0.28 (-0.12, 0.68)	0.30 (-0.11, 0.70)	0.11 (-0.31, 0.52)	0.04 (-0.28, 0.36)	Upa 30mg										
0.13 (-0.25, 0.51)	-0.14 (-0.46, 0.18)	-0.12 (-0.41, 0.17)	-0.31 (-0.63, 0.00)	-0.38 (-0.73, -0.04)	-0.42 (-0.80, -0.03)	ADA									
0.40 (-0.01, 0.81)	0.13 (-0.23, 0.49)	0.15 (-0.20, 0.50)	-0.04 (-0.41, 0.32)	-0.11 (-0.5, 0.27)	-0.15 (-0.57, 0.26)	0.27 (-0.06, 0.60)	ETN								
0.06 (-0.50, 0.64)	-0.21 (-0.75, 0.32)	-0.19 (-0.72, 0.34)	-0.38 (-0.92, 0.15)	-0.45 (-1.00, 0.10)	-0.49 (-1.06, 0.08)	-0.07 (-0.59, 0.45)	-0.34 (-0.88, 0.20)	IFX							
0.29 (-0.22, 0.80)	0.02 (-0.45, 0.48)	0.04 (-0.43, 0.51)	-0.15 (-0.63, 0.33)	-0.22 (-0.70, 0.27)	-0.26 (-0.77, 0.26)	0.16 (-0.30, 0.62)	-0.11 (-0.59, 0.36)	0.23 (-0.39, 0.85)	GOL						
0.62 (0.21, 1.03)	0.35 (0.00, 0.71)	0.37 (0.02, 0.72)	0.18 (-0.19, 0.55)	0.11 (-0.27, 0.49)	0.07 (-0.34, 0.49)	0.49 (0.16, 0.82)	0.22 (-0.15, 0.58)	0.56 (0.02, 1.10)	0.33 (-0.15, 0.81)	CZP					
1.11 (0.73, 1.51)	0.84 (0.51, 1.19)	0.86 (0.53, 1.20)	0.67 (0.32, 1.02)	0.60 (0.24, 0.97)	0.56 (0.17, 0.96)	0.98 (0.66, 1.31)	0.71 (0.37, 1.06)	1.05 (0.53, 1.58)	0.82 (0.36, 1.29)	0.49 (0.14, 0.84)	TCZ				
0.45 (-0.06, 0.95)	0.18 (-0.29, 0.65)	0.20 (-0.27, 0.67)	0.01 (-0.47, 0.48)	-0.06 (-0.54, 0.43)	-0.10 (-0.61, 0.41)	0.32 (-0.14, 0.78)	0.05 (-0.43, 0.53)	0.39 (-0.22, 1.00)	0.16 (-0.40, 0.72)	-0.17 (-0.65, 0.31)	-0.66 (-1.11, -0.20)	RTX			
0.37 (-0.03, 0.77)	0.10 (-0.25, 0.44)	0.11 (-0.22, 0.45)	-0.08 (-0.43, 0.28)	-0.15 (-0.51, 0.22)	-0.19 (-0.58, 0.22)	0.23 (-0.07, 0.55)	-0.04 (-0.39, 0.32)	0.31 (-0.19, 0.80)	0.07 (-0.38, 0.54)	-0.25 (-0.61, 0.11)	-0.75 (-1.06, -0.43)	-0.09 (-0.55, 0.38)	ABT		
-0.33 (-0.88, 0.22)	-0.60 (-1.12, -0.09)	-0.58 (-1.10, -0.06)	-0.77 (-1.30, -0.25)	-0.84 (-1.37, -0.31)	-0.88 (-1.43, -0.32)	-0.46 (-0.95, 0.04)	-0.73 (-1.10, -0.36)	-0.39 (-1.05, 0.27)	-0.62 (-1.22, -0.02)	-0.95 (-1.47, -0.43)	-1.44 (-1.95, -0.93)	-0.78 (-1.38, -0.18)	-0.70 (-1.22, -0.17)	csDM ARDs	
-0.79 (-1.11, -0.48)	-1.07 (-1.31, -0.82)	-1.05 (-1.29, -0.80)	-1.24 (-1.50, -0.98)	-1.31 (-1.59, -1.03)	-1.35 (-1.66, -1.02)	-0.93 (-1.15, -0.70)	-1.20 (-1.46, -0.94)	-0.86 (-1.34, -0.39)	-1.09 (-1.48, -0.69)	-1.42 (-1.68, -1.15)	-1.91 (-2.14, -1.68)	-1.25 (-1.65, -0.85)	-1.16 (-1.41, -0.92)	-0.47 (-0.93, -0.01)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of DAS28. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment).
CrI, credible interval; DAS28, Disease Activity Score in 28 joints; WMDs, weighted mean differences; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.2.3 HAQ-DI

Bar 2mg															
0.06 (-0.02, 0.14)	Bar 4mg														
0.07 (-0.02, 0.17)	0.01 (-0.07, 0.09)	Tof 5mg													
0.13 (0.03, 0.23)	0.07 (-0.01, 0.15)	0.06 (0.00, 0.12)	Tof 10mg												
0.11 (0.01, 0.21)	0.05 (-0.04, 0.13)	0.04 (-0.05, 0.12)	-0.02 (-0.11, 0.07)	Upa 15mg											
0.11 (0.00, 0.22)	0.05 (-0.04, 0.15)	0.04 (-0.06, 0.14)	-0.02 (-0.11, 0.08)	0.00 (-0.07, 0.08)	Upa 30mg										
0.04 (-0.05, 0.13)	-0.02 (-0.09, 0.05)	-0.03 (-0.09, 0.03)	-0.09 (-0.16, -0.02)	-0.07 (-0.14, 0.01)	-0.07 (-0.16, 0.02)	ADA									
0.07 (-0.04, 0.18)	0.01 (-0.09, 0.10)	0.00 (-0.10, 0.09)	-0.06 (-0.16, 0.04)	-0.04 (-0.14, 0.06)	-0.04 (-0.16, 0.07)	0.03 (-0.06, 0.12)	ETN								
0.07 (-0.06, 0.21)	0.01 (-0.11, 0.14)	0.00 (-0.12, 0.13)	-0.05 (-0.18, 0.08)	-0.03 (-0.16, 0.10)	-0.04 (-0.18, 0.10)	0.03 (-0.09, 0.15)	0.01 (-0.13, 0.14)	IFX							
0.06 (-0.04, 0.17)	0.00 (-0.09, 0.10)	-0.01 (-0.10, 0.09)	-0.06 (-0.16, 0.03)	-0.04 (-0.14, 0.06)	-0.05 (-0.15, 0.06)	0.02 (-0.06, 0.11)	0.00 (-0.11, 0.10)	-0.01 (-0.14, 0.13)	GOL						
0.10 (0.01, 0.20)	0.04 (-0.04, 0.12)	0.03 (-0.05, 0.11)	-0.02 (-0.11, 0.06)	-0.01 (-0.09, 0.08)	-0.01 (-0.1, 0.09)	0.06 (-0.01, 0.13)	0.04 (-0.06, 0.13)	0.03 (-0.10, 0.16)	0.04 (-0.05, 0.13)	CZP					
0.05 (-0.05, 0.15)	-0.01 (-0.09, 0.07)	-0.02 (-0.10, 0.06)	-0.08 (-0.16, 0.01)	-0.06 (-0.15, 0.04)	-0.06 (-0.16, 0.04)	0.01 (-0.07, 0.09)	-0.02 (-0.12, 0.08)	-0.03 (-0.15, 0.10)	-0.02 (-0.11, 0.08)	-0.05 (-0.13, 0.03)	TCZ				
0.13 (-0.01, 0.27)	0.07 (-0.06, 0.20)	0.07 (-0.07, 0.19)	0.01 (-0.13, 0.14)	0.03 (-0.11, 0.16)	0.02 (-0.12, 0.16)	0.10 (-0.04, 0.22)	0.07 (-0.08, 0.21)	0.06 (-0.11, 0.22)	0.07 (-0.07, 0.21)	0.03 (-0.10, 0.16)	0.09 (-0.05, 0.22)	RTX			
0.08 (-0.02, 0.18)	0.02 (-0.06, 0.10)	0.01 (-0.07, 0.09)	-0.05 (-0.13, 0.04)	-0.03 (-0.12, 0.06)	-0.03 (-0.13, 0.07)	0.04 (-0.03, 0.11)	0.01 (-0.08, 0.11)	0.01 (-0.11, 0.13)	0.01 (-0.08, 0.11)	-0.02 (-0.11, 0.06)	0.03 (-0.06, 0.12)	-0.06 (-0.18, 0.08)	ABT		
-0.11 (-0.26, 0.05)	-0.17 (-0.31, -0.02)	-0.18 (-0.32, -0.04)	-0.23 (-0.38, -0.09)	-0.22 (-0.36, -0.07)	-0.22 (-0.38, -0.07)	-0.15 (-0.29, -0.01)	-0.17 (-0.28, -0.07)	-0.18 (-0.36, -0.01)	-0.17 (-0.32, -0.02)	-0.21 (-0.35, -0.07)	-0.16 (-0.30, -0.01)	-0.24 (-0.42, -0.06)	-0.19 (-0.33, -0.04)	csDM ARDs	
-0.21 (-0.28, -0.13)	-0.27 (-0.32, -0.21)	-0.27 (-0.33, -0.22)	-0.33 (-0.39, -0.27)	-0.31 (-0.38, -0.25)	-0.32 (-0.39, -0.24)	-0.24 (-0.29, -0.20)	-0.27 (-0.35, -0.19)	-0.28 (-0.39, -0.17)	-0.27 (-0.35, -0.20)	-0.31 (-0.36, -0.25)	-0.25 (-0.32, -0.19)	-0.34 (-0.45, -0.22)	-0.28 (-0.35, -0.22)	-0.10 (-0.23, 0.04)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of HAQ-DI. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; HAQ-DI, Health Assessment Questionnaire–Disability Index; WMDs, weighted mean differences; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.2.4 Discontinuations for AEs

Bar 2mg															
0.74 (0.33, 1.63)	Bar 4mg														
0.65 (0.24, 1.66)	0.88 (0.40, 1.84)	Tof 5mg													
0.63 (0.23, 1.65)	0.86 (0.38, 1.83)	0.97 (0.59, 1.62)	Tof 10mg												
0.90 (0.32, 2.47)	1.23 (0.51, 2.84)	1.39 (0.64, 3.11)	1.43 (0.63, 3.34)	Upa 15mg											
0.48 (0.16, 1.41)	0.65 (0.25, 1.63)	0.75 (0.32, 1.76)	0.77 (0.31, 1.87)	0.53 (0.26, 1.07)	Upa 30mg										
0.71 (0.29, 1.68)	0.96 (0.50, 1.83)	1.09 (0.65, 1.87)	1.12 (0.63, 2.06)	0.78 (0.39, 1.57)	1.46 (0.67, 3.26)	ADA									
0.92 (0.31, 2.60)	1.24 (0.51, 2.99)	1.41 (0.63, 3.31)	1.45 (0.62, 3.51)	1.01 (0.41, 2.58)	1.90 (0.71, 5.10)	1.30 (0.63, 2.68)	ETN								
0.53 (0.18, 1.52)	0.72 (0.29, 1.75)	0.82 (0.36, 1.93)	0.85 (0.36, 2.03)	0.59 (0.23, 1.50)	1.10 (0.42, 3.02)	0.76 (0.35, 1.62)	0.59 (0.22, 1.51)	IFX							
0.95 (0.30, 2.94)	1.29 (0.47, 3.43)	1.47 (0.57, 3.79)	1.51 (0.58, 3.95)	1.05 (0.38, 2.86)	1.97 (0.67, 5.76)	1.34 (0.56, 3.22)	1.04 (0.36, 2.92)	1.78 (0.62, 5.11)	GOL						
0.57 (0.22, 1.42)	0.78 (0.37, 1.59)	0.88 (0.46, 1.72)	0.91 (0.45, 1.84)	0.63 (0.29, 1.37)	1.18 (0.51, 2.78)	0.81 (0.46, 1.37)	0.63 (0.28, 1.38)	1.07 (0.47, 2.41)	0.60 (0.24, 1.51)	CZP					
0.71 (0.27, 1.72)	0.96 (0.44, 1.95)	1.09 (0.56, 2.10)	1.13 (0.55, 2.22)	0.78 (0.35, 1.66)	1.46 (0.62, 3.37)	1.00 (0.55, 1.73)	0.77 (0.33, 1.72)	1.32 (0.59, 2.94)	0.74 (0.30, 1.80)	1.24 (0.65, 2.28)	TCZ				
0.32 (0.07, 1.21)	0.43 (0.11, 1.47)	0.49 (0.13, 1.60)	0.51 (0.14, 1.70)	0.35 (0.09, 1.21)	0.66 (0.16, 2.38)	0.45 (0.13, 1.39)	0.34 (0.09, 1.23)	0.59 (0.15, 2.08)	0.33 (0.08, 1.29)	0.56 (0.16, 1.76)	0.45 (0.13, 1.45)	RTX			
1.15 (0.44, 2.93)	1.57 (0.74, 3.25)	1.79 (0.91, 3.59)	1.84 (0.90, 3.83)	1.28 (0.59, 2.81)	2.39 (1.02, 5.73)	1.64 (0.93, 2.84)	1.26 (0.55, 2.87)	2.16 (1.00, 4.84)	1.22 (0.47, 3.14)	2.02 (1.06, 3.92)	1.63 (0.90, 3.13)	3.64 (1.11, 13.25)	ABT		
0.32 (0.08, 1.23)	0.44 (0.13, 1.48)	0.50 (0.16, 1.66)	0.52 (0.15, 1.76)	0.36 (0.10, 1.27)	0.67 (0.18, 2.49)	0.46 (0.15, 1.40)	0.36 (0.15, 0.81)	0.61 (0.17, 2.16)	0.34 (0.09, 1.32)	0.57 (0.18, 1.83)	0.46 (0.14, 1.52)	1.03 (0.23, 5.19)	0.28 (0.09, 0.91)	csDM ARDs	
0.98 (0.42, 2.18)	1.33 (0.73, 2.35)	1.51 (0.92, 2.52)	1.55 (0.91, 2.72)	1.08 (0.57, 2.05)	2.03 (0.99, 4.18)	1.38 (0.95, 1.99)	1.07 (0.53, 2.10)	1.84 (0.93, 3.59)	1.03 (0.47, 2.28)	1.71 (1.10, 2.69)	1.38 (0.92, 2.19)	3.08 (1.06, 10.19)	0.85 (0.52, 1.36)	3.01 (1.02, 8.90)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of discontinuations for AEs. ORs lower than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). AEs, adverse events; CrI, credible interval; ORs, odds ratios; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.3 Sensitivity analysis by excluding open-label trials

12.3.1 ACR20

Bar 2mg															
0.87 (0.52, 1.49)	Bar 4mg														
0.81 (0.42, 1.57)	0.93 (0.52, 1.62)	Tof 5mg													
0.70 (0.36, 1.36)	0.80 (0.45, 1.44)	0.86 (0.57, 1.32)	Tof 10mg												
0.79 (0.40, 1.55)	0.90 (0.50, 1.63)	0.98 (0.55, 1.75)	1.13 (0.62, 2.06)	Upa 15mg											
0.73 (0.35, 1.48)	0.83 (0.44, 1.56)	0.89 (0.48, 1.68)	1.03 (0.54, 1.98)	0.92 (0.56, 1.49)	Upa 30mg										
1.02 (0.56, 1.81)	1.15 (0.71, 1.85)	1.25 (0.80, 1.95)	1.44 (0.89, 2.34)	1.28 (0.78, 2.07)	1.39 (0.80, 2.42)	ADA									
0.63 (0.30, 1.30)	0.72 (0.38, 1.39)	0.78 (0.41, 1.50)	0.90 (0.46, 1.75)	0.80 (0.41, 1.55)	0.87 (0.43, 1.75)	0.62 (0.35, 1.12)	ETN								
0.90 (0.46, 1.79)	1.03 (0.57, 1.85)	1.11 (0.62, 2.01)	1.28 (0.70, 2.37)	1.13 (0.62, 2.09)	1.24 (0.64, 2.38)	0.89 (0.53, 1.49)	1.43 (0.72, 2.77)	IFX							
0.91 (0.44, 1.88)	1.04 (0.55, 1.97)	1.12 (0.59, 2.14)	1.29 (0.67, 2.50)	1.15 (0.60, 2.21)	1.25 (0.62, 2.54)	0.89 (0.50, 1.60)	1.44 (0.70, 2.96)	1.01 (0.52, 1.95)	GOL						
0.60 (0.32, 1.11)	0.68 (0.40, 1.14)	0.73 (0.44, 1.24)	0.85 (0.49, 1.46)	0.75 (0.44, 1.29)	0.82 (0.45, 1.48)	0.59 (0.39, 0.90)	0.94 (0.51, 1.73)	0.67 (0.38, 1.14)	0.66 (0.36, 1.20)	CZP					
0.67 (0.36, 1.26)	0.77 (0.45, 1.31)	0.83 (0.49, 1.42)	0.96 (0.55, 1.67)	0.85 (0.49, 1.46)	0.93 (0.51, 1.70)	0.67 (0.42, 1.05)	1.07 (0.58, 1.99)	0.75 (0.43, 1.31)	0.74 (0.40, 1.35)	1.13 (0.70, 1.85)	TCZ				
0.77 (0.37, 1.62)	0.88 (0.45, 1.71)	0.95 (0.49, 1.85)	1.11 (0.56, 2.16)	0.98 (0.49, 1.91)	1.07 (0.52, 2.16)	0.76 (0.42, 1.39)	1.23 (0.60, 2.49)	0.86 (0.43, 1.69)	0.86 (0.41, 1.75)	1.30 (0.69, 2.39)	1.15 (0.61, 2.15)	RTX			
0.81 (0.42, 1.54)	0.92 (0.53, 1.61)	1.00 (0.58, 1.74)	1.15 (0.65, 2.04)	1.02 (0.57, 1.81)	1.12 (0.60, 2.09)	0.80 (0.51, 1.26)	1.29 (0.68, 2.45)	0.90 (0.52, 1.56)	0.89 (0.47, 1.66)	1.36 (0.81, 2.26)	1.20 (0.71, 2.04)	1.04 (0.55, 2.02)	ABT		
0.64 (0.21, 2.01)	0.74 (0.25, 2.18)	0.79 (0.27, 2.35)	0.92 (0.31, 2.75)	0.81 (0.27, 2.45)	0.89 (0.29, 2.71)	0.64 (0.23, 1.80)	1.02 (0.41, 2.58)	0.72 (0.24, 2.14)	0.71 (0.23, 2.17)	1.08 (0.38, 3.09)	0.95 (0.33, 2.76)	0.83 (0.29, 2.44)	0.80 (0.27, 2.33)	csDM ARDs	
3.04 (1.81, 5.13)	3.48 (2.34, 5.19)	3.75 (2.53, 5.67)	4.33 (2.84, 6.70)	3.84 (2.51, 5.90)	4.20 (2.57, 6.86)	3.00 (2.25, 4.02)	4.83 (2.89, 8.06)	3.39 (2.20, 5.21)	3.35 (2.02, 5.50)	5.10 (3.67, 7.14)	4.51 (3.19, 6.42)	3.91 (2.36, 6.74)	3.76 (2.56, 5.54)	4.71 (1.75, 12.79)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of ACR20. ORs higher than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). ACR20, American College of Rheumatology 20% response; CrI, credible interval; ORs, odds ratios; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.3.2 DAS28

Bar 2mg															
0.27 (-0.02, 0.57)	Bar 4mg														
0.25 (-0.11, 0.61)	-0.02 (-0.33, 0.29)	Tof 5mg													
0.44 (0.07, 0.81)	0.17 (-0.15, 0.49)	0.19 (-0.04, 0.42)	Tof 10mg												
0.52 (0.13, 0.90)	0.24 (-0.09, 0.57)	0.26 (-0.07, 0.59)	0.08 (-0.27, 0.41)	Upa 15mg											
0.55 (0.14, 0.96)	0.28 (-0.09, 0.64)	0.30 (-0.06, 0.67)	0.11 (-0.26, 0.49)	0.04 (-0.25, 0.34)	Upa 30mg										
0.12 (-0.23, 0.48)	-0.15 (-0.44, 0.14)	-0.13 (-0.4, 0.14)	-0.32 (-0.61, -0.02)	-0.39 (-0.70, -0.07)	-0.43 (-0.79, -0.07)	ADA									
0.27 (-0.15, 0.70)	0.00 (-0.38, 0.39)	0.02 (-0.37, 0.41)	-0.17 (-0.56, 0.23)	-0.24 (-0.64, 0.17)	-0.28 (-0.71, 0.15)	0.15 (-0.24, 0.54)	ETN								
0.06 (-0.45, 0.58)	-0.21 (-0.70, 0.27)	-0.19 (-0.68, 0.29)	-0.38 (-0.87, 0.11)	-0.46 (-0.96, 0.05)	-0.49 (-1.02, 0.03)	-0.07 (-0.55, 0.41)	-0.22 (-0.76, 0.32)	IFX							
0.29 (-0.18, 0.75)	0.02 (-0.41, 0.44)	0.04 (-0.39, 0.46)	-0.15 (-0.58, 0.28)	-0.22 (-0.67, 0.22)	-0.26 (-0.73, 0.21)	0.17 (-0.27, 0.60)	0.02 (-0.47, 0.50)	0.23 (-0.34, 0.80)	GOL						
0.58 (0.20, 0.95)	0.30 (-0.03, 0.64)	0.33 (-0.01, 0.65)	0.14 (-0.20, 0.47)	0.06 (-0.29, 0.41)	0.02 (-0.36, 0.40)	0.45 (0.12, 0.78)	0.30 (-0.10, 0.70)	0.52 (0.02, 1.01)	0.28 (-0.16, 0.72)	CZP					
1.10 (0.74, 1.47)	0.83 (0.52, 1.15)	0.85 (0.54, 1.16)	0.66 (0.34, 0.99)	0.59 (0.25, 0.93)	0.55 (0.18, 0.91)	0.98 (0.67, 1.29)	0.82 (0.44, 1.22)	1.04 (0.56, 1.53)	0.81 (0.39, 1.24)	0.52 (0.19, 0.86)	TCZ				
0.46 (0.01, 0.89)	0.18 (-0.22, 0.59)	0.20 (-0.20, 0.61)	0.02 (-0.40, 0.42)	-0.06 (-0.47, 0.36)	-0.10 (-0.54, 0.35)	0.33 (-0.07, 0.73)	0.18 (-0.25, 0.60)	0.39 (-0.15, 0.95)	0.16 (-0.33, 0.66)	-0.12 (-0.54, 0.30)	-0.64 (-1.06, -0.24)	RTX			
0.36 (-0.01, 0.73)	0.09 (-0.23, 0.41)	0.11 (-0.20, 0.42)	-0.08 (-0.40, 0.25)	-0.15 (-0.50, 0.19)	-0.19 (-0.56, 0.18)	0.24 (-0.06, 0.54)	0.09 (-0.31, 0.48)	0.30 (-0.15, 0.76)	0.07 (-0.37, 0.51)	-0.22 (-0.56, 0.13)	-0.74 (-1.07, -0.42)	-0.10 (-0.51, 0.32)	ABT		
0.01 (-0.57, 0.60)	-0.26 (-0.82, 0.30)	-0.24 (-0.80, 0.31)	-0.43 (-0.99, 0.13)	-0.50 (-1.07, 0.07)	-0.55 (-1.13, 0.06)	-0.11 (-0.67, 0.45)	-0.26 (-0.74, 0.22)	-0.05 (-0.72, 0.62)	-0.28 (-0.91, 0.35)	-0.57 (-1.14, 0.01)	-1.09 (-1.65, -0.53)	-0.44 (-0.97, 0.08)	-0.35 (-0.92, 0.22)	csDM ARDs	
-0.79 (-1.08, -0.50)	-1.06 (-1.28, -0.84)	-1.04 (-1.27, -0.82)	-1.23 (-1.47, -1.00)	-1.30 (-1.56, -1.05)	-1.34 (-1.64, -1.05)	-0.91 (-1.14, -0.69)	-1.07 (-1.38, -0.75)	-0.85 (-1.29, -0.41)	-1.08 (-1.45, -0.72)	-1.37 (-1.61, -1.12)	-1.89 (-2.12, -1.67)	-1.25 (-1.58, -0.91)	-1.15 (-1.39, -0.92)	-0.80 (-1.32, -0.30)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of DAS28. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; DAS28, Disease Activity Score in 28 joints; WMDs, weighted mean differences; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.3.3 HAQ-DI

Bar 2mg															
0.06 (-0.02, 0.14)	Bar 4mg														
0.07 (-0.02, 0.17)	0.01 (-0.07, 0.09)	Tof 5mg													
0.13 (0.03, 0.23)	0.07 (-0.01, 0.15)	0.06 (0.00, 0.12)	Tof 10mg												
0.11 (0.01, 0.21)	0.05 (-0.04, 0.13)	0.04 (-0.05, 0.12)	-0.02 (-0.11, 0.07)	Upa 15mg											
0.11 (0.00, 0.22)	0.05 (-0.04, 0.15)	0.04 (-0.06, 0.13)	-0.02 (-0.12, 0.08)	0.00 (-0.07, 0.08)	Upa 30mg										
0.04 (-0.04, 0.13)	-0.02 (-0.09, 0.06)	-0.03 (-0.09, 0.04)	-0.08 (-0.15, -0.01)	-0.06 (-0.14, 0.02)	-0.07 (-0.16, 0.02)	ADA									
0.05 (-0.08, 0.18)	-0.01 (-0.13, 0.11)	-0.02 (-0.14, 0.10)	-0.08 (-0.20, 0.04)	-0.06 (-0.18, 0.07)	-0.06 (-0.19, 0.07)	0.01 (-0.11, 0.12)	ETN								
0.07 (-0.06, 0.21)	0.01 (-0.11, 0.14)	0.00 (-0.12, 0.13)	-0.05 (-0.18, 0.08)	-0.03 (-0.16, 0.10)	-0.04 (-0.17, 0.10)	0.03 (-0.09, 0.15)	0.03 (-0.13, 0.18)	IFX							
0.06 (-0.04, 0.17)	0.00 (-0.09, 0.1)	-0.01 (-0.10, 0.09)	-0.06 (-0.16, 0.03)	-0.04 (-0.14, 0.06)	-0.05 (-0.15, 0.06)	0.02 (-0.07, 0.11)	0.01 (-0.11, 0.14)	-0.01 (-0.15, 0.13)	GOL						
0.10 (0.00, 0.20)	0.04 (-0.04, 0.12)	0.03 (-0.05, 0.11)	-0.03 (-0.11, 0.05)	-0.01 (-0.10, 0.08)	-0.01 (-0.11, 0.08)	0.05 (-0.02, 0.13)	0.05 (-0.07, 0.17)	0.02 (-0.10, 0.15)	0.03 (-0.06, 0.13)	CZP					
0.05 (-0.05, 0.15)	-0.01 (-0.1, 0.08)	-0.02 (-0.11, 0.06)	-0.08 (-0.17, 0.01)	-0.06 (-0.15, 0.04)	-0.06 (-0.16, 0.04)	0.00 (-0.07, 0.08)	0.00 (-0.12, 0.12)	-0.03 (-0.16, 0.11)	-0.02 (-0.11, 0.08)	-0.05 (-0.13, 0.04)	TCZ				
0.13 (-0.01, 0.27)	0.07 (-0.06, 0.20)	0.06 (-0.07, 0.19)	0.01 (-0.13, 0.14)	0.03 (-0.11, 0.16)	0.02 (-0.12, 0.16)	0.09 (-0.04, 0.21)	0.08 (-0.07, 0.24)	0.06 (-0.11, 0.22)	0.07 (-0.07, 0.21)	0.04 (-0.10, 0.16)	0.08 (-0.06, 0.22)	RTX			
0.08 (-0.01, 0.18)	0.02 (-0.06, 0.11)	0.01 (-0.07, 0.09)	-0.05 (-0.13, 0.04)	-0.03 (-0.11, 0.07)	-0.03 (-0.13, 0.07)	0.04 (-0.04, 0.11)	0.03 (-0.09, 0.15)	0.01 (-0.11, 0.13)	0.02 (-0.08, 0.11)	-0.02 (-0.10, 0.07)	0.03 (-0.06, 0.12)	-0.05 (-0.18, 0.09)	ABT		
-0.02 (-0.26, 0.22)	-0.08 (-0.31, 0.15)	-0.09 (-0.32, 0.14)	-0.15 (-0.38, 0.09)	-0.13 (-0.36, 0.12)	-0.13 (-0.37, 0.11)	-0.07 (-0.30, 0.17)	-0.07 (-0.27, 0.13)	-0.10 (-0.35, 0.16)	-0.09 (-0.32, 0.16)	-0.12 (-0.35, 0.12)	-0.07 (-0.30, 0.17)	-0.16 (-0.41, 0.11)	-0.10 (-0.34, 0.14)	csDM ARDs	
-0.21 (-0.28, -0.13)	-0.27 (-0.32, -0.21)	-0.28 (-0.33, -0.22)	-0.33 (-0.40, -0.27)	-0.31 (-0.38, -0.25)	-0.32 (-0.39, -0.24)	-0.25 (-0.30, -0.20)	-0.25 (-0.36, -0.15)	-0.28 (-0.39, -0.17)	-0.27 (-0.34, -0.20)	-0.30 (-0.36, -0.25)	-0.25 (-0.32, -0.19)	-0.34 (-0.45, -0.22)	-0.29 (-0.35, -0.23)	-0.18 (-0.41, 0.04)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of HAQ-DI. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; HAQ-DI, Health Assessment Questionnaire–Disability Index; WMDs, weighted mean differences; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.3.4 Discontinuations for AEs

Bar 2mg															
0.73 (0.32, 1.63)	Bar 4mg														
0.63 (0.23, 1.58)	0.85 (0.40, 1.81)	Tof 5mg													
0.62 (0.22, 1.61)	0.84 (0.37, 1.84)	0.99 (0.60, 1.63)	Tof 10mg												
0.90 (0.30, 2.44)	1.23 (0.51, 2.84)	1.43 (0.66, 3.17)	1.45 (0.65, 3.35)	Upa 15mg											
0.48 (0.16, 1.41)	0.66 (0.26, 1.64)	0.77 (0.33, 1.83)	0.78 (0.33, 1.89)	0.54 (0.26, 1.09)	Upa 30mg										
0.64 (0.26, 1.54)	0.88 (0.45, 1.68)	1.02 (0.61, 1.77)	1.04 (0.58, 1.90)	0.71 (0.36, 1.44)	1.33 (0.60, 2.90)	ADA									
1.29 (0.25, 5.95)	1.77 (0.40, 7.43)	2.06 (0.49, 8.52)	2.10 (0.49, 8.80)	1.45 (0.32, 6.28)	2.68 (0.57, 11.82)	2.01 (0.50, 7.99)	ETN								
0.54 (0.18, 1.51)	0.74 (0.30, 1.73)	0.86 (0.37, 1.99)	0.87 (0.37, 2.06)	0.60 (0.24, 1.52)	1.11 (0.42, 2.96)	0.84 (0.39, 1.78)	0.42 (0.09, 1.88)	IFX							
0.95 (0.30, 2.98)	1.30 (0.48, 3.49)	1.53 (0.60, 3.94)	1.55 (0.59, 4.10)	1.06 (0.39, 2.97)	1.98 (0.67, 5.72)	1.49 (0.61, 3.58)	0.73 (0.16, 3.47)	1.77 (0.62, 5.02)	GOL						
0.61 (0.23, 1.51)	0.83 (0.39, 1.71)	0.97 (0.50, 1.91)	0.98 (0.49, 2.04)	0.68 (0.31, 1.47)	1.25 (0.54, 2.92)	0.95 (0.54, 1.65)	0.47 (0.12, 1.97)	1.13 (0.50, 2.56)	0.64 (0.25, 1.60)	CZP					
0.75 (0.28, 1.84)	1.03 (0.47, 2.10)	1.20 (0.59, 2.33)	1.21 (0.59, 2.41)	0.84 (0.38, 1.76)	1.55 (0.66, 3.49)	1.17 (0.63, 2.07)	0.58 (0.14, 2.40)	1.39 (0.61, 3.06)	0.79 (0.31, 1.93)	1.23 (0.63, 2.32)	TCZ				
0.31 (0.07, 1.20)	0.42 (0.11, 1.44)	0.49 (0.14, 1.64)	0.50 (0.13, 1.69)	0.35 (0.09, 1.18)	0.64 (0.16, 2.33)	0.49 (0.14, 1.51)	0.24 (0.04, 1.37)	0.58 (0.14, 2.04)	0.33 (0.08, 1.25)	0.51 (0.14, 1.65)	0.42 (0.12, 1.34)	RTX			
1.08 (0.40, 2.74)	1.47 (0.67, 3.14)	1.72 (0.86, 3.53)	1.75 (0.84, 3.72)	1.20 (0.55, 2.73)	2.22 (0.94, 5.40)	1.68 (0.93, 3.07)	0.84 (0.20, 3.59)	2.00 (0.90, 4.51)	1.13 (0.44, 2.96)	1.77 (0.90, 3.55)	1.43 (0.74, 2.92)	3.48 (1.06, 12.87)	ABT		
0.50 (0.06, 3.83)	0.68 (0.09, 4.86)	0.79 (0.11, 5.47)	0.81 (0.10, 5.74)	0.56 (0.07, 4.01)	1.03 (0.13, 7.47)	0.78 (0.11, 5.22)	0.38 (0.09, 1.45)	0.92 (0.12, 6.78)	0.52 (0.06, 3.95)	0.82 (0.11, 5.52)	0.66 (0.09, 4.62)	1.63 (0.17, 14.96)	0.46 (0.06, 3.19)	csDM ARDs	
0.99 (0.42, 2.21)	1.35 (0.75, 2.40)	1.58 (0.96, 2.65)	1.60 (0.93, 2.82)	1.11 (0.59, 2.08)	2.05 (1.00, 4.16)	1.55 (1.05, 2.26)	0.76 (0.21, 2.97)	1.83 (0.94, 3.60)	1.04 (0.47, 2.31)	1.63 (1.03, 2.59)	1.32 (0.86, 2.12)	3.17 (1.09, 10.64)	0.92 (0.54, 1.53)	1.97 (0.31, 13.92)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of discontinuations for AEs. ORs lower than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). AEs, adverse events; CrI, credible interval; ORs, odds ratios; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.4 Sensitivity analysis by excluding unpublished trials

12.4.1 ACR20

Bar 2mg															
0.86 (0.54, 1.38)	Bar 4mg														
0.80 (0.45, 1.42)	0.93 (0.56, 1.51)	Tof 5mg													
0.69 (0.38, 1.25)	0.80 (0.48, 1.33)	0.86 (0.59, 1.26)	Tof 10mg												
0.83 (0.45, 1.54)	0.96 (0.56, 1.65)	1.03 (0.61, 1.79)	1.21 (0.70, 2.09)	Upa 15mg											
0.75 (0.39, 1.47)	0.87 (0.48, 1.58)	0.94 (0.52, 1.71)	1.10 (0.60, 1.98)	0.91 (0.56, 1.45)	Upa 30mg										
0.90 (0.53, 1.51)	1.04 (0.68, 1.58)	1.12 (0.76, 1.67)	1.30 (0.86, 2.00)	1.08 (0.68, 1.69)	1.19 (0.70, 2.03)	ADA									
0.54 (0.30, 0.98)	0.63 (0.38, 1.06)	0.68 (0.41, 1.14)	0.79 (0.47, 1.35)	0.66 (0.38, 1.13)	0.72 (0.40, 1.31)	0.61 (0.39, 0.95)	ETN								
0.90 (0.49, 1.64)	1.04 (0.61, 1.76)	1.13 (0.67, 1.91)	1.31 (0.76, 2.24)	1.09 (0.62, 1.87)	1.20 (0.65, 2.20)	1.01 (0.63, 1.60)	1.66 (0.97, 2.83)	IFX							
0.91 (0.48, 1.72)	1.05 (0.59, 1.87)	1.14 (0.65, 2.03)	1.32 (0.74, 2.38)	1.09 (0.60, 1.99)	1.21 (0.63, 2.33)	1.01 (0.61, 1.71)	1.67 (0.94, 2.97)	1.00 (0.56, 1.84)	GOL						
0.59 (0.34, 1.03)	0.69 (0.43, 1.1)	0.74 (0.47, 1.18)	0.86 (0.53, 1.40)	0.72 (0.43, 1.17)	0.79 (0.45, 1.39)	0.66 (0.46, 0.96)	1.09 (0.68, 1.75)	0.66 (0.41, 1.07)	0.65 (0.38, 1.13)	CZP					
0.68 (0.39, 1.19)	0.79 (0.49, 1.26)	0.86 (0.53, 1.37)	0.99 (0.60, 1.61)	0.83 (0.49, 1.35)	0.91 (0.51, 1.60)	0.76 (0.51, 1.13)	1.25 (0.78, 2.03)	0.76 (0.46, 1.25)	0.75 (0.43, 1.29)	1.15 (0.75, 1.76)	TCZ				
0.80 (0.42, 1.56)	0.93 (0.51, 1.68)	1.00 (0.56, 1.83)	1.17 (0.64, 2.12)	0.97 (0.52, 1.80)	1.07 (0.55, 2.09)	0.89 (0.53, 1.54)	1.47 (0.83, 2.66)	0.89 (0.48, 1.66)	0.88 (0.46, 1.69)	1.35 (0.77, 2.37)	1.18 (0.67, 2.08)	RTX			
0.80 (0.45, 1.42)	0.93 (0.57, 1.52)	1.00 (0.62, 1.64)	1.16 (0.70, 1.93)	0.97 (0.57, 1.64)	1.06 (0.59, 1.92)	0.89 (0.60, 1.34)	1.47 (0.89, 2.45)	0.89 (0.55, 1.44)	0.88 (0.50, 1.56)	1.35 (0.86, 2.11)	1.17 (0.74, 1.87)	1.00 (0.55, 1.78)	ABT		
1.23 (0.58, 2.65)	1.43 (0.71, 2.88)	1.55 (0.77, 3.13)	1.79 (0.88, 3.70)	1.49 (0.72, 3.06)	1.65 (0.76, 3.53)	1.38 (0.71, 2.70)	2.27 (1.37, 3.74)	1.37 (0.67, 2.83)	1.36 (0.63, 2.92)	2.08 (1.06, 4.10)	1.81 (0.92, 3.59)	1.53 (0.73, 3.24)	1.54 (0.76, 3.11)	csDM ARDs	
3.03 (1.91, 4.82)	3.52 (2.46, 5.02)	3.80 (2.68, 5.46)	4.41 (3.03, 6.46)	3.67 (2.46, 5.45)	4.04 (2.52, 6.52)	3.38 (2.62, 4.38)	5.57 (3.87, 8.04)	3.36 (2.30, 4.98)	3.34 (2.15, 5.25)	5.11 (3.79, 6.88)	4.44 (3.28, 6.09)	3.78 (2.37, 6.05)	3.79 (2.67, 5.37)	2.46 (1.33, 4.50)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of ACR20. ORs higher than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). ACR20, American College of Rheumatology 20% response; CrI, credible interval; ORs, odds ratios; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.4.2 DAS28

Bar 2mg															
0.27 (-0.02, 0.57)	Bar 4mg														
0.25 (-0.11, 0.61)	-0.02 (-0.33, 0.29)	Tof 5mg													
0.44 (0.07, 0.81)	0.17 (-0.15, 0.49)	0.19 (-0.04, 0.42)	Tof 10mg												
0.52 (0.13, 0.90)	0.24 (-0.09, 0.57)	0.26 (-0.07, 0.59)	0.08 (-0.27, 0.41)	Upa 15mg											
0.55 (0.14, 0.96)	0.28 (-0.09, 0.64)	0.30 (-0.06, 0.67)	0.11 (-0.26, 0.49)	0.04 (-0.25, 0.34)	Upa 30mg										
0.12 (-0.23, 0.48)	-0.15 (-0.44, 0.14)	-0.13 (-0.40, 0.14)	-0.32 (-0.61, -0.02)	-0.39 (-0.70, -0.07)	-0.43 (-0.79, -0.07)	ADA									
0.27 (-0.15, 0.70)	0.00 (-0.38, 0.39)	0.02 (-0.37, 0.41)	-0.17 (-0.56, 0.23)	-0.24 (-0.64, 0.17)	-0.28 (-0.71, 0.15)	0.15 (-0.24, 0.54)	ETN								
0.06 (-0.45, 0.58)	-0.21 (-0.70, 0.27)	-0.19 (-0.68, 0.29)	-0.38 (-0.87, 0.11)	-0.46 (-0.96, 0.05)	-0.49 (-1.02, 0.03)	-0.07 (-0.55, 0.41)	-0.22 (-0.76, 0.32)	IFX							
0.29 (-0.18, 0.75)	0.02 (-0.41, 0.44)	0.04 (-0.39, 0.46)	-0.15 (-0.58, 0.28)	-0.22 (-0.67, 0.22)	-0.26 (-0.73, 0.21)	0.17 (-0.27, 0.60)	0.02 (-0.47, 0.50)	0.23 (-0.34, 0.80)	GOL						
0.58 (0.20, 0.95)	0.30 (-0.03, 0.64)	0.33 (-0.01, 0.65)	0.14 (-0.20, 0.47)	0.06 (-0.29, 0.41)	0.02 (-0.36, 0.40)	0.45 (0.12, 0.78)	0.30 (-0.10, 0.70)	0.52 (0.02, 1.01)	0.28 (-0.16, 0.72)	CZP					
1.10 (0.74, 1.47)	0.83 (0.52, 1.15)	0.85 (0.54, 1.16)	0.66 (0.34, 0.99)	0.59 0.25, 0.93)	0.55 (0.18, 0.91)	0.98 (0.67, 1.29)	0.82 (0.44, 1.22)	1.04 (0.56, 1.53)	0.81 (0.39, 1.24)	0.52 (0.19, 0.86)	TCZ				
0.46 (0.01, 0.89)	0.18 (-0.22, 0.59)	0.20 (-0.20, 0.61)	0.02 (-0.40, 0.42)	-0.06 (-0.47, 0.36)	-0.10 (-0.54, 0.35)	0.33 (-0.07, 0.73)	0.18 (-0.25, 0.60)	0.39 (-0.15, 0.95)	0.16 (-0.33, 0.66)	-0.12 (-0.54, 0.30)	-0.64 (-1.06, -0.24)	RTX			
0.36 (-0.01, 0.73)	0.09 (-0.23, 0.41)	0.11 (-0.20, 0.42)	-0.08 (-0.40, 0.25)	-0.15 (-0.50, 0.19)	-0.19 (-0.56, 0.18)	0.24 (-0.06, 0.54)	0.09 (-0.31, 0.48)	0.30 (-0.15, 0.76)	0.07 (-0.37, 0.51)	-0.22 (-0.56, 0.13)	-0.74 (-1.07, -0.42)	-0.10 (-0.51, 0.32)	ABT		
0.01 (-0.57, 0.60)	-0.26 (-0.82, 0.30)	-0.24 (-0.80, 0.31)	-0.43 (-0.99, 0.13)	-0.50 (-1.07, 0.07)	-0.55 (-1.13, 0.06)	-0.11 (-0.67, 0.45)	-0.26 (-0.74, 0.22)	-0.05 (-0.72, 0.62)	-0.28 (-0.91, 0.35)	-0.57 (-1.14, 0.01)	-1.09 (-1.65, -0.53)	-0.44 (-0.97, 0.08)	-0.35 (-0.92, 0.22)	csDM ARDs	
-0.79 (-1.08, -0.50)	-1.06 (-1.28, -0.84)	-1.04 (-1.27, -0.82)	-1.23 (-1.47, -1.00)	-1.30 (-1.56, -1.05)	-1.34 (-1.64, -1.05)	-0.91 (-1.14, -0.69)	-1.07 (-1.38, -0.75)	-0.85 (-1.29, -0.41)	-1.08 (-1.45, -0.72)	-1.37 (-1.61, -1.12)	-1.89 (-2.12, -1.67)	-1.25 (-1.58, -0.91)	-1.15 (-1.39, -0.92)	-0.80 (-1.32, -0.30)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of DAS28. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; DAS28, Disease Activity Score in 28 joints; WMDs, weighted mean differences; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.4.3 HAQ-DI

Bar 2mg															
0.06 (-0.02, 0.14)	Bar 4mg														
0.07 (-0.02, 0.17)	0.01 (-0.07, 0.09)	Tof 5mg													
0.13 (0.03, 0.23)	0.07 (-0.01, 0.15)	0.06 (0.00, 0.12)	Tof 10mg												
0.10 (0.00, 0.21)	0.04 (-0.04, 0.13)	0.03 (-0.05, 0.12)	-0.02 (-0.11, 0.07)	Upa 15mg											
0.10 (-0.01, 0.21)	0.04 (-0.06, 0.14)	0.03 (-0.07, 0.13)	-0.03 (-0.13, 0.07)	-0.01 (-0.09, 0.08)	Upa 30mg										
0.04 (-0.04, 0.13)	-0.02 (-0.08, 0.05)	-0.03 (-0.09, 0.04)	-0.08 (-0.15, -0.01)	-0.06 (-0.14, 0.02)	-0.06 (-0.15, 0.04)	ADA									
0.07 (-0.04, 0.18)	0.00 (-0.09, 0.10)	0.00 (-0.1, 0.09)	-0.06 (-0.16, 0.04)	-0.04 (-0.14, 0.07)	-0.03 (-0.15, 0.08)	0.02 (-0.07, 0.11)	ETN								
0.07 (-0.06, 0.21)	0.01 (-0.11, 0.14)	0.00 (-0.12, 0.13)	-0.05 (-0.18, 0.07)	-0.03 (-0.16, 0.10)	-0.03 (-0.17, 0.11)	0.03 (-0.09, 0.15)	0.01 (-0.13, 0.14)	IFX							
0.06 (-0.04, 0.17)	0.00 (-0.09, 0.10)	-0.01 (-0.10, 0.09)	-0.06 (-0.16, 0.03)	-0.04 (-0.14, 0.06)	-0.03 (-0.15, 0.08)	0.02 (-0.06, 0.11)	0.00 (-0.11, 0.10)	-0.01 (-0.14, 0.12)	GOL						
0.10 (0.00, 0.19)	0.03 (-0.04, 0.12)	0.03 (-0.05, 0.11)	-0.03 (-0.11, 0.05)	-0.01 (-0.10, 0.08)	0.00 (-0.10, 0.10)	0.05 (-0.02, 0.13)	0.03 (-0.07, 0.12)	0.02 (-0.10, 0.15)	0.03 (-0.06, 0.12)	CZP					
0.03 (-0.07, 0.13)	-0.03 (-0.11, 0.06)	-0.04 (-0.12, 0.04)	-0.10 (-0.18, -0.01)	-0.07 (-0.17, 0.02)	-0.07 (-0.17, 0.03)	-0.01 (-0.09, 0.06)	-0.04 (-0.14, 0.06)	-0.04 (-0.17, 0.08)	-0.03 (-0.13, 0.06)	-0.07 (-0.15, 0.02)	TCZ				
0.14 (0.00, 0.27)	0.08 (-0.05, 0.20)	0.07 (-0.06, 0.19)	0.01 (-0.12, 0.14)	0.03 (-0.10, 0.16)	0.04 (-0.11, 0.17)	0.09 (-0.03, 0.21)	0.07 (-0.07, 0.20)	0.06 (-0.10, 0.22)	0.07 (-0.07, 0.20)	0.04 (-0.09, 0.16)	0.11 (-0.03, 0.23)	RTX			
0.08 (-0.01, 0.18)	0.02 (-0.06, 0.10)	0.01 (-0.07, 0.09)	-0.05 (-0.13, 0.04)	-0.02 (-0.11, 0.07)	-0.02 (-0.12, 0.08)	0.04 (-0.03, 0.11)	0.01 (-0.08, 0.11)	0.01 (-0.11, 0.12)	0.02 (-0.08, 0.11)	-0.02 (-0.10, 0.07)	0.05 (-0.04, 0.14)	-0.06 (-0.18, 0.07)	ABT		
-0.11 (-0.26, 0.04)	-0.17 (-0.31, -0.03)	-0.18 (-0.32, -0.04)	-0.24 (-0.38, -0.09)	-0.21 (-0.36, -0.07)	-0.21 (-0.36, -0.05)	-0.15 (-0.29, -0.01)	-0.17 (-0.28, -0.07)	-0.18 (-0.36, -0.01)	-0.17 (-0.32, -0.03)	-0.21 (-0.35, -0.06)	-0.14 (-0.28, 0.01)	-0.25 (-0.41, -0.07)	-0.19 (-0.33, -0.04)	csDM ARDs	
-0.21 (-0.28, -0.13)	-0.27 (-0.32, -0.21)	-0.27 (-0.33, -0.22)	-0.33 (-0.39, -0.27)	-0.31 (-0.38, -0.24)	-0.30 (-0.39, -0.22)	-0.25 (-0.30, -0.20)	-0.27 (-0.35, -0.20)	-0.28 (-0.39, -0.17)	-0.27 (-0.34, -0.20)	-0.30 (-0.36, -0.25)	-0.24 (-0.30, -0.17)	-0.34 (-0.45, -0.23)	-0.28 (-0.35, -0.23)	-0.10 (-0.23, 0.03)	PBO

The numbers below the diagonal represent the WMDs and 95% CrI of HAQ-DI. WMDs less than 0 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). CrI, credible interval; HAQ-DI, Health Assessment Questionnaire–Disability Index; WMDs, weighted mean differences; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.

12.4.4 Discontinuations for AEs

Bar 2mg															
0.74 (0.33, 1.65)	Bar 4mg														
0.65 (0.24, 1.63)	0.88 (0.40, 1.85)	Tof 5mg													
0.64 (0.23, 1.65)	0.86 (0.38, 1.87)	0.98 (0.59, 1.61)	Tof 10mg												
0.95 (0.33, 2.65)	1.30 (0.54, 3.00)	1.46 (0.66, 3.29)	1.50 (0.66, 3.47)	Upa 15mg											
0.62 (0.20, 1.82)	0.84 (0.32, 2.12)	0.94 (0.39, 2.33)	0.97 (0.39, 2.46)	0.65 (0.31, 1.37)	Upa 30mg										
0.69 (0.29, 1.66)	0.94 (0.49, 1.79)	1.06 (0.63, 1.83)	1.09 (0.61, 2.02)	0.73 (0.36, 1.47)	1.12 (0.50, 2.55)	ADA									
0.92 (0.32, 2.65)	1.25 (0.51, 2.97)	1.42 (0.62, 3.29)	1.45 (0.61, 3.48)	0.97 (0.39, 2.44)	1.49 (0.55, 4.08)	1.33 (0.65, 2.69)	ETN								
0.54 (0.19, 1.53)	0.74 (0.30, 1.77)	0.84 (0.37, 1.95)	0.85 (0.37, 2.04)	0.57 (0.23, 1.43)	0.88 (0.33, 2.38)	0.78 (0.36, 1.65)	0.59 (0.22, 1.52)	IFX							
0.97 (0.31, 3.01)	1.31 (0.48, 3.46)	1.48 (0.57, 3.80)	1.52 (0.57, 3.96)	1.01 (0.36, 2.79)	1.55 (0.52, 4.64)	1.38 (0.56, 3.31)	1.05 (0.36, 2.98)	1.77 (0.62, 5.02)	GOL						
0.61 (0.24, 1.52)	0.83 (0.39, 1.73)	0.94 (0.48, 1.86)	0.96 (0.48, 1.95)	0.64 (0.29, 1.41)	0.99 (0.41, 2.38)	0.88 (0.50, 1.53)	0.66 (0.29, 1.48)	1.13 (0.51, 2.54)	0.64 (0.26, 1.61)	CZP					
0.75 (0.29, 1.84)	1.02 (0.47, 2.08)	1.15 (0.59, 2.21)	1.18 (0.59, 2.34)	0.79 (0.36, 1.69)	1.22 (0.51, 2.83)	1.08 (0.60, 1.89)	0.82 (0.36, 1.82)	1.38 (0.62, 3.07)	0.78 (0.32, 1.88)	1.22 (0.64, 2.29)	TCZ				
0.31 (0.07, 1.18)	0.42 (0.11, 1.43)	0.48 (0.13, 1.56)	0.50 (0.13, 1.67)	0.33 (0.08, 1.13)	0.51 (0.12, 1.86)	0.46 (0.12, 1.38)	0.34 (0.08, 1.19)	0.58 (0.14, 2.02)	0.32 (0.08, 1.22)	0.51 (0.14, 1.62)	0.42 (0.11, 1.31)	RTX			
1.16 (0.46, 2.99)	1.57 (0.74, 3.33)	1.79 (0.92, 3.56)	1.84 (0.91, 3.82)	1.22 (0.55, 2.71)	1.88 (0.78, 4.58)	1.67 (0.96, 2.94)	1.26 (0.55, 2.88)	2.14 (0.99, 4.74)	1.21 (0.48, 3.09)	1.90 (0.99, 3.70)	1.55 (0.85, 2.95)	3.69 (1.17, 14.33)	ABT		
0.33 (0.08, 1.24)	0.44 (0.13, 1.51)	0.50 (0.15, 1.67)	0.51 (0.15, 1.75)	0.34 (0.10, 1.24)	0.53 (0.14, 2.00)	0.47 (0.15, 1.44)	0.36 (0.15, 0.82)	0.60 (0.17, 2.12)	0.34 (0.09, 1.30)	0.53 (0.17, 1.73)	0.43 (0.13, 1.43)	1.05 (0.23, 5.50)	0.28 (0.08, 0.90)	csDM ARDs	
0.99 (0.43, 2.20)	1.34 (0.74, 2.39)	1.52 (0.93, 2.53)	1.56 (0.90, 2.72)	1.04 (0.54, 1.96)	1.60 (0.76, 3.35)	1.42 (0.98, 2.05)	1.07 (0.54, 2.13)	1.82 (0.94, 3.57)	1.03 (0.46, 2.30)	1.61 (1.03, 2.56)	1.32 (0.86, 2.08)	3.13 (1.10, 10.97)	0.85 (0.52, 1.37)	3.04 (1.02, 9.03)	PBO

The numbers below the diagonal represent the ORs and 95% CrI of discontinuations for AEs. ORs lower than 1 indicate that the drug in that column is better than the drug in that row (ie, favour the upper treatment). AEs, adverse events; CrI, credible interval; ORs, odds ratios; ABT, abatacept; ADA, adalimumab; Bar, baricitinib; csDMARDs, conventional synthetic disease-modifying antirheumatic drugs; CZP, certolizumab pegol; ETN, etanercept; GOL, golimumab; IFX, infliximab; PBO, placebo; RTX, rituximab; TCZ, tocilizumab; Tof, tofacitinib; Upa, upadacitinib.